General Relativity Sandbox

Design and Implementation of a Linearized GR + Electromagnetism Simulation

Comparing Full Linearized Einstein–Maxwell vs. GEM/EM Approaches

Revision v38

v37 derives the GEMPIC canonical-momentum push step in detail, then – via Fourier analysis and direct empirical comparison (Stage 41) – reaches the conclusion that GEMPIC is NOT a structural fix for the Stage 37 uniform-motion heating. The analytic argument's load-bearing assumption (that HE matched-shape kills the discrete $\nabla\phi$ at a *moving* particle) does not hold; HE only delivers zero gradient at $v = 0$. The structural problem lives in the deposit-and-gather pair, not the pusher. See the addendum §2.8. GEMPIC is committed as a default-off diagnostic (`3e0b209`) – the infrastructure is reusable if a future scheme combines it with a deposit-side construction that does target the actual structural property. Inherited from v36: **Stage 37 uniform-motion diagnosis exposes a structural defect in the discrete $-\nabla\phi - \partial_t\mathbf{A}$ cancellation on a moving deposit, and v36 specifies the benchmark suite and the canonical-momentum (GEMPIC) reformulation required to address it.** Prior orbit-test "fixes" (half-step $\mathbf{A}$ potentials, inductive sign flip) are re-labeled as partial mitigations of the orbit-level symptom rather than structural fixes. See §3 for the diagnosis, §4 for the benchmark scope, and §5 for the canonical-momentum reformulation.

Inherited from v35: implementation alignment with §15's Yee specification, plus opt-in Esirkepov current deposition. Section 15 already specifies the Yee staggered grid and the discrete continuity equation as the gauge-preservation defense; the v32–v34 implementation departed from that spec by using cell-centered storage, which broke discrete continuity and produced "grid-heating" self-forces strong enough that Stage 11 (mutual gravity via FDTD) is unreachable. v35 brings the storage layout, deposition kernels, and force-evaluation operators into line with what Section 15 prescribes, and adds Esirkepov current deposition [20] as an opt-in path beyond the truncation-order accuracy of Section 15's CIC-on-Yee recipe, giving exact discrete continuity and structural cancellation of the moving- particle self-force.

Section 6 contains the motivation and migration plan (save points S1–S7); intermediate cell-centered version preserved on the `v34-cell-centered` git branch.

All v34 physics corrections (EIH 1PN sign+Shapiro term, proper-time Eq. 105, analytic-background mode) carry forward unchanged: they are independent of the grid layout.

Contents

Changes from v32

This revision makes two changes to the v32 document:

1. **Implementation plan, §18**: a new stage is inserted between the original Stages 5 and 6, introducing *sampled background field arrays* as a prerequisite for the particle-dynamics tests. All subsequent stages are renumbered (old Stages 6–16 become new Stages 7–17). The motivation is testability: with sampled backgrounds available before the perturbation-source tests, a single test particle can be placed in any prescribed background (Keplerian orbit around a softened point mass, gyroscope in a frame-dragging field, charge in an external Coulomb potential, etc.) without first having to set up a second body and the FDTD source-deposition path. The full background-modes coverage formerly in old Stage 14 (Schwarzschild, Kerr, Reissner–Nordström, Kerr–Newman, binary) remains in the new Stage 15, where it expands the library of generators introduced in the new Stage 6.

2. **§15.7 clarification**: the round-trip reflection $R \approx 10^{-3}$ derived there is the plane-wave normal-incidence limit; the realized in-interior residual for a 2D circular wavefront is observed to be several percent due to two effects — the $(1 - \sigma\,\mathrm{d}t)$ linearization differing from $\exp(-\sigma\,\mathrm{d}t)$ at the $\sigma_{\max}\,\mathrm{d}t \sim 0.46$ used in practice, and oblique-incidence waves into the grid corners being weakly damped by the axis-aligned $d_{i,j} = \max(d_x, d_y)$ profile. Both effects would be mitigated by a PML; for the qualitative pedagogical purpose of the sandbox, the simple damping layer is adequate. See the new paragraph at the end of §15.7 for the empirical figures.

All physics chapters (§8.2 through §15.7 and beyond) are unchanged from v32. Cross-references to `eq:` labels remain valid.

Changes from v37 (this revision)

After v37's negative result on GEMPIC, the direction pivoted from pusher rewrites to deposit-side mitigation: suppress the spurious self-force on a moving deposit by making the macroparticle wider or smoother. v38 scopes a parameterizable kernel design that addresses three concerns Dan articulated:

1. Kernel size should be a runtime knob, not a code path per order. B-splines $W_4, W_5, \ldots$ as separate integer orders each need their own Esirkepov implementation. A parameterizable kernel with continuous scale R gives one implementation, runtime-configurable.
2. Truncated B-splines (and truncated Gaussians) are not the optimal compactly-supported kernels. The variational result: minimising $\int k^2|\hat{W}(k)|^2 \mathrm{d}k$ subject to compact support and normalisation gives the lowest Dirichlet-Laplacian eigenmode of the support domain. In 1D this is a half-cosine; in 2D circular it is the first lobe of J_0.
3. The C^∞ bump $\exp(-1/(1-x^2))$ has sub-exponential Fourier decay $\sim k^{-3/4}\exp(-c\sqrt{k})$ – faster than any polynomial $1/k^n$. Best tail-decay candidate at compact support.

New section §1 scopes the design space, the implementation tiers, and the comparison plan. Code in this revision: `GR_SHAPE_BUMP` added behind shape-function dispatch on the rho / phi-gradient-gather side of the HE adjoint; J / Esirkepov stay on TSC for this first cut. Stage 43 measures whether kernel shape matters at fixed 3-cell support. If yes, follow-up work extends to parameterizable R and bump-side Esirkepov.

All v36 / v37 content carries forward unchanged. The v37 addendum §2.8 (GEMPIC empirically disproven) stands; v38 is the next attempt at the deposit-side mitigation it pointed to.

1 Parameterizable Macroparticle Kernel

1.1 Motivation: deposit-side fix for Stage 37

The Fourier analysis in v37 §2.8 pinned the Stage 37 spurious force to the discrete $\partial_y^h \phi$ at the moving particle position. In Fourier:

$$\partial_y^h \phi(\mathbf{x}_p) \;\propto\; \int k_y\, \hat{W}(\mathbf{k}) \left[\hat{G}_h(\mathbf{k}, \mathbf{k}\cdot\mathbf{v}_0) - \hat{G}_h(\mathbf{k}, -\mathbf{k}\cdot\mathbf{v}_0)\right] \mathrm{d}^2k.$$

The factor of k_y shifts the integrand's dominance toward higher k than the field itself would (which is $\int \hat{W}\hat{G}$ with no k_y). Suppressing $\hat{W}$ at high k is the relevant lever.

1.2 Kernel candidates and their high-k behaviour

Kernel (1D, compact $[-R, R]$)	Boundary smoothness	$\hat{W}(k)$ tail
Rectangle (W_1)	C^{-1} (jump)	$1/k$
CIC (W_2, tent)	C^0 (corner)	$1/k^2$
TSC (W_3, quadratic B-spline)	C^1	$1/k^3$
Cubic B-spline (W_4)	C^2	$1/k^4$
Half-cosine $\cos(\pi x/(2R))$	C^0 (slope jump)	$1/k^2$
Raised cosine $\frac{1}{2}(1+\cos(\pi x/R))$	C^1	$1/k^3$
Bump $\exp(-1/(1-(x/R)^2))$	C^∞	$k^{-3/4}\exp(-c\sqrt{k})$ *
Bessel $J_0(\alpha r/R)$ (2D, $r < R$)	C^0	$1/k^2$

The variational optimum for $\min \int k^2 |\hat{W}|^2 \mathrm{d}k$ on compact support with Dirichlet $W(\pm R) = 0$ is the half-cosine in 1D (Dirichlet Laplacian ground state $W'' = -\lambda W$, $W = \cos(\pi x/(2R))$, $\lambda = (\pi/(2R))^2$) and $J_0(\alpha r/R)$ on the 2D disk ($\alpha \approx 2.405$). These minimise the mean-squared frequency spread but have only C^0 extension at the boundary, so their tail decay is only $1/k^2$.

The bump function has C^∞ extension at the boundary, giving sub-exponential decay – faster than any polynomial. Best tail-decay candidate.

Which criterion matters for Stage 37 depends on where the asymmetric part of $\hat{G}_h$ lives in k-space. Worth testing empirically.

1.3 Implementation tiers

Tier 1 (this revision, minimal viable test): add `GR_SHAPE_BUMP` as a new `shape_function` option, with bump weight + derivative evaluation, used in:

- rho deposit on corners,
- LB-style $\nabla\phi$ gather (the spurious-force channel).

Keep TSC for J / Esirkepov and **A** interpolation. This breaks the rho-J shape symmetry but preserves HE adjoint on the rho/ϕ-gather pair, so Stage 39 (static reciprocity at $v = 0$) should still pass. Stage 43 measures bump-vs-TSC on the phi-only force. Bump support: 3 cells (matched to TSC for first cut).

Tier 2: extend bump kernel to J Esirkepov so rho-J shape is consistent. Requires a parameterizable Esirkepov over a variable-width cell window. 200 lines plus tests.

Tier 3: introduce continuous scale parameter R (`gr_sim_set_kernel_radius`). Cell-window size adapts to R; auto-scale $R \propto 1/\mathrm{d}x$ for fixed physical width.

Tier 4: add comparison kernels (half-cosine, Bessel J_0 first lobe) as additional `shape_function` options. Empirical comparison of all options.

Tier 5: if any prior tier validates the principle, promote the chosen kernel + auto-scaled R to production default. Mark Boris-pusher and TSC path as legacy.

1.4 Tier 1: bump weight on 3-cell support

Map the natural support of $\exp(-1/(1-u^2))$ ($u \in [-1, 1]$) onto cell-distance range $[-1.5, 1.5]$ to match TSC. Define

$$b(d) = \frac{1}{N_b} \exp\left(-\frac{1}{1-(d/1.5)^2}\right) \quad \text{for } |d| < 1.5,$$

zero otherwise. N_b is the normalisation $\int_{-1.5}^{1.5} \exp(-1/(1-(d/1.5)^2))\mathrm{d}d \approx 0.6660$ so that $\int b(d)\mathrm{d}d = 1$.

Derivative:

$$b'(d) = b(d) \cdot \frac{-2(d/1.5)}{1.5\,(1-(d/1.5)^2)^2}.$$

Both b and b' go smoothly to zero at $d = \pm 1.5$ in all derivatives (C^∞ extension), and $b'(0) = 0$ by symmetry.

Cell-sum normalisation

For a particle at sub-cell offset $u \in [-0.5, +0.5]$ from its nearest cell, weights at cell offsets $\{-1, 0, +1\}$ are $w_i = b(i - u)$, normalised so $\sum w_i = 1$ at every u. Note: the function b is normalised so its INTEGRAL is 1, but the discrete SUM at cell positions varies with u. In practice we either (a) renormalise the discrete weights to sum to 1 (preserves charge conservation but changes the effective kernel shape away from pure bump) or (b) accept $O(10^{-2})$ charge non-conservation per particle and document.

Recommended: option (a) – renormalise. The shape on the grid is then "discretised bump" which loses some of the bump's C^∞ boundary smoothness but keeps the right qualitative high-k behaviour.

Stage 43 acceptance

Compare TSC vs bump on Stage 37's v-scan at `mass=1.0, rho_smooth=0`. For each v, report:

- `accel(phi-only, TSC)`,
- `accel(phi-only, BUMP)`,
- `ratio = accel(BUMP) / accel(TSC)`.

Also verify Stage 39 reciprocity for BUMP (should pass at $v = 0$).

Threshold for "kernel shape matters at fixed support": `|ratio| < 0.5` at at least half the v values. If this holds, Tier 2 / 3 are worth pursuing. If `|ratio|` stays near 1 at every v, kernel shape doesn't matter at fixed support and we move to widening (Tier 3) instead of shape-comparison.

Threshold for "Tier 1 should not regress anything": Stage 39 reciprocity ratio still $< 10^{-6}$ at every separation (BUMP preserves HE adjoint on the rho/ϕ-gather pair).

1.5 Empirical findings (2026-05-25): equal-distance methodology

The Tier 1 implementation (commit 425ed05) was extended to a *parameterized* bump kernel with continuous radius R in cell units (commit on this revision). The radius controls a variable cell window: 2*ceil(R+0.5)+1 cells per axis, capped at 17 (R=8). API: `gr_sim_set_kernel_radius(sim, R)` – default 1.5 matches TSC's 3-cell support.

Measurement methodology correction

Initial measurements (Stage 41, Stage 43 v1) used a fixed 200-step force-measurement window regardless of v. This meant at low v the particle moved < 1 cell during the window, and the measured force reflected the spurious force at a single (nearly fixed) sub-cell offset rather than averaging over the full sub-cell distribution. This produced a spurious "v=0.005 spike" that was purely a methodology artifact.

Stage 43 v2 (equal-distance) fixes this: window steps scale as $5/(v\,\mathrm{d}t)$ so the particle traverses 5 cells at every v, sampling the full sub-cell-offset distribution. Heavy particle mass ($m = 100$) keeps the spurious force from changing v materially over the longer windows.

TSC baseline – corrected absolute spurious force

v/c	accel (TSC bare, cells/dt^2)	n_window
0.001	-1.12×10^{-8}	7071
0.005	-7.40×10^{-9}	1414
0.010	-4.67×10^{-9}	707
0.050	-2.37×10^{-9}	200
0.100	$+1.42 \times 10^{-9}$	200
0.200	$+2.11 \times 10^{-10}$	200
0.300	$+4.43 \times 10^{-9}$	200
0.500	$+3.20 \times 10^{-8}$	200

The absolute spurious force is $\sim 10^{-9}$ to 10^{-8} across the whole v range, much smaller than the $\sim 10^{-6}$ to 10^{-5} numbers reported under the old methodology. The nonrelativistic regime IS stable in absolute terms; previous characterisations were contaminated by sub-cell-offset bias in the fixed-window measurement.

Bump-R vs TSC + smoothing: equivalent effectiveness

Ratio table (|accel/accel(TSC)|, equal-distance methodology):

v/c	bump R=2.5	R=4	R=6	R=8	TSC N=4	N=16	N=64
0.001	1.02	0.74	0.88	0.91	0.93	0.91	0.91
0.005	0.97	0.98	0.98	0.98	0.98	0.98	0.98
0.010	0.99	1.01	1.02	1.01	1.02	1.02	1.01
0.050	0.33	0.06	0.02	**0**	0.04	**0**	**0**
0.100	2.93	0.37	0.07	**0**	0.30	**0**	**0**
0.200	14.0	8.0	**0**	**0**	**0**	**0**	**0**
0.300	0.19	0.33	0.10	**0**	**0**	**0**	**0**
0.500	0.24	0.11	**0**	**0**	**0**	**0**	**0**

0 entries indicate the spurious force has dropped below float32 noise floor.

What this tells us

1. TSC's bare spurious force is already $\sim 10^{-8}$ at typical pedagogical v, much smaller than previous methodology suggested. The artifact is not as severe as we'd been treating it – it was inflated by measurement noise.
2. For $v \geq 0.05$, both bump-R at $R \geq 6$ and TSC + smoothing at $N \geq 16$ suppress the spurious force below float32 noise floor. Effectively equivalent in result.
3. For $v \leq 0.01$, neither approach helps because the TSC artifact is already at the noise floor at low v. Nothing further to suppress.
4. Choice between bump-R and smoothing-N is now about *implementation cost*, not effectiveness:
    - Smoothing: per-step grid convolution, scales with grid size $W \cdot H$, independent of particle count. At $N = 16$, roughly quadruples FDTD step cost. Affects ρ and $\vec{J}$ consistently.
    - Bump-R: per-particle deposit only, scales as $N_{\text{particles}} \cdot R^2$, independent of grid size. For pedagogical few-particle sims at $R = 6$ (~ 169 cells per particle), cost is negligible. Currently only affects ρ deposit and ϕ gather; full coverage needs Tier 2.
5. For real-time pedagogical demos (few particles, target 60 fps), bump-R is the better choice on cost grounds. For plasma or high-particle-count regimes, smoothing wins.

Implementation status and Tier 2 decision

Bump-R implementation in tree (commits 425ed05 + this revision):

- `GR_SHAPE_BUMP` enum, `gr_sim_set_kernel_radius`.
- Variable-width ρ deposit on corners, parameterised by R.
- Variable-width ϕ-gradient LB gather on corners.
- Stage 39 reciprocity passes (ratio = 0 to float precision).

NOT yet in tree:

- Bump Esirkepov for $\vec{J}$ on edge sublattices.
- Bump interp on X-edge / Y-edge for $\vec{A}$ reads.
- Bump LB gradient gathers on edge sublattices (`gr_bump_lb_dx_xedge` etc.).

For phi-only Stage 37 tests the current implementation is complete and sufficient. For *full-force* EM PIC orbit tests (Stage 27 type) with the bump kernel, Tier 2 is required to maintain shape consistency between ρ and $\vec{J}$ wakes.

Tier 2 LANDED (commit 98b4fe9): bump Esirkepov + edge interp

Tier 2 implementation:

- `gr_bump_interp_xedge` / `_yedge`: variable- window bump interp on edge sublattices. Used by `dt_A_em_at_total` when shape=BUMP.
- `gr_bump_esirkepov_deposit_jxy`: variable-window Esirkepov current decomposition with renormalised bump weights. Same separable algebra as existing CIC Esirkepov.
- Dispatch in `sim.c` (Esirkepov path) and `particle.c` ($\vec{A}$ edge interp).

Validation – new Stage 44:

- Part A: bump Esirkepov continuity verified at 5 values of R x 5 trajectory shapes = 25 cases. All max|residual| $\sim 10^{-8}$ (float32 noise). Discrete continuity $\partial_t \rho + \nabla \cdot \vec{J} = 0$ holds at every corner.
- Part B: full force (inductive ON), bump R=8 and TSC + N=16 smoothing are EMPIRICALLY EQUIVALENT. Both drop spurious accel below float32 noise floor at $v \geq 0.05$. Confirms the deposit-side mitigation works equivalently in either form.

Production recommendation (v38, post-Tier-2)

For real-time pedagogical EM PIC closed-loop demos:

```
gr_sim_set_shape_function(sim, GR_SHAPE_BUMP);
gr_sim_set_kernel_radius (sim, 6.0f);  /* or 8.0f */
```

Cost: O($N_{\text{particles}} \cdot (2\text{hw}+1)^2$) per step for the deposit; negligible vs FDTD for few-particle pedagogical sims.

For high-particle-count plasma simulations: switch to TSC + smoothing `rho_smooth_passes = 16` (or higher), since per-step convolution cost is constant in particle count.

Both paths are now production-ready and validated.

Tier-2 gap closure (commit e571c54)

After the initial Tier-2 commit, an audit revealed three partial- coverage gaps:

1. `B_em_z_at_total` and `B_g_z_at_total` (curl $\mathbf{A}$ for the $v \times \mathbf{B}$ Lorentz force) used a centered-difference + CIC interp gather (`curl_Agz_sampled_at`) with no shape dispatch.
2. `grav_grad_at_lb` (gravity-side $\nabla\Phi_g$ gather) had TSC and CIC branches but no BUMP.
3. `dt_A_g_at_total` (gravity-side $\partial_t \mathbf{A}_g$ edge interp) had only CIC.

All three closed by:

- Four new bump-LB mixed-axis edge gathers in deposit.c: `gr_bump_lb_dx_xedge`, `_dy_xedge`, `_dx_yedge`, `_dy_yedge`.
- New `bump_curl_Az_at` helper in particle.c that computes $B_z = \partial_x A_y - \partial_y A_x$ via the bump-LB mixed gathers. Used by both `B_em_z_at_total` and `B_g_z_at_total` when shape=BUMP.
- BUMP branches added to `grav_grad_at_lb` and `dt_A_g_at_total`, parallel to the existing EM-side BUMP branches.

After gap closure (Stage 45 EM, Stage 46 GR Kepler):

config	EM (before)	EM (after)	GR (before)	GR (after)
TSC bare	−5.60%	−5.60%	−8.78%	−8.78%
TSC + $N = 16$	−2.46%	−2.46%	−2.32%	−2.32%
BUMP $R = 6$	−3.24%	−2.43%	−3.00%	−2.11%
BUMP $R = 8$	−2.80%	−2.34%	−2.30%	−2.05%

Bump $R = 8$ now slightly beats TSC + $N = 16$ on both sectors – the HE adjoint pair is matched end-to-end and the bump approach is the better choice in addition to being cheaper per step.

Convenience API: `gr_sim_use_pedagogical_defaults`

After the cleanup of debugging knobs (commits 5735ef7, d4f1f5a), a single convenience function applies the recommended pedagogical configuration in one call:

```
gr_sim_t* sim = gr_sim_create(W, H, dx, c_eff, cfl);
gr_sim_use_pedagogical_defaults(sim);
// ... add particles, background, etc.
```

The bundle sets: `shape_function = GR_SHAPE_BUMP`, `kernel_radius = 6.0`, `force_interp = LEWIS_BIRDSALL`, `particle_source_deposition = 1`, `damping = 16`. Tests that need to exercise a specific non-default knob (Stage 11 for CIC Esirkepov continuity, Stage 18 for LB regression, etc.) should NOT call this and instead set required knobs explicitly.

`gr_sim_create` defaults are NOT changed; this is a strictly additive convenience. Existing tests' calibrations hold. Migration to the bundle is incremental and per-test as appropriate.

Stage 47: equal-coupling EM binary – bump kernel still works

Stages 45 (EM Kepler) and 46 (GR Kepler) use a *test particle* configuration: $q_{\text{test}}/Q_{\text{central}} = 10^{-3}$ or $m_{\text{test}}/M_{\text{central}} = 10^{-3}$. Stage 47 is the analog with BOTH bodies as full PIC sources at equal coupling: $m = Q = 0.01$, opposite charges. In this regime self/mutual force ratio is 1 – the self-force per particle is comparable to the mutual force.

Methodology caveat from initial run (corrected below): The first Stage 47 attempt did not explicitly disable gravity ($G_{\text{eff}} = 1$ at default). With $m = Q$ and $G_{\text{eff}} = k_e$, 2D-log gravitational attraction is COMPARABLE to EM attraction at the same parameters, so the orbital dynamics were actually mixed EM + gravity, not pure EM. Mixed dynamics produced large chaotic drift (−60% to −90% over 4 orbits) that misled the initial conclusion.

Corrected setup ($G_{\text{eff}} = 0$, gravitomagnetic disabled):

config	final radius drift after 4 orbits
TSC bare	**UNBOUND** (only 1/4 orbits completed)
TSC + $N = 16$	+5.6% (bounded)
BUMP $R = 6$	−51.2% (marginal – possibly kernel-edge effect)
BUMP $R = 8$	+5.3% (bounded)

Bump $R = 8$ delivers +5.3% drift – comparable to Stage 45/46's test-particle results. Smoothing $N = 16$ gives essentially the same behavior. TSC bare is fully unbound after 1 orbit; the equal-coupling self-force without mitigation destroys the orbit immediately.

The marginal BUMP $R = 6$ result deserves note: at $r_{\text{orb}} = 20$, the kernel width $2\,\text{ceil}(R + 0.5) + 1 = 15$ cells. Two particles on opposite sides of the orbit have separation $2r = 40$ cells. Kernel overlap is $2 \cdot 7 < 40$, so no real-space overlap. But the long tails of the bump deposits may overlap subtly in ways that perturb the orbit; $R = 8$ (17-cell width) appears more robust here. Could be investigated further but isn't urgent for production.

Pedagogical scope implication: the v38 production setup (bump kernel, $R = 6$–8) is reliable for BOTH:

- Test particle around analytic background (Stages 45/46, $\sim$ 2–3% inspiral)
- Equal-coupling fully-PIC binaries (Stage 47, $\sim$ 5% drift at $R = 8$, EM-only)

Both demos are stable and quantitatively interpretable.

Stage 48: Bertrand-aware conservation diagnostic

Dan, 2026-05-26: "With a 1/r force law, the interaction is strong, there may not be orbits that are closed (Bertrand's theorem), they may oscillate or precess. There are a number of things to check: that the absorbing boundary conditions are doing their job, momentum/energy conservation of the two bodies, and what happens in the limit of nonrelativistic velocities."

Bertrand's theorem says only $1/r^2$ (3D Kepler) and r^2 (harmonic) central forces give closed orbits. GRlite's 2D log Coulomb gives force $\propto 1/d$, which is NOT Bertrand: any deviation from exactly circular initial condition gives a non-closed rosette with radial oscillation between $r_{\min}$ and $r_{\max}$ while θ advances. This means Stage 47's "% drift in radius at each θ-wrap" metric conflates physical rosette behavior with numerical heating.

Stage 48 replaces that with CONSERVATION LAWS, which must hold regardless of orbit shape:

- $E_{\text{tot}} = E_{\text{kin}} + U_{\text{int}}$ with $U_{\text{int}}(d) = +2k_eQ^2\ln(d)$ (attractive 2D Coulomb)
- L_z (total) about COM
- P_x, P_y (linear momentum, ~ 0 by symmetry)
- max $|\phi_{\text{em}}|$ at PML inner edge vs. bulk

Three configurations probe Dan's concerns:

(A) m=0.01, v_{factor}=1 (v_{orb}=0.1, mildly rel. circular): TSC bare chaotic; TSC+N=16, BUMP R=6, BUMP R=8 all preserve E to ±3% and L_z to ±15–21% over 4 orbits. Separation oscillates ±10% – this is the physical rosette, not numerical drift, since E is well-conserved. PML edge/bulk field ratio $\sim$ 10%.

(B) m=0.01, v_{factor}=0.1 (v_{orb}=0.01, low-L NOT circular): This is NOT a non-relativistic test – it sets L_z to 1/10 the circular value at d=40. In the 2D-log effective potential $V_{\text{eff}}(d) = L^2/(2m_{\text{red}}d^2) + 2k_eQ^2\ln(d)$, the inner turning point sits at $d_{\min} \approx 1.55$ cells – below all kernel scales. The orbit physically dives toward $d_{\min}$. At close approach, deposit/gather kernels overlap and PIC heating dominates; particles get kicked apart. All four configs fail catastrophically (separation grows 2000–8000%, P_x drifts to 5×10^{-3}). PML ratio 24–44% (worse static- field absorption at low v). Not a kernel-mitigation failure – a fundamental limit of PIC at sub-kernel close approach.

(C) m=1.0, v_{factor}=1 (v_{orb}=0.01, PROPER non-rel circular): True non-relativistic limit with circular IC. Heavier particles give $v_{\text{orb}} = Q\sqrt{k_e/m} = 0.01$ at d=40, and V_{eff} has its minimum at d=40 – so any perturbation gives only small rosette amplitude. Results over 4 orbits:

config	$\Delta E/\lvert E_0\rvert$	$\Delta L_z/\lvert L_0\rvert$	$\lvert d-d_0\rvert/d_0$	$\lvert P_x\rvert$ max
TSC bare	±2%	−198% (reversed!)	0–+68%	5×10^{-4}
TSC + N=16	±0.45%	±0.8%	±13%	1×10^{-7}
BUMP R=6	±0.37%	±1.3%	±12%	2×10^{-4}
BUMP R=8	±0.35%	±1.1%	±13%	1×10^{-4}

Mitigation configs preserve E to better than 0.5% and L_z to better than 1.5% in proper non-rel circular. TSC+N=16 has remarkable momentum conservation (10^{-7}); the bump kernels give $\sim 10^{-4}$ which is small but suggests a sub-cell asymmetry in the bump weight evaluation worth probing.

Answers to Dan's three checks:

- **PML doing job**: yes, $\sim 10\%$ absorbing ratio at circular orbits. Worse (24–44%) when orbit collapses to small d, but that's a different failure mode (close-approach near-field swamps).
- **Conservation of E and L_z**: $< 0.5\%$ in proper non-rel circular for all mitigation configs. $\sim 3\%$ for E and $\sim 15\%$ for L_z at $v/c = 0.1$ (Case A); plausibly relativistic corrections to L_z definition (we used canonical angular momentum which mixes mechanical and field momentum at higher v/c).
- **Non-rel limit**: clean. Case C shows the v38 production setup works correctly at $v/c = 10^{-2}$.

Stage 47 % drift retraction (further): The $\sim 5\%$ "drift" reported in Stage 47 BUMP $R = 8$ over 4 θ-wraps was mostly the 2D-log rosette physics, not numerical error. Real numerical drift (from E conservation) is $< 5\%$ at $v_{\rm orb} = 0.1$ and $< 0.5\%$ at $v_{\rm orb} = 0.01$. The bump kernel is performing better than the radius-drift metric implied.

Changes from v36 (the v37 revision)

This revision works out the GEMPIC canonical-momentum time-stepping flagged as "to be locked via literature check" in v36 §5. Three concrete additions:

1. A clean Hamilton's-equations derivation of the canonical EOMs, including the algebraic identity that converts $q\nabla(\mathbf{v}\cdot\mathbf{A})$ in the canonical force law to the Lorentz-form $-q\partial_t\mathbf{A} + q\mathbf{v}\times\mathbf{B}$ at the continuum level. Both forms are exact; GEMPIC uses the first because it requires only scalar-gradient discrete operators that the Yee staggering naturally provides.
2. A concrete *relativistic leapfrog* scheme for the canonical push, with explicit time-staggering for p_c, x, and the field reads. Treats the implicit dependence of $\mathbf{v}$ on p_c at the relativistic level via a single fixed-point step that we show converges in O(few) iterations for $v < 0.9c$ and is exact for non-relativistic v.
3. An *analytic verification* that the proposed discrete force law gives identically zero net force on a uniformly-moving point deposit, in the limit of a static field source and with matched TSC deposit / TSC-derivative force gather. This is the structural guarantee the Stage 37 diagnosis showed is absent from the current $-\nabla\phi - \partial_t\mathbf{A}$ pairing.

All v36 content (Stage 37 diagnosis, benchmark suite scope, GEMPIC high-level scope) carries forward unchanged.

Important update (2026-05-24, addendum §2.8): after writing the derivation above, the load-bearing claim of the analytic argument was checked via direct Fourier analysis of the discrete Yee operator and via an empirical comparison (Stage 41) of Boris vs GEMPIC pushers on the Stage 37 setup. Both checks agree: **GEMPIC is not a structural fix.** At every v in the Stage 37 sweep, the GEMPIC spurious acceleration is within a few percent of the Boris value. The structural problem lives in the discrete deposit's $\nabla^h\phi$ at the moving particle position (not in the pusher form), and neither pusher addresses it.

GEMPIC remains in the tree as a default-off diagnostic (commit `3e0b209`); the next scope item is a deposit-side construction that targets the actual structural property at fault. The original derivation in §2 below is preserved verbatim for the historical record; the addendum §2.8 corrects its load-bearing claim.

2 GEMPIC Time-Stepping: Derivation and Discrete Scheme

2.1 Recap of the structural problem (v36 abridged)

Stage 37 (§3) showed that the discrete operator pair $-\nabla_h\phi$ and $-\partial_t^h\mathbf{A}$ does not constitute a discrete adjoint cancellation for a moving deposit: the residual force is UV-divergent in Δx, dt-independent, and grows worst at low v. The current Boris pusher tracks the *kinetic* momentum $\mathbf{p}_{\rm kin} = \gamma m\mathbf{v}$ and computes $\mathrm{d}\mathbf{p}_{\rm kin}/\mathrm{d}t$ via the Lorentz force law, which puts the broken $-\partial_t\mathbf{A}$ operator directly in the push. GEMPIC removes that operator by tracking canonical momentum instead.

2.2 Continuum derivation: Hamilton's equations

A relativistic charged particle in an external electromagnetic field has the canonical Hamiltonian

$$H(\mathbf{x}, \mathbf{p}_c; t) = c\sqrt{m^2c^2 + |\mathbf{p}_c - q\mathbf{A}(\mathbf{x}, t)|^2} \; + \; q\,\phi(\mathbf{x}, t), \tag{1}$$

where $\mathbf{p}_c$ is the canonical momentum, related to the kinetic momentum by

$$\mathbf{p}_c \;=\; \mathbf{p}_{\rm kin} + q\mathbf{A}(\mathbf{x}, t) \;=\; \gamma m\mathbf{v} + q\mathbf{A}(\mathbf{x}, t). \tag{2}$$

Hamilton's equations are

$$\dot{\mathbf{x}} \;=\; \frac{\partial H}{\partial \mathbf{p}_c} \;=\; \frac{c^2(\mathbf{p}_c - q\mathbf{A})}{\sqrt{m^2c^4 + c^2|\mathbf{p}_c - q\mathbf{A}|^2}} \;=\; \mathbf{v}, \tag{3}$$

$$\dot{\mathbf{p}}_c \;=\; -\frac{\partial H}{\partial \mathbf{x}}. \tag{4}$$

Computing $\partial H/\partial x_i$ at fixed $\mathbf{p}_c$:

$$\begin{aligned}\frac{\partial H}{\partial x_i} &= \frac{c^2(p_{c,j} - qA_j)}{H_{\rm kin}} \cdot (-q)\,\frac{\partial A_j}{\partial x_i} + q\,\frac{\partial \phi}{\partial x_i} \\ &= -q\,v_j\,\frac{\partial A_j}{\partial x_i} + q\,\frac{\partial \phi}{\partial x_i},\end{aligned} \tag{5}$$

where $H_{\rm kin} = c\sqrt{m^2c^2 + |\mathbf{p}_c - q\mathbf{A}|^2} = \gamma mc^2$. Therefore

$$\dot{p}_{c,i} \;=\; q\,v_j\,\frac{\partial A_j}{\partial x_i} \;-\; q\,\frac{\partial \phi}{\partial x_i}. \tag{6}$$

The sum convention is Einstein. The term $qv_j(\partial A_j/\partial x_i)$ is $q\,\nabla_i(\mathbf{v}\cdot\mathbf{A})$ in the convention where $\mathbf{v}$ is held constant during the gradient (which it is in the Hamiltonian derivation, since the partial derivative is at fixed $\mathbf{p}_c$ and hence at fixed $\mathbf{v}$ except through $\mathbf{A}(\mathbf{x})$).

2.3 Equivalence to the Lorentz form (continuum identity)

Eq. (6) is on the canonical momentum. The relation to the kinetic momentum, from Eq. (2), is

$$\dot{\mathbf{p}}_{\rm kin} = \dot{\mathbf{p}}_c - q\,\dot{\mathbf{A}}(\mathbf{x}(t), t) = \dot{\mathbf{p}}_c - q\,\partial_t\mathbf{A} - q\,(\mathbf{v}\cdot\nabla)\mathbf{A}. \tag{7}$$

Substituting Eq. (6) and the vector identity

$$\nabla(\mathbf{v}\cdot\mathbf{A}) - (\mathbf{v}\cdot\nabla)\mathbf{A} = \mathbf{v}\times(\nabla\times\mathbf{A}), \tag{8}$$

where $\nabla(\mathbf{v}\cdot\mathbf{A})$ in this identity has the same "$\mathbf{v}$ constant" convention as Eq. (6), gives the standard Lorentz form:

$$\dot{\mathbf{p}}_{\text{kin}} = -q\nabla\phi - q\partial_t\mathbf{A} + q\mathbf{v}\times\mathbf{B} \quad \text{with} \quad \mathbf{B} = \nabla\times\mathbf{A}. \tag{9}$$

So at the continuum level, the canonical Eq. (6) and the kinetic Lorentz form are algebraically equivalent. They differ in *which discrete operators* appear when we discretise. GEMPIC uses Eq. (6) because it requires only scalar-gradient operations on ϕ and $\mathbf{A}$ at the particle position; no discrete $\partial_t\mathbf{A}$ and no discrete $\nabla\times\mathbf{A}$. The discrete $\partial_t\mathbf{A}$ is the operator the Stage 37 diagnosis showed does not cancel against the discrete $\nabla\phi$ on a moving deposit, and is what is structurally absent from the GEMPIC force law.

2.4 2D specialisation of the force law

In 2D ($\mathbf{x} = (x, y)$, $\mathbf{A} = (A_x, A_y)$, no z-dependence) the GEMPIC force has two components:

$$F_x = \dot{p}_{c,x} = q\,v_x\frac{\partial A_x}{\partial x} + q\,v_y\frac{\partial A_y}{\partial x} - q\,\frac{\partial\phi}{\partial x}, \tag{10}$$

$$F_y = \dot{p}_{c,y} = q\,v_x\frac{\partial A_x}{\partial y} + q\,v_y\frac{\partial A_y}{\partial y} - q\,\frac{\partial\phi}{\partial y}. \tag{11}$$

The four mixed-component derivatives $\partial A_i/\partial x_j$ ($i, j \in \{x, y\}$) are evaluated at the particle position by shape-function-weighted gathers from the Yee-edge sublattices on which A_x (X-edge) and A_y (Y-edge) live.

2.5 Discrete operators – what we have, what we need

The current code (post-Stage 37 additions) already implements:

- Corner-stored ϕ with TSC value interpolation `gr_tsc_interp_corner` and LB-style $\partial_x\phi$, $\partial_y\phi$ gathers (`phi_em_grad_at_lb` in `core/src/particle.c`).
- X-edge stored A_x with TSC value interpolation `gr_tsc_interp_xedge` and LB-style $\partial_x A_x$ gather `gr_tsc_lb_dx_xedge`.
- Y-edge stored A_y with TSC value interpolation `gr_tsc_interp_yedge` and LB-style $\partial_y A_y$ gather `gr_tsc_lb_dy_yedge`.

What GEMPIC additionally needs (the "mixed" gradients):

- $\partial_y A_x$ at the particle position: gather from X-edge with a kernel that is the y-derivative of TSC. New: `gr_tsc_lb_dy_xedge`.
- $\partial_x A_y$ at the particle position: gather from Y-edge with a kernel that is the x-derivative of TSC. New: `gr_tsc_lb_dx_yedge`.

Both new gathers reuse the existing dW_3/du kernel (`tsc_dw_1d` in `core/src/deposit.c`); only the sublattice indexing changes. Estimate: 50 lines each plus tests.

2.6 Discrete time-stepping: leapfrog on the canonical state

The state we evolve is $(\mathbf{x}^n, \mathbf{p}_c^{n-1/2})$ at integer spatial step n and half-step canonical momentum, paired with the existing field state ($A_{\text{prev}} = \mathbf{A}^{n-1/2}, A_{\text{curr}} = \mathbf{A}^{n+1/2}, \phi^n$) – the half-step $\mathbf{A}$ convention from commit 9769217 carries through unchanged.

The proposed leapfrog step from n to $n+1$:

Step 1. Read fields at $(\mathbf{x}^n, t^n)$. Compute at the particle position via the existing TSC interpolators and the two new mixed-derivative gathers: $\mathbf{A}^n$ (the time-average $(A_{\text{prev}} + A_{\text{curr}})/2$, then spatially interpolated), $\partial_i \phi^n$, and $\partial_j A_i$ for all four (i, j) pairs.

Step 2. Compute $\mathbf{v}^{n-1/2}$ from $\mathbf{p}_c^{n-1/2}$ and $\mathbf{A}^{n-1/2}$. Read $\mathbf{A}^{n-1/2}$ at $\mathbf{x}^n$ (the time-staggered field at the time-symmetric position). Define $\mathbf{p}_{\text{kin}}^{n-1/2} = \mathbf{p}_c^{n-1/2} - q\mathbf{A}^{n-1/2}$. Compute $\gamma^{n-1/2} = \sqrt{1 + |\mathbf{p}_{\text{kin}}^{n-1/2}|^2/(mc)^2}$ and $\mathbf{v}^{n-1/2} = \mathbf{p}_{\text{kin}}^{n-1/2}/(\gamma^{n-1/2} m)$.

Step 3. Kick. Compute the discrete force using Eqs. (10)–(11) with $\mathbf{v} = \mathbf{v}^{n-1/2}$ (first-order in time on the velocity but second-order on the field reads since the field gradients are at t^n):

$$\mathbf{F}^n \;=\; q\,(\mathbf{v}^{n-1/2} \cdot \nabla)\mathbf{A}|_n - q\,\nabla\phi|_n. \tag{12}$$

Wait – this is $q(\mathbf{v} \cdot \nabla)\mathbf{A}$, NOT $q\nabla(\mathbf{v} \cdot \mathbf{A})$. Which is correct?

From Eq. (6): $\dot{p}_{c,i} = q v_j \partial_i A_j - q \partial_i \phi$. The free index i is the component of $\mathbf{F}$; the summed index j is the component of $\mathbf{v}$ that pairs with the component of $\mathbf{A}$ being differentiated. So we need $\partial_i A_j$ where i is the FORCE direction and j is the velocity component.

For 2D:

$$F_x = q\,v_x\,\partial_x A_x + q\,v_y\,\partial_x A_y - q\,\partial_x \phi, \tag{13}$$
$$F_y = q\,v_x\,\partial_y A_x + q\,v_y\,\partial_y A_y - q\,\partial_y \phi, \tag{14}$$

matching Eqs. (10)–(11).

Apply the kick:

$$\mathbf{p}_c^{n+1/2} \;=\; \mathbf{p}_c^{n-1/2} + \mathbf{F}^n\,\Delta t. \tag{15}$$

Step 4. Drift. Compute $\mathbf{v}^{n+1/2}$ from $\mathbf{p}_c^{n+1/2}$ and $\mathbf{A}^{n+1/2}$ via the same algebra as Step 2 (read $\mathbf{A}^{n+1/2}$ at $\mathbf{x}^n$, subtract, compute γ and $\mathbf{v}$). Then

$$\mathbf{x}^{n+1} \;=\; \mathbf{x}^n + \mathbf{v}^{n+1/2}\,\Delta t. \tag{16}$$

This is explicit – no fixed-point iteration is required at the non-relativistic level because γ is computed from $\mathbf{p}_{\text{kin}}$ directly, and $\mathbf{p}_{\text{kin}}$ is just $\mathbf{p}_c - q\mathbf{A}$.

Note on time-centering of v in the force

The Hamilton's-equations derivation places $\mathbf{v}$ at the same instant as the force, namely t^n. The discrete scheme above uses $\mathbf{v}^{n-1/2}$, which is half a step early. For a leapfrog structure, the canonical move is to use the time-symmetric average $\mathbf{v}^n \approx (\mathbf{v}^{n-1/2} + \mathbf{v}^{n+1/2})/2$, but $\mathbf{v}^{n+1/2}$ is not available until after the kick.

Three options:

1. **Iterate** – predictor with $\mathbf{v}^{n-1/2}$ to get a tentative $\mathbf{p}_c^{n+1/2}$ and hence $\mathbf{v}^{n+1/2}$, then recompute the kick with the averaged velocity. This converges in one Newton-like step at non-relativistic v and a few at relativistic. Cost: $\sim 2\times$ the bare scheme.
2. **Accept the half-step lag** – use $\mathbf{v}^{n-1/2}$ in the force. Loses second-order time accuracy on the $\mathbf{v} \cdot \nabla\mathbf{A}$ contribution (still second-order on $-\nabla\phi$). For Stage 37's uniform-motion test the difference is irrelevant ($\mathbf{v}$ is constant).

3. **Use a different splitting** – DKD (drift-kick-drift) where $\mathbf{x}$ advances by $\Delta t/2$ first, then a full kick at the half-step position, then another $\Delta t/2$ drift. This is the velocity-Verlet structure and is symplectic for separable Hamiltonians. Our Hamiltonian is not separable, but DKD with appropriate field reads should still give second-order accuracy and good energy behaviour.

Default proposal: **option 2** (accept the half-step lag) as the first GEMPIC implementation, since it is the simplest and already gives correct uniform-motion behaviour (verified analytically below). Iterate later if the orbit-test radiation reaction is qualitatively wrong.

Note on A time-staggering

The half-step $\mathbf{A}$ convention (Aprev $= \mathbf{A}^{n-1/2}$, Acurr $= \mathbf{A}^{n+1/2}$) lets us read $\mathbf{A}$ at three relevant times via simple combinations:

$$\mathbf{A}^{n-1/2} \text{ from } \texttt{prev}, \tag{17}$$

$$\mathbf{A}^{n+1/2} \text{ from } \texttt{curr}, \tag{18}$$

$$\mathbf{A}^{n} = \tfrac{1}{2}(\mathbf{A}^{n-1/2} + \mathbf{A}^{n+1/2}) \quad \text{(second-order time interpolation).} \tag{19}$$

For the canonical-to-kinetic momentum conversion in Step 2 we use $\mathbf{A}^{n-1/2}$; in Step 4 we use $\mathbf{A}^{n+1/2}$. The force itself in Step 3 reads field gradients at t^n, which means using $\mathbf{A}^n$ (the average) for the $\partial_j A_i$ gathers.

2.7 Analytic verification: zero force on uniform motion

The structural claim is that the GEMPIC discrete force law gives identically zero force on a uniformly-moving deposit. Verify by the following argument.

Setup

A single particle moves with constant velocity $\mathbf{v}_0$. Its position at time t is $\mathbf{x}_p(t) = \mathbf{x}_0 + \mathbf{v}_0 t$. The fields ϕ and $\mathbf{A}$ at any time t are produced by the particle's own past trajectory via the discrete wave equations sourced by ρ and $\mathbf{J}$ deposited by the particle. For constant motion, the wake of the deposit is a steadily-translating pattern that travels with the particle in the long-time limit (the boosted-Coulomb / boosted-vector-potential pattern).

In this steady state, the field at the particle position is constant in the particle's rest frame: $\phi(\mathbf{x}_p(t), t)$ and $\mathbf{A}(\mathbf{x}_p(t), t)$ are time-independent.

Equation of motion in the steady state

For uniform motion, we need $\dot{\mathbf{p}}_c = 0$ and $\dot{\mathbf{x}} = \mathbf{v}_0$. The second follows from $\mathbf{p}_c = \gamma m \mathbf{v}_0 + q\mathbf{A}(\mathbf{x}_p(t), t)$ being constant (so $\mathbf{v}$ is constant). The first requires the force in Eq. (6) to vanish:

$$q\, v_{0,j}\, \partial_i A_j \; - \; q\, \partial_i \phi \; = \; 0 \quad \text{for } i = x, y. \tag{20}$$

In the continuum, this is a property of the boosted-Coulomb / boosted-vector-potential solution: $\phi(\mathbf{x}_p)$ and $\mathbf{A}(\mathbf{x}_p)$ satisfy $\mathbf{v} \cdot \mathbf{A}(\mathbf{x}_p) = \phi(\mathbf{x}_p) +$ (boost-invariant constant), and at the symmetric point of the wake pattern ∇ gives the same gradient on both sides modulo a sign.

The question is: does the *discrete* scheme preserve Eq. (20)?

The discrete cancellation

In the discrete scheme, the deposit-to-field map is the same operator for ϕ (from ρ) and for $\mathbf{A}$ (from $\mathbf{J}$). Both pass through the SAME discrete wave equation (with the same c_{eff}, the same Yee Laplacian, the same PML), and both are sourced by deposits from the same particle position. The only difference: $\rho = q\,S(\mathbf{x}-\mathbf{x}_p)$ while $\mathbf{J} = q\mathbf{v}\,S(\mathbf{x}-\mathbf{x}_p)$, where S is the deposit shape (TSC, with Esirkepov continuity for $\mathbf{J}$).

Hence in the long-time steady state, the field patterns satisfy

$$\mathbf{A}(\mathbf{x},t) \;=\; \mathbf{v}_0\,\phi(\mathbf{x},t) \tag{21}$$

at the level of the discrete operators, provided the deposit shape for ρ and the deposit shape for $\mathbf{J}$ are the same (they are: both TSC in this scheme), the wave equation is the same (it is), and the PML treats ϕ and $\mathbf{A}$ symmetrically (it does).

[**Open item:** this argument is heuristic; the rigorous discrete statement is that the discrete wave-equation Green's function is the same scalar operator applied to ρ or to each component of $\mathbf{J}$. A pencil-and-paper Fourier analysis of the discrete Yee wave equation should verify the proportionality $\hat{A}_j(\mathbf{k},\omega) = v_{0,j}\,\hat{\phi}(\mathbf{k},\omega)$ for a deposit moving with constant velocity. This is the formal work that should accompany the implementation.]

Given the proportionality $\mathbf{A} = \mathbf{v}_0\,\phi$, the zero-force condition Eq. (20) becomes

$$q\,v_{0,j}\,\partial_i(v_{0,j}\,\phi) - q\,\partial_i\phi \;=\; q\,(|\mathbf{v}_0|^2 - 1)\,\partial_i\phi, \tag{22}$$

which (in units with $c=1$) is $q\,(v^2-1)\,\partial_i\phi$. At the particle position by the Hockney-Eastwood matched-shape property the spatial gradient $\partial_i\phi$ at the particle is the same as the gradient of the deposit by itself, which *is non-zero* unless we use a centred-symmetric deposit/gather pair.

This is the structural cancellation we want. Two important observations:

- The cancellation does *not* depend on $\partial_i\phi(\mathbf{x}_p) = 0$ (the v=0 Hockney-Eastwood condition). It depends on $\mathbf{A} = \mathbf{v}_0\phi$ as fields, which is a property of the deposit and wave equation.
- The relativistic prefactor $(v^2-1)/c^2 = -1/\gamma^2$ in the relativistic case modifies the result; in the non-relativistic limit $v \ll c$ the factor approaches -1 and the force is exactly $-q\partial_i\phi$ instead of zero. This says the canonical force law on a uniformly- moving non-relativistic charge is just $-q\nabla\phi$ at the moving-deposit's ϕ. The HE adjoint condition then delivers the zero net force.

So GEMPIC's structural cancellation routes through the *discrete* property that ρ and $\mathbf{J}$ source the same wave equation, plus the existing matched-shape HE adjoint between ρ deposit and ϕ-gradient gather at $v=0$. Both of these are already in the scheme; we are just leveraging them through a different particle EOM.

Why the current Lorentz-form scheme fails the same test

The current scheme uses

$$\mathbf{F}_{\text{Lorentz}} = -q\nabla_h\phi - q\,\partial_t^h\mathbf{A} + q\mathbf{v}\times(\nabla_h\times\mathbf{A}). \tag{23}$$

The discrete $\partial_t^h\mathbf{A}$ is a centred (or one-sided) time-difference on the $\mathbf{A}$ buffers. On a uniformly-moving deposit, the steady-state $\mathbf{A}$ at the particle is *time-independent in the particle's frame* but *moving relative to the grid*. The discrete $\partial_t^h\mathbf{A}$ at the lab-frame particle position picks up the convective contribution $(\mathbf{v}\cdot\nabla)\mathbf{A}$, but discretised on a STAGGERED time stencil while $\nabla\phi$ is on a SPATIAL stencil at a different time slice. There is no algebraic identity at the discrete level that makes these two discrete operations cancel to the residual order Stage 37 needs.

GEMPIC routes around this by never computing $\partial_t^h\mathbf{A}$ at all.

2.8 Addendum (2026-05-24): the analytic argument above is incomplete; empirical confirmation that GEMPIC is not a structural fix

The "Analytic verification" subsection above asserts the discrete GEMPIC force on a uniformly-moving deposit vanishes by leveraging the HE matched-shape adjoint. That claim is INCOMPLETE. This addendum corrects it.

The corrected Fourier analysis

The discrete Yee wave operator symbol is the same on every sublattice:

$$\hat{L}(\mathbf{k},\omega) = \frac{4}{\mathrm{d}t^2}\sin^2\!\left(\tfrac{\omega\,\mathrm{d}t}{2}\right) - \frac{4c^2}{\mathrm{d}x^2}\left[\sin^2\!\left(\tfrac{k_x\,\mathrm{d}x}{2}\right) + \sin^2\!\left(\tfrac{k_y\,\mathrm{d}x}{2}\right)\right].$$

The sources for a uniformly-moving deposit (TSC ρ on corners, Esirkepov-TSC $\mathbf{J}$ on edges) satisfy $\hat{J}_j(\mathbf{k},\omega) \approx v_{0,j}\,\hat{\rho}(\mathbf{k},\omega)$ to leading order (sinc corrections appear at $O((kv_0\mathrm{d}t)^2)$ from the path-integration over the time step). Passing both through the identical discrete wave operator gives the proportionality

$$\hat{A}_j(\mathbf{k},\omega) = \frac{v_{0,j}}{c^2}\,\hat{\phi}(\mathbf{k},\omega)\;[1 + O(kv_0\mathrm{d}t)^2].$$

This part of the v37 argument is correct.

Substituting into the GEMPIC force at the particle position gives

$$F_i \;=\; -q\left(1 - \frac{v_0^2}{c^2}\right)\partial_i^h\phi\big|_{\mathbf{x}_p} \;=\; -\frac{q}{\gamma^2}\,\partial_i^h\phi\big|_{\mathbf{x}_p}.$$

For zero net force this requires $\partial_i^h\phi|_{\mathbf{x}_p} = 0$. The original v37 argument claimed this followed from the HE matched-shape adjoint. **That was wrong**: HE matched-shape only guarantees $\partial_i^h\phi|_{\mathbf{x}_p} = 0$ at $v = 0$ (the stationary case). For a moving deposit, $\partial_i^h\phi|_{\mathbf{x}_p}$ is precisely the quantity Stage 37's "phi-only" force measures, and it is non-zero on the grid – in fact UV-divergent in $\mathrm{d}x$ per the Stage 37 $\mathrm{d}x$ convergence scan.

Empirical confirmation (Stage 41)

GEMPIC was implemented behind a default-off flag (`gr_sim_set_pusher(sim, GR_PUSHER_GEMPIC)`) and `stage41_gempic_check.c` ran the Stage 37 setup with both pushers side by side, reporting per-step spurious acceleration:

v/c	accel (Boris)	accel (GEMPIC)	ratio (GEMPIC / Boris)
0.001	$+1.04\times10^{-6}$	$+1.03\times10^{-6}$	0.986
0.01	-4.75×10^{-6}	-4.75×10^{-6}	0.999
0.05	$+4.87\times10^{-8}$	$+4.93\times10^{-8}$	1.012
0.1	$+2.31\times10^{-7}$	$+2.24\times10^{-7}$	0.969
0.2	-2.71×10^{-7}	-2.14×10^{-7}	0.790
0.3	$+4.95\times10^{-8}$	-2.66×10^{-7}	5.4× (sign flip)

At every v in the sweep, the GEMPIC residual is within a few percent of the Boris residual. At $v = 0.3$ the discrete Boris cancellation happens to be unusually clean, so GEMPIC is actually slightly worse there.

Implication

The structural problem behind Stage 37 lives in the *deposit-and-gather pair geometry*, NOT in the pusher form:

- Both pushers' residuals reduce to a prefactor times $\partial_y^h \phi|_{\mathbf{x}_p}$ on the discrete grid.
- The discrete $\partial_y^h \phi|_{\mathbf{x}_p}$ for a moving deposit is non-zero, with the magnitude Stage 37 measures, and grows as dx shrinks (UV-divergent).
- The HE matched-shape adjoint kills this gradient only at $v = 0$. At $v \neq 0$ a stronger condition is needed (matched- shape adjoint at non-zero velocity), and we do not have a construction that delivers it on TSC + Esirkepov + Yee.

The empirical paths that DO shrink the artifact, per Stage 37's smoothing scan, all act on the effective deposit width:

- Smoothing the deposit (Stage 37: 16 binomial passes reduces the phi-only artifact $\sim 200\times$).
- A wider macroparticle (equivalent to smoothing in the high-k content).
- Higher-order Yee schemes (better k-space isotropy, marginal improvement expected).
- Spectral / pseudo-spectral wave equations (no grid dispersion; much more expensive).

GEMPIC is therefore not the principled fix Stage 37 calls for. The pusher is correct (no $-\partial_t^h \mathbf{A}$ to break the cancellation), but the cancellation that needs to happen is on the *field* side at the particle position, not on the *pusher* side, and GEMPIC leaves the field-side cancellation unchanged.

Status of the implementation

GEMPIC stays in the tree as a default-off diagnostic (`gr_sim_set_pusher` API, `gempic_em_kick` in `core/src/particle.c`, two new LB gathers in `core/src/deposit.c`, `stage41_gempic_check.c`). The infrastructure is reusable if a future scheme combines a canonical-momentum push with a different deposit-side construction. Commit `3e0b209` carries all of this.

The "Implementation order" section below (and the v36 "Implementation order" it refines) is superseded by the new direction: **focus on the deposit-side construction, not the pusher.** The next scope item should be a deposit-and-gather pair that satisfies the matched-shape adjoint condition at $v \neq 0$ – or, failing a structural construction, a deposit-side mitigation (fixed-physical-width macroparticle) with clear accuracy bounds documented.

2.9 Open items before implementation

1. **Rigorous discrete proof of $\hat{A}_j = v_{0,j}\,\hat{\phi}$ for moving deposit.** The continuum statement is trivially true (the wave equations are identical scalar operators on ρ and each J_j, with $J_j = \rho v_j$). The discrete statement requires that the Yee Laplacian on edge-stored $\mathbf{A}$ commutes with the Esirkepov current decomposition in the right sense. Pencil- and-paper Fourier analysis.
2. **Implementation of the two new mixed-derivative LB gathers.** `gr_tsc_lb_dy_xedge` and `gr_tsc_lb_dx_yedge` in `core/src/deposit.c`. Mirror the existing `gr_tsc_lb_dx_xedge` / `_dy_yedge` with the sublattice indexing swapped.
3. **Particle state**: add $\mathbf{p}_c$ alongside $\mathbf{p}_{\text{kin}}$ in `gr_particle_t`, or replace. Recommendation per v36: keep $\mathbf{p}_{\text{kin}}$ as the "output" field (used for diagnostics, KE computation, etc.) and compute it on demand from $\mathbf{p}_c$ via Eq. (2). The stored canonical field is the new addition.
4. **Feature-flag plumbing**: add `gr_pusher_kind_t {GR_PUSHER_BORIS, GR_PUSHER_GEMPIC}` and a setter on `gr_sim_t`. Default Boris. When GEMPIC is selected, the particle push branches in `core/src/particle.c`.

5. **Gravity-side parallel.** The gravity sector has an analogous canonical structure (with $q \to m$, $\mathbf{A} \to \mathbf{A}_g$) and presumably the same structural defect under Stage 28's heating diagnostic. Plan: do EM GEMPIC first as a proof of concept; replicate to gravity afterwards. The benchmark suite (Stages 37–40) should be extended with gravity-analog cases at that time.

6. **Time-step / dispersion considerations.** The non-Boris pusher does not have Boris's exact-rotation structure for the magnetic force. At high $v \times B$ (strong $\mathbf{B}$, large gyroradius), the leapfrog GEMPIC may have larger gyrofrequency dispersion than Boris. For our pedagogical regime (γ typically below 1.1) this is likely a non-issue, but should be measured.

2.10 Implementation order (refining v36's order)

Per v36 §4, the benchmark suite (B-1 through B-6) is in hand as of commits `67e22d7` (Stage 37 baseline-ready) and `0a7c2b7` (Stages 38–40 and `make benchmark-pic`). The next steps:

1. Pencil-and-paper Fourier verification of $\hat{A}_j = v_{0,j}\,\hat{\phi}$ on the discrete Yee scheme. Out-of-band of this document; should be a one-page note.

2. Implement the two new LB gathers (`gr_tsc_lb_dy_xedge`, `gr_tsc_lb_dx_yedge`). Add unit tests that read a known polynomial $\mathbf{A}$ field and verify the gather equals the analytic derivative.

3. Implement the GEMPIC pusher behind a default-off flag `gr_sim_set_pusher(sim, GR_PUSHER_GEMPIC)`. Keep Boris as default and as legacy regression path.

4. Run `make benchmark-pic` with GEMPIC enabled. Capture the table. Compare to baseline.

5. If the benchmark passes acceptance (per v36 sec gempic_benchmark): flip the default, mark Boris as legacy. Update memory files.

6. If the benchmark FAILS acceptance: do NOT flip the default. Diagnose specifically which sub-test fails and which open item from this section was the cause. Iterate.

Changes from v35 (this revision)

This revision documents the Stage 37 diagnostic findings, scopes the benchmark suite that any EM PIC scheme change must pass, and scopes the canonical-momentum (GEMPIC) reformulation that is the principled fix for the structural defect Stage 37 exposed.

1. **Stage 37 (§3)**: a constant- velocity charged particle in vacuum should conserve energy and momentum exactly (continuum Maxwell self-force on uniformly moving charge is zero). The current scheme fails this catastrophically: at $v/c = 0.001$ the spurious self-force captures the particle at a stable fixed point $v_y \approx -0.28c$ within ~ 50 steps. The artifact is dt-independent, UV-divergent in Δx (grows by $1.5\times$ on phi-only per Δx halving), and shrinks only when the deposit is smoothed (effectively widening the macroparticle). This is a structural failure of the discrete $-\nabla\phi - \partial_t\mathbf{A}$ cancellation, not a tunable bug.

2. **Prior "fixes" re-labeled**: the half-step $\mathbf{A}$ potentials change (commit `9769217`, 2026-05-21) and the Stage 36 inductive sign flip diagnostic (2026-05-22) both reduced *orbit-test* drift. Stage 37 shows neither addressed the underlying cancellation failure. The half-step $\mathbf{A}$ reinterpretation is still a real improvement and is kept; the inductive sign API is now labeled diagnostic-only.

3. **Benchmark suite scope (§4)**: six tests with explicit acceptance criteria, designed to FAIL the current scheme along the dimensions where it is broken. The primary acceptance gate is uniform-motion conservation of E and p to integrator-noise level; radiation-rate quantitative tests are deferred until a 2D Liénard–Wiechert analytic is built (2D retarded fields decay as $1/r$, not $1/r^2$).

4. **GEMPIC canonical-momentum reformulation scope (§5)**: track $\mathbf{p}_c = \gamma m\mathbf{v} + q\mathbf{A}$ instead of $\mathbf{p}_{\text{kin}}$; force law becomes $\mathrm{d}\mathbf{p}_c/\mathrm{d}t = q\nabla(\mathbf{v}\cdot\mathbf{A}) - q\nabla\phi$ with no $\partial_t\mathbf{A}$ and no $\mathbf{v}\times\mathbf{B}$ explicitly (both emerge in the $\mathbf{p}_c \to \mathbf{p}_{\text{kin}}$ conversion). The structurally-broken operator is eliminated.

5. **Test catalog blind spot identified**: pre-Stage-37 PIC tests were all either orbit-based (averages over direction) or $v = 0$ (Hockney–Eastwood). No test of "moving charged particle, measure its KE drift" existed. Stage 37 is now mandatory regression for every future EM PIC change.

All v35 content (implementation alignment, Yee + Esirkepov migration, the full v32 physics framework) carries forward unchanged. Cross- references to `eq:` and `sec:` labels remain valid.

3 Stage 37: Uniform-Motion Self-Force Diagnosis

3.1 The problem statement

A free charged particle moving at constant velocity in vacuum does *not* radiate (Larmor radiation requires acceleration). Equivalently, in continuous Maxwell electrodynamics the self-force on a uniformly-moving charge is identically zero: the $-\nabla\phi$ piece (longitudinal field of the boosted Coulomb potential) is *exactly* cancelled by the $-\partial_t \mathbf{A}$ piece (from the time-changing $\mathbf{A}$ in the boosted frame). In a faithful PIC discretisation the kinetic energy of such a particle must be conserved up to grid- aliasing noise — there is no physically-allowed channel for KE to leak into the field.

Stage 37 (`core/tests/stage37_em_constant_velocity.c`) is the test that exercises this condition: a 128×1024 asymmetric domain, a single $+Q$ charge launched at low y with constant $+y$ velocity, no background, PML absorbers on both transverse edges.

3.2 Convergence triangulation

Four orthogonal refinement scans were run. All at $v = 0.30c$ with the default production knobs (`rho_smooth=4`, TSC+LB), except the smoothing scan which scans the smoothing. Results are KE drift percentages over ~ 4000 steps.

Smoothing scan (passes applied to BOTH ρ and J)

passes	phi-only	full
0	+1587.74%	−22.82%
1	+65.37%	−9.13%
2	+31.49%	−7.46%
4	+16.19%	−6.31%
8	+9.74%	−5.78%
16	+7.29%	−5.67%

Smoothing widens the macroparticle and damps the high-k content of the deposit; the artifact follows it. The original “baseline” phi-only of +16% was actually 4-pass smoothed; the bare phi-only self-force in this scheme is +1587%.

CFL / dt scan (`dx`, `c_eff`, v all held fixed; only dt varies)

CFL	dt	phi-only	full
0.7071	0.7071	+16.16%	−61.79%
0.5000	0.5000	+15.10%	−62.92%
0.2500	0.2500	+14.11%	−62.46%
0.1250	0.1250	+14.68%	−63.94%

dt cut by 5.66× produced under 10% change in drift. Not a time-truncation error. Refining dt does *not* help.

Δx convergence scan (physical setup held fixed; W, H, n_{damp}, dt all scale with Δx)

Δx	n_steps	phi-only	full	halving ratio
1.0	4223	+16.10%	−62.11%	—
0.5	8447	+24.69%	−72.55%	phi= 1.53, full= 1.17

Halving Δx *grows* the phi-only artifact by 53%. Ratio > 1 means UV-divergent: the artifact is the discrete shadow of an IR-divergent point-source self-energy that the implicit $\sim \Delta x$ deposit regularisation was hiding. As Δx shrinks the regularisation shrinks with it, exposing more of the divergence.

v_{drift} scan (with low-y bail-out)

v/c	Δv_y (phi)	Δv_y (full)	cells	KE% (phi)
0.001	−0.277	−0.273	−242	+8 466 374%
0.01	−0.287	−0.283	−55	+81 234%
0.1	−0.019	−0.026	664	−34.3%
0.3	+0.021	−0.110	674	+16.1%

At $v = 0.001$ AND $v = 0.01$ the spurious force *reverses* the particle's velocity within ~ 50 steps and drives it to a stable equilibrium near $v_y \approx -0.28c$. The dvy values at low v are not "force at $v = v_{\text{init}}$" — they are "force has a stable fixed point at $v \approx -0.28c$ that captures any low-v particle." The artifact is *worst* at non-relativistic v, not mildest.

3.3 Combined picture

Knob	Direction	Effect
Refine dt	CFL down 5.7×	unchanged
Refine Δx	halve	grows $\sim 50\%$ (phi)
Smooth deposit	passes $0 \to 16$	shrinks 200×
Shrink v (c_eff ↑)	prior c_eff scan	shrinks to 0
Shrink v_{drift}	v scan, this section	gets MUCH worse

This is a textbook UV-sensitive structural failure of the discrete $-\nabla\phi + (-\partial_t \mathbf{A})$ cancellation on a moving deposit. The two operators do not constitute a discrete adjoint pair at finite sub-cell motion; the residual is the self-force of the regularised delta function moving across the grid. Refining Δx tightens the implicit regularisation, the effective point-source gets sharper, and the spurious self-force *grows*.

3.4 What this implies for prior "fixes"

Half-step A potentials (commit 9769217, 2026-05-21). The reinterpretation of $\mathbf{A}_{\text{prev}} = \mathbf{A}^{n-1/2}$, $\mathbf{A}_{\text{curr}} = \mathbf{A}^{n+1/2}$ giving $\partial_t \mathbf{A}^n = (\mathbf{A}_{\text{curr}} - \mathbf{A}_{\text{prev}})/\mathrm{d}t$ on a 1-step stencil naturally aligned with $\mathbf{J}^{n-1/2}$ was a real improvement. It reduced the Stage 27 EM PIC orbit drift from +18.6% outspiral to −9.2% inspiral, and corresponding numbers in gravity. At the orbit-test level, the inductive heating was gone. Stage 37 shows the *operator pair* still fails to cancel on a moving deposit; the orbit-test improvement masked but did not remove the structural defect. The half-step change is kept (better time-staggering, 1-step stencil); it is now correctly labeled as *partial mitigation*, not a structural fix.

Stage 36 inductive sign flip (2026-05-22). Setting `em_inductive_sign = -1` cosmetically rotates Stage 27/30 orbit drift from +25% outspiral to −3.5% inspiral. The quantitative evidence from Stage 37 (uniform motion) shows that inverting the sign of the inductive contribution would change the −78% inductive-only drift to +78% — equally broken, opposite sign. No choice of sign on a broken operator pair fixes a cancellation that does not exist. The API knob `gr_sim_set_em_inductive_sign` remains as a diagnostic probe with its docstring updated to label it diagnostic-only and warn against using it to "fix" orbit drift. Default sign stays +1 (the variationally-derived value).

3.5 Test catalog blind spot

None of the existing ~ 36 stage tests exercised the failure mode Stage 37 exposes. The blind spot is coherent and worth recording so future test design avoids it:

- **Field-only tests (Stages 1–6, 13–17)**: no particles deposit anything. Cannot probe self-force.
- **Particle-in-applied-field tests (Stages 7–9, 20–25)**: a particle feels a known background field without depositing. Tests the force law given a field, not the deposit/field/-force loop.
- **Stage 10 "perturbation FDTD with moving particle source"**: closest existing test. Tests A and C use a stationary particle. Test B uses a moving particle but measures only the boosted field shape ($|\mathbf{A}_g|/|\Phi_g| \approx v/c^2$) at a distance, not the force back on the source. Test E measures orbital period accuracy. No test of "moving charged particle, measure its KE drift" was present. The `pic_constant_v` scenario used by Test B sets `charge = 0.0f` (a neutral particle) and therefore cannot test EM self-force at all.
- **Stage 11 Esirkepov continuity**: verifies $\partial\rho/\partial t + \nabla\cdot\mathbf{J} = 0$. Necessary condition for gauge consistency but does NOT imply $-\nabla\phi$ and $-\partial_t\mathbf{A}$ cancel on a moving deposit. Different invariant; we were checking the wrong thing for self-force.
- **Orbit tests (Stages 12, 18, 18b, 27, 30, 32)**: the velocity vector rotates through all 360° per period. A directional spurious force partially averages out around a loop. The orbital (central) force is $\sim 10^2\times$ the spurious force, so the artifact appears as a small percentage-per-orbit instead of a flip-the-velocity-in-50-steps as it does on the uniform-motion test. The Stage 27/30 +25% outspiral WAS this artifact, but was interpreted as a sign-convention puzzle.
- **Hockney–Eastwood static self-force tests (Stage 10C, Stage 33)**: verify zero self-force at $v = 0$. HE only guarantees zero self-force at zero velocity; nobody tested $v \neq 0$ until Stage 37.

Implication for the suite design in §4: tests must probe the deposit→field→force-on-moving-source chain as an independent invariant, not rely on orbit-level cumulative drift, not rely on $v = 0$ HE, not rely on charge conservation as a proxy for force balance.

4 Benchmark Suite for EM PIC Scheme Changes

This section formalises the regression and acceptance suite that any EM PIC scheme change — the planned GEMPIC canonical-momentum reformulation (§5) or any alternative — must pass before being declared a fix. The suite is designed to FAIL the current scheme along the dimensions where it is genuinely broken, so an acceptance run gives clean signal about whether a candidate fix actually addresses the structural problem (vs. rotating the symptom).

4.1 Design principles

1. Tests are designed to BREAK candidates, not to confirm them. Each test states the failure modes it is built to expose.
2. Acceptance criteria are quantitative absolute numbers, not pass/fail strings or "drift looks reasonable" judgements.
3. Tests report results in BOTH percent and absolute units (KE in energy units, momentum in mc, force in $q^2/\Delta x^2$). At low v the percent metric is uninformative; absolute is the gate.
4. Every test runs in two sweep configurations: current scheme (baseline, expected fail) and candidate fix. Both sets of numbers go in the report. A fix that breaks an existing-passing regression is not acceptable.
5. No acceptance number depends on a quantity that any of the diagnostic knobs (`em_inductive_sign`, `em_inductive_disc`, `j_deposit_shift`, `em_magnetic_enabled`) can rotate. These knobs are diagnostic-only.
6. Runner defaults are $\rho_{\text{smooth}} = 0$, $j_{\text{smooth}} = 0$. The bare deposit is the honest baseline. Smoothed runs are explicit add-on configs, never the default.

4.2 Primary acceptance gate (necessary condition)

A uniformly-moving free particle MUST conserve its energy and momentum to integrator-noise level, at every v in the test sweep, with the bare unsmoothed deposit ($\rho_{\text{smooth}} = 0$, $j_{\text{smooth}} = 0$).

This is the *necessary* condition. Anything that fails it is not a candidate fix, regardless of how well it does on orbit tests. Tests **B-1** (single particle) and **B-2** (neutral pair) gate this condition.

4.3 Secondary acceptance (sufficient for production trust)

Static reciprocity (**B-3**), basic two-body dynamics (**B-4**), and the existing PIC regressions (**B-5**) must also hold. Together with the primary gate, this is the sufficient set for declaring the scheme production-trusted.

4.4 Exploratory / report-only (NOT acceptance gates)

Radiation-reaction quantitative tests (**B-6**) are exploratory. Quantitative agreement with a 2D Larmor formula is NOT a hard gate because:

- In 2D the retarded fields decay as $1/r$, not $1/r^2$ as in 3D. Total radiated power has a different scaling and prefactor from the textbook 3D Larmor formula.
- The correct analytic benchmark is the 2D Liénard–Wiechert potentials. These are not currently implemented in GRlite, and deriving them is itself a substantial separate task.

- A scheme that passes the necessary conditions above might still give the "wrong" absolute radiation rate in 2D for reasons unrelated to the uniform-motion artifact (grid dispersion, PML residual, etc.).

B-6 is run AFTER B-1–B-5 are passing, and produces a qualitative report ("orbit inspirals monotonically; loss rate is non-zero and scales with v^k for some empirical k") rather than a hard threshold. A separate scope item, deferred, is to derive the 2D Liénard–Wiechert potential, add a 2D Larmor analytic, and then promote B-6 to a real acceptance gate.

4.5 Tests

B-1. Stage 37 uniform motion, expanded

Status: partially exists; needs the following extensions before the baseline run.

- Strip the $\rho_{\text{smooth}} = 4$ default from the runner; run at $\rho_{\text{smooth}} = 0$ for the primary scan.
- v-sweep at $v/c \in \{0.001, 0.01, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7\}$.
- Add per-step force measurement, not just integrated KE drift: average $\Delta v_y/\Delta t$ over a short window after warm-up, reported in absolute units ($\mathrm{d}v/\mathrm{d}t \cdot \tau$, where τ is a fixed physical reference timescale). This factors out the "particle captured at $v_y \approx -0.28c$" pathology.
- Report Δx convergence and CFL invariance on the v-sweep, not just $v = 0.3$.

Acceptance for a fix:

- $|\Delta v_y/\Delta t| <$ integrator-noise floor (measured from the no-force config in the same Stage 37) at every v in the sweep, with $\rho_{\text{smooth}} = 0$. Absolute threshold to be set from the baseline run.
- $|KE_{\text{final}} - KE_{\text{initial}}|/KE_{\text{initial}} < 1\%$ at $v \geq 0.1$ and $|\Delta v_y/v_{y,\text{init}}| < 1\%$ at $v < 0.1$.
- Δx halving ratio $\in (0.25, 1.0]$ — convergent or constant, never growing.
- No sign-of-force flip across the v sweep. A working scheme has no net longitudinal force at any v.

B-2. Neutral pair co-moving (NEW, Stage 38)

Setup: two equal-and-opposite charges ($+Q$, $-Q$), both at velocity $(0, +v)$ along the same y-axis line, separated by a transverse distance Δx_{sep} in the x-direction. Net ρ is zero modulo the finite extent of the deposit shape; net $\mathbf{J}$ is zero ($\mathbf{J}_{+Q} + \mathbf{J}_{-Q} = 0$). At long range the joint field is dipole, not monopole.

Physics expectation: each particle's KE is conserved up to dipole-wake radiation (very small at modest v). NO spurious self-force: the joint field is structurally smaller than either particle's individual wake.

Why this probe: isolates self-force-on-moving-deposit from "interaction with own wake". The current scheme's UV-divergent self-force will still hit each particle independently, and both will drift to the same fixed point at $v_y \approx -0.28c$ since the fixed point does not depend on field source.

Acceptance for a fix:

- $|\Delta v_y| < 10^{-6}$ for both particles over 5000 steps at $v_{\text{init}} = 0.001$.
- $|\Delta v_y| < 10^{-4}$ for both particles over 5000 steps at $v_{\text{init}} = 0.1$.
- Total momentum $\mathbf{P}_{+Q} + \mathbf{P}_{-Q}$ conserved to machine precision modulo PML absorption.

B-3. Reciprocal force action-reaction (NEW, Stage 39)

Setup: two like charges ($+Q$, $+Q$), one at rest at (c_x, c_y), one at rest at $(c_x, c_y + r)$. Static configuration. Measure the force on each particle via the existing force-evaluation hooks.

Physics expectation: $\mathbf{F}_1 + \mathbf{F}_2 = 0$ to numerical noise (continuous Newton's third law). Each particle sees the other's static Coulomb field plus its own self-force (= 0 by Hockney– Eastwood).

Acceptance for a fix:

- $|\mathbf{F}_1 + \mathbf{F}_2|/|\mathbf{F}_1| < 10^{-6}$ at all separations $r \geq 4$ cells.
- This already holds for the current scheme at $v = 0$ (Hockney– Eastwood). The point is to ensure GEMPIC's modified force law does not accidentally break static reciprocity.

B-4. Co-moving like charges (NEW, addition to Stage 38 or new file)

Setup: two like charges at the same v in the y-direction, separated transverse-wise. Identical to B-2 except both are $+Q$.

Physics expectation: mutual Coulomb repulsion (perpendicular to motion) plus self-force on each. With perfect self-force cancellation, only mutual repulsion remains, and the test reduces to "uniform-motion two-particle, predictable lateral expansion" matching boosted Coulomb.

Why this probe: distinguishes "self-force still broken" from "mutual force still broken". If lateral expansion matches the analytic prediction but particles are dragged backward in y, self-force is broken and mutual is fine. The current scheme will fail in both channels.

Acceptance for a fix:

- Lateral expansion within 5% of the boosted-Coulomb prediction at $v = 0.1$ over 5000 steps.
- y-velocity drift of each particle $< 1\%$ over the run.

B-5. Existing PIC regressions

The following must NOT get worse than current numbers under any candidate fix. The baseline run records current values.

- Stage 11 (Esirkepov continuity): $\nabla \cdot \mathbf{J} + \partial\rho/\partial t = 0$ to machine precision. GEMPIC must preserve this.
- Stage 18 / 18b (Lewis–Birdsall multi-orbit, gravity): GEMPIC for EM should not perturb gravity.
- Stage 27 (EM PIC orbit): current value +25% outspiral at sign= +1 with $\rho_{\text{smooth}} = 4$. Acceptance for a fix: physically-correct inspiral (back-reaction direction; magnitude not gated due to the 2D Larmor caveat), with no need for sign-flip or special smoothing.
- Stage 28 (gravity inductive heating): re-run with the gravity-is-attractive caveat (the original "gravity doesn't heat" conclusion was downstream of the always-attractive sign of gravity). Acceptance: monotone inspiral; no sensitivity to coupling strength.
- Stage 32 (EM binary PIC): bound for hundreds of orbits with monotone (small) inspiral.
- Stage 34 (EM chain signs): bit-clean.
- Stage 35 (EM magnetic-only): parallel-current attraction bit-clean.

B-6. Radiation reaction sanity check (NEW, Stage 40 – EXPLORATORY)

NOT an acceptance gate. See the "Exploratory / report-only" discussion above for the 2D physics reasoning.

Setup: orbiting charge at known r and v. Measure energy and angular-momentum loss rates and the orbit trajectory.

Report contents (no thresholds):

- Energy loss rate vs orbit number.
- Angular momentum loss rate vs orbit number.
- Empirical v^k scaling exponent over $v \in [0.05, 0.3]$.
- Pre-fix vs post-fix loss-rate comparison.

Follow-up (separate scope, deferred): derive 2D Liénard–Wiechert potential, build 2D Larmor analytic, promote B-6 to acceptance gate.

4.6 Pre-GEMPIC baseline run (mandatory)

Before any GEMPIC implementation begins, the suite B-1 through B-6 runs on the current scheme. Two reasons:

1. Some acceptance thresholds are quoted absolute (the integrator-noise floor); others are relative ("better than current"). The relative ones need the baseline number on file.
2. Currently-passing tests may be marginal. If Stage 11 continuity holds to 10^{-12} now and GEMPIC drops it to 10^{-9}, that is worse but possibly acceptable. Without the baseline number we cannot judge.

4.7 Implementation order

1. Strip ρ_{smooth} default from Stage 37 runner, add v-sweep and per-step force. Run baseline numbers.
2. Write Stage 38 (neutral pair). Run baseline.
3. Write Stage 39 (reciprocal force). Run baseline.
4. Write Stage 40 (radiation reaction, exploratory). Run baseline.
5. Bundle B-1–B-6 into a `make benchmark-pic` target that prints the table.
6. ONLY THEN begin GEMPIC implementation.

5 GEMPIC: Canonical-Momentum Reformulation Scope

5.1 Reference and provenance

The reformulation is the canonical-momentum (geometric / variational) PIC approach following Kraus, Kormann, Morrison and Sonnendrücker (2017), often abbreviated GEMPIC. The relevant idea for our purposes is not the full symplectic structure of the field-particle phase space but the specific consequence: the discrete force law on the particle becomes a gradient of a scalar at the particle position $(-\nabla\phi + q\nabla(\mathbf{v}\cdot\mathbf{A}))$, with no $-\partial_t\mathbf{A}$ and no $\mathbf{v}\times\mathbf{B}$ explicitly — both emerge in the conversion between canonical and kinetic momentum.

This eliminates the broken operator pair Stage 37 diagnosed: the $-\partial_t\mathbf{A}$ operator that fails to be the discrete adjoint of $-\nabla\phi$ on a moving deposit is structurally absent from the force law. The cancellation is no longer a property the discretisation must preserve; it is a property the formulation does not require.

5.2 Variables tracked on the particle

Current scheme tracks $\mathbf{p}_{\mathrm{kin}} = \gamma m\mathbf{v}$. GEMPIC tracks

$$\mathbf{p}_c = \gamma m\mathbf{v} + q\mathbf{A}(\mathbf{x}_p, t). \tag{24}$$

This is the canonical (conjugate) momentum: the variable for which Hamilton's equations apply cleanly when the field is present.

5.3 The push step

The Hamilton equations of motion for the particle in the canonical variables are

$$\frac{\mathrm{d}\mathbf{x}_p}{\mathrm{d}t} = \mathbf{v} = \frac{\mathbf{p}_c - q\mathbf{A}(\mathbf{x}_p, t)}{\gamma m}, \tag{25}$$

$$\frac{\mathrm{d}\mathbf{p}_c}{\mathrm{d}t} = -q\nabla\phi(\mathbf{x}_p, t) + q\nabla\big(\mathbf{v}\cdot\mathbf{A}(\mathbf{x}_p, t)\big). \tag{26}$$

The right-hand side of Eq. (26) contains no $\partial_t\mathbf{A}$ and no explicit $\mathbf{v}\times\mathbf{B}$. The vector identity

$$\mathbf{v}\times(\nabla\times\mathbf{A}) = \nabla(\mathbf{v}\cdot\mathbf{A}) - (\mathbf{v}\cdot\nabla)\mathbf{A} \tag{27}$$

and the chain-rule expansion of $\mathrm{d}\mathbf{p}_c/\mathrm{d}t = \mathrm{d}(\gamma m\mathbf{v})/\mathrm{d}t + q[\partial_t\mathbf{A} + (\mathbf{v}\cdot\nabla)\mathbf{A}]$ together imply that the kinetic momentum obeys the standard Lorentz force law,

$$\frac{\mathrm{d}(\gamma m\mathbf{v})}{\mathrm{d}t} = q\big(-\nabla\phi - \partial_t\mathbf{A} + \mathbf{v}\times(\nabla\times\mathbf{A})\big), \tag{28}$$

which is the continuum equation we want. GEMPIC differs by *which discrete operators* produce that continuum result: only $\nabla\phi$ and $\nabla(\mathbf{v}\cdot\mathbf{A})$ are discretised on the particle's force side; $\partial_t\mathbf{A}$ and $\mathbf{v}\times(\nabla\times\mathbf{A})$ are implicit, emerging via Eq. (24) when $\mathbf{p}_c \to \mathbf{p}_{\mathrm{kin}}$.

5.4 Discretisation choices

Time-stepping

Eqs. (25)–(26) are a coupled ODE in $(\mathbf{x}_p, \mathbf{p}_c)$. At leading order, a leapfrog-style scheme is:

Step 1. Read $\mathbf{A}^n$ at the particle position via shape-function interpolation. Compute $\mathbf{v}^{n-1/2} = (\mathbf{p}_c^{n-1/2} - q\mathbf{A}(\mathbf{x}_p^n, t^n))/(\gamma m)$. (Note: this is implicit in γ; solve via fixed-point or via the standard relativistic algebra.)

Step 2. Advance position: $\mathbf{x}_p^{n+1} = \mathbf{x}_p^n + \mathbf{v}^{n-1/2}\mathrm{d}t$. *Open question: is this the right half-step indexing? Likely we want* $\mathbf{v}^n$*, which means a single half-step kick before the drift. Details TBD by literature check.*

Step 3. Read $\nabla\phi^n$ and $\nabla(\mathbf{v}^n \cdot \mathbf{A}^n)$ at the particle position via LB-style gradient gather (the $\mathrm{d}W_3/\mathrm{d}x$ kernels already implemented in `core/src/deposit.c`).

Step 4. Advance canonical momentum: $\mathbf{p}_c^{n+1/2} = \mathbf{p}_c^{n-1/2} + [-q\nabla\phi^n + q\nabla(\mathbf{v}^n \cdot \mathbf{A}^n)]\mathrm{d}t$.

Step 5. Compute $\mathbf{v}^{n+1/2}$ for diagnostics from $\mathbf{v}^{n+1/2} = (\mathbf{p}_c^{n+1/2} - q\mathbf{A}(\mathbf{x}_p^{n+1}, t^{n+1/2}))/(\gamma m)$.

Open issue: the precise time-staggering of the $\mathbf{A}$ reads (integer step vs half-step), the $\mathbf{v}$ implicit solve, and the consistency with the existing half-step $\mathbf{A}$ buffer convention all need a literature pass plus a unit derivation. Do not lock these details until the GEMPIC implementation scope is written out in its own follow-up document.

Field-side stays the same

The wave-equation update for ϕ and $\mathbf{A}$ is unchanged. The ρ and $\mathbf{J}$ deposition stays unchanged (Esirkepov + TSC+LB). Esirkepov continuity is still required for gauge consistency. The only structural change is the particle push.

What about $\mathbf{v} \cdot \mathbf{A}$?

This is a scalar at the particle position, equal to $v_x A_x + v_y A_y$ in 2D. Its gradient $\nabla(\mathbf{v} \cdot \mathbf{A}) = v_x \nabla A_x + v_y \nabla A_y$ (treating $\mathbf{v}$ as held fixed during the gradient since the gradient is at the particle position, not in $\mathbf{v}$-space) is two scalar-gradient gathers on the edge-staggered $\mathbf{A}$ components. The existing TSC LB infrastructure (`gr_tsc_lb_dx_xedge`, `gr_tsc_lb_dy_yedge`, added in the Stage 37 session) provides these gathers directly.

5.5 What GEMPIC does NOT change

- Field representation (storage layout, Yee staggering, sublattices).
- Wave equation update (leapfrog ϕ, $\mathbf{A}$).
- Deposition: ρ from particle to CORNER, $\mathbf{J}$ from particle via Esirkepov to X_EDGE / Y_EDGE.
- Esirkepov continuity (still required).
- Half-step $\mathbf{A}$ convention (still useful for the diagnostic $\partial_t\mathbf{A}$ if anyone ever wants to compare).
- Shape function and force interpolation infrastructure (TSC+LB on the deposit side, LB gradient gather on the force side).

5.6 What GEMPIC DOES change

- Particle state vector: $\mathbf{p}_c$ replaces (or shadows) $\mathbf{p}_{\mathrm{kin}}$. Storage: one new field on `gr_particle_t`, or a flag toggling interpretation.

- Particle push: new path `gr_particle_push_gempic` alongside the existing Boris path. Selected via `gr_sim_set_particle_pusher(sim, GR_PUSHER_GEMPIC)` behind a feature flag.
- Force evaluation: $\nabla(\mathbf{v} \cdot \mathbf{A})$ gather instead of $-\partial_t \mathbf{A}$ and $\mathbf{v} \times \mathbf{B}$.
- Output / diagnostics: KE and velocity must still be reported from $\mathbf{v}$, computed from $\mathbf{p}_c$ via Eq. (24) with the current $\mathbf{A}$.

5.7 Feature flag and rollout

GEMPIC lands behind a default-OFF feature flag. The default scheme remains the current $\mathbf{p}_{\mathrm{kin}}$ Boris pusher until the benchmark suite (§4) confirms GEMPIC passes the necessary and sufficient acceptance conditions. When GEMPIC passes, default flips and the old path becomes a legacy comparison mode (kept indefinitely for regression).

5.8 Open items before implementation begins

1. Lock the time-staggering and the $\mathbf{v}$ implicit-solve convention via literature check. This will be its own follow-up scope document (gr_sandbox_v37 or similar) once benchmark baselines are in hand.
2. Confirm with a pencil-and-paper derivation that GEMPIC's discretisation gives *exactly zero* force on a uniformly- moving deposit, at least in the continuum limit and ideally on the discrete grid. This is the analytic check that should precede the numerical acceptance run.
3. Identify whether the gravity-side $-m\partial_t \mathbf{A}_g$ inductive piece has the same structural issue and whether the analogous canonical-momentum reformulation applies (carry over with $q \to m$, $\mathbf{A} \to \mathbf{A}_g$).
4. Decide whether legacy and GEMPIC pushers share particle state (with a flag determining interpretation) or have parallel storage. Storage-flag is simpler; parallel storage is more robust against an interpretation-flag bug. Default recommendation: storage-flag, with a runtime assertion that the flag matches the pusher selection.

6 Implementation Alignment with §15 (v35): Yee + Esirkepov

6.1 Why this work is being done

The Yee staggered grid is already specified in Section 15 ("Field grid" and "Why this staggering: exact gradient and curl operations"). Source deposition placed at the correct sublattices is also specified there, as is the discrete continuity equation as the primary defense against gauge drift ("The discrete continuity equation: the primary defense" in Section 15's gauge-condition subsection). **This section is not introducing a new architecture; it is documenting the migration of the v32–v34 implementation to match what the doc already specifies, plus one addition (Esirkepov current deposition) that goes beyond the truncation-order accuracy of Section 15's CIC-on-Yee recipe.**

What the v32–v34 implementation actually did. Rather than the Yee layout specified in Section 15, the implementation stored all six potentials, all six source arrays, and all derived field quantities on a single cell-centered grid at $(i + \frac{1}{2}, j + \frac{1}{2})\Delta x$, with CIC deposition into a single ρ_{matter} array at cell centers and a CIC-of-FD operator for the force gradient. This was the simpler path to a working scalar wave equation (Stages 1–4), and Stages 5–9 each rely on at most a *static* particle in a *fixed* background where the Hockney–Eastwood theorem still gives zero self-force regardless of the underlying grid placement. Stage 10's self-force-cancellation test (drift $< 10^{-5}$ cells over 10^4 timesteps) passes in the cell-centered code at $m = 10^{-3}$.

Where the cell-centered implementation breaks down. Once particles *move* and *deposit dynamic sources*, the cell-centered code shows pathological behavior. Both ρ and $\mathbf{J}$ live at the same cell-centered nodes there, so the discrete continuity equation that Section 15 writes for the Yee layout (with ρ at corners and J_i at edges and a staggered divergence stencil) is not even satisfied to truncation order in the cell-centered code. The 2-point centered FD divergence at the cell-center ρ position uses J values two cells away, breaking the consistency that would damp gauge drift. The numerical residual takes the form of a "grid-heating" self-force on the moving particle, with the wrong sign — energy is pumped *into* the orbit rather than radiated out. Scaling, in our natural units:

$$\frac{a_{\text{heat}}}{a_{\text{phys}}} \sim \alpha \frac{r}{m_{\text{src}}}, \qquad \alpha = \mathcal{O}(1) \text{ dimensionless heating coefficient,} \tag{29}$$

where r is the orbital scale and m_{src} is the *source* mass on the other side of the mutual interaction.

For a single test particle around a fixed analytic background of mass M, this gives $a_{\text{heat}}/a_{\text{phys}} \sim \alpha r/M$, which *is* suppressed by going to small m_{test} (Stage 10 worked at $m_{\text{test}} = 10^{-6}$ in the weak-coupling regime). But for *mutual* gravity between two particles of mass m at separation r, both bodies' self-heating scales like the physical mutual acceleration, and the ratio is $\sim \alpha r/m$ *independent of* m. Scaling masses down does not help. Linearization breaks down before heating becomes negligible. Stage 11 (two-body gravitational interaction via FDTD) is therefore unreachable in the cell-centered framework.

What this implies for the project. The cell-centered implementation effectively limits FDTD's role to visualization of fields and to quantitative tests with analytic backgrounds; the mutual-gravity Stage 11 case is unreachable. That undermines the project's framing as a dynamic-field PIC simulation, since the FDTD ends up carrying weight only as scaffolding around an analytic backbone. v35 fixes this by aligning the implementation with the Section 15 Yee specification.

6.2 Esirkepov current deposition: the one addition beyond §15

Section 15's gauge subsection notes that "using a consistent CIC kernel for both ρ and $\mathbf{J}$ deposition satisfies [the discrete continuity equation] to the accuracy of the kernel," i.e. to $\mathcal{O}((k\Delta x)^2)$ truncation error. That truncation is the source of the residual grid-heating self-force on moving particles; on the Yee grid the heating is much smaller than on cell-centered storage (the discrete divergence stencil now matches the source-deposition staggering), but not zero.

v35 adds *Esirkepov current deposition* [20] as an opt-in code path beyond what Section 15 specifies. Given a particle's positions $\mathbf{x}^n$ and $\mathbf{x}^{n+1}$, Esirkepov's local recurrence computes the deposited J_x values at horizontal-edge cells and J_y values at vertical-edge cells such that the discrete continuity equation of Section 15

$$\frac{\rho^{n+1}_{i,j} - \rho^{n}_{i,j}}{\Delta t} + \frac{(J_x)^{n+1/2}_{i+1/2,j} - (J_x)^{n+1/2}_{i-1/2,j}}{\Delta x} + \frac{(J_y)^{n+1/2}_{i,j+1/2} - (J_y)^{n+1/2}_{i,j-1/2}}{\Delta y} = 0 \tag{30}$$

holds *exactly* (not just to truncation order) at every corner (i, j). The recurrence is local (involves only the cells touched by the particle's trajectory in one timestep) and is roughly 50 lines in 2D. Once Eq. (30) is satisfied exactly, the Hockney–Eastwood-style residual self-force on a moving particle is structurally zero, and the parabolic / hyperbolic cleaning steps described in Section 15 are not needed to keep gauge drift bounded.

The opt-in flag lets us keep simple CIC-on-Yee deposition (as specified in Section 15) as the default for tests that don't involve dynamic mutual sources, and switch on Esirkepov for stages that do (Stages 10–11+). This matches Section 15's framing of parabolic/hyperbolic cleaning as optional refinements: the simple recipe works for the common case; the exact-continuity recipe is brought in when needed.

6.3 What this enables

- **Stage 11 becomes feasible.** Two equal-mass bodies in mutual orbit, each sourcing the other via the FDTD perturbation field (and reading the resulting force via the Yee gradient operator at the particle's CIC-interpolated position), should give a stable bound orbit of the correct period. This is the original sandbox vision.
- **Quantitative radiation reaction.** With charge-conserving deposition, the dynamic self-force has the correct sign (energy LOSS, not energy gain) and an amplitude of the right order. Orbital decay for an inspiraling binary becomes measurable, not buried by spurious heating.
- **Mass-scale independence.** Tests no longer require artificially small m_{test} in the weak-coupling regime; the heating residual is structurally zero. The relevant constraint becomes the linearization condition $r_s/r \ll 1$, which is much more permissive.
- **Genuinely dynamic-field PIC.** The FDTD is again load-bearing for quantitative dynamics, not just visualization.

6.4 Migration plan

The refactor proceeds in save points, each of which leaves a building tree with all previously-passing tests still passing. Tag at every save point so a regression can be bisected cleanly.

S	What lands	Should pass
S0	v35 doc revision (this section), architectural pivot agreed and documented; cell-centered v34 preserved on `v34-cell-centered` branch.	doc only
S1	Add Yee sublattice constants and documentation to internal headers (`sim_internal.h`, comments). Storage unchanged. Code unchanged.	1–10
S2	`src/field.c`: per-field Laplacian uses correct sublattice neighbors and boundary indexing. No functional change for test 1–4 (scalar-only) at the discretization-order accuracy.	1–4
S3	`src/deposit.c`: split CIC kernel into corner-deposit (rho), x-edge-deposit (J_x), y-edge-deposit (J_y) variants. No Esirkepov yet — just W_2 on each sublattice.	1–5
S4	`src/background.c`: per-field sampled-background placement. `gr_bg_eval_analytic` and `gr_bg_eval_A_g` return values at the correct sublattice.	1–6
S5	`src/particle.c`: replace CIC-of-FD gradient with Yee gradient (each $\partial_i \Phi$ on its native sublattice, then CIC-interpolated to the particle). Replace ad-hoc curl with Yee curl (Eq. 114).	1–9
S6	Esirkepov current deposition as a separate code path (opt-in flag, default off so existing tests continue using simple W_2). Verify that with Esirkepov on, the Stage 10 self-force test (now at $m_{\text{test}} = 1$, not 10^{-6}) gives drift $< 10^{-5}$ cells over 10^4 steps.	1–10
S7	Stage 11: `pic_binary` scenario (two equal masses, mutual gravity through the FDTD path, Esirkepov on). Verify orbital period matches the analytic two-body period to $\sim 1\%$.	1–11
S8 (defer)	Stage 12+: EM Effective GEM sources, spin sources (W_3 deposition), background-spacetime modes.	12+

Compatibility. The public API surface (`grlite.h`) changes minimally: field-pointer queries continue to return `float*` to the underlying storage, with the caller responsible for knowing the field's sublattice. Scenarios that use only the analytic-background path (Stages 7–9 in v34) are re-validated unchanged. The WASM EXPORTED_FUNCTIONS list grows by one entry (`gr_sim_set_esirkepov`).

Risks. Anticipated debugging hotspots:

- Index off-by-one between corner-indexed and edge-indexed arrays (each edge array has $N_x \times (N_y$ or $N_y - 1)$ valid cells depending on direction).
- Sign conventions in the Esirkepov recurrence. Cross-check against the published 2D pseudocode in [20] and at least one reference implementation.
- Damping layer's sublattice consistency: each field needs its $d_{i,j}$ multiplier evaluated at its

own sublattice position, otherwise the damping looks different to different fields at the boundary.

- Background-array sampling formerly placed all generators at cell centers. With Yee, the same analytic formula must be sampled at corner, x-edge, and y-edge positions, with different sub-cell offsets each time.

What does NOT change. All v34 physics corrections from Section 12 (EIH 1PN with sign+Shapiro term), Section 13 (gravitomagnetic proper-time formula), and Section 18.6 (analytic-background evaluation mode) remain valid and carry forward unchanged: the physics is independent of the grid layout used to discretize the field equations.

7 Project Goal

The aim is to build an interactive general relativity sandbox that runs in real time on a desktop computer. A full numerical relativity simulation using the ADM (Arnowitt–Deser–Misner) framework with a Cauchy boundary is computationally infeasible for this purpose, as it typically requires supercomputer resources. Instead, the simulation is based on *linearized* general relativity, which captures the most physically interesting effects while remaining tractable.

In the linearized theory, the metric perturbation satisfies a wave equation—the d'Alembert equation—which can be time-stepped efficiently on a grid. Massive particles and charged test particles then follow trajectories determined by the geodesic equation in this perturbed spacetime, with electromagnetic forces added via the standard Lorentz force.

The sandbox also incorporates electromagnetism, using the full coupled Einstein–Maxwell system in the linearized limit. This means the electromagnetic field contributes to the gravitational field via its stress-energy tensor, and the gravitational field in turn affects the propagation of electromagnetic waves.

8 Physical Framework

8.1 Linearized General Relativity

In linearized GR, the metric is written as a small perturbation around flat (Minkowski) spacetime:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1. \tag{31}$$

In the Lorenz gauge, the trace-reversed perturbation $\bar{h}^{\mu\nu} \equiv h^{\mu\nu} - \frac{1}{2}\eta^{\mu\nu}h$ satisfies a wave equation sourced by the full stress-energy tensor:

$$\Box\,\bar{h}^{\mu\nu} = -\frac{16\pi G}{c^4}\,T^{\mu\nu}_{\text{total}}. \tag{32}$$

This is structurally identical to the wave equation for the electromagnetic four-potential, so FDTD methods from computational electromagnetics apply directly.

8.2 The GEM Metric

In the Lorenz gauge, choosing the GEM potentials Φ_g and $\mathbf{A}_g$ to represent the relevant components of $\bar{h}^{\mu\nu}$, the full linearized metric perturbation is:

$$h_{00} = -\frac{2\Phi_g}{c^2}, \qquad h_{0i} = h_{i0} = \frac{2A_{g,i}}{c}, \qquad h_{ij} = -\frac{2\Phi_g}{c^2}\,\delta_{ij}, \tag{33}$$

giving the line element:

$$\mathrm{d}s^2 = -\left(1 + \frac{2\Phi_g}{c^2}\right)c^2\mathrm{d}t^2 + \frac{4}{c}\,\mathbf{A}_g \cdot \mathrm{d}\mathbf{x}\,\mathrm{d}t + \left(1 - \frac{2\Phi_g}{c^2}\right)\left(\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\right). \tag{34}$$

The trace-reversed perturbation $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$ has components:

$$\bar{h}_{00} = -\frac{4\Phi_g}{c^2}, \qquad \bar{h}_{0i} = \frac{4A_{g,i}}{c}, \qquad \bar{h}_{ij} = 0, \tag{35}$$

which is why the GEM wave equations decouple cleanly: $\Box\Phi_g \propto \rho$ and $\Box\mathbf{A}_g \propto \mathbf{J}$, with no spatial stress sourcing the spatial metric at this order.

What each metric component produces

Each of the three distinct components of the metric in Eq. (33) is responsible for a different class of physical effects:

$h_{00} = -2\Phi_g/c^2$ **(time-time component):** This is the Newtonian gravitational potential in relativistic clothing. It is the only component present for a static non-rotating mass. Its physical consequences include:

- **Gravitational attraction**: the gradient $-\nabla\Phi_g$ gives the Newtonian force.
- **Gravitational time dilation**: clocks in a gravitational well ($\Phi_g < 0$) run slow by factor $\sqrt{1 + 2\Phi_g/c^2}$, contributing $+\Phi_g/c^2$ to $\mathrm{d}\tau/\mathrm{d}t$ at first order.
- **Shapiro delay**: light (and any massless signal) is slowed in a gravitational potential, producing the delay $\Delta t = -2\int \Phi_g \,\mathrm{d}\ell/c^3$.
- **Half of light bending**: the h_{00} component alone would deflect light by $2GM/(c^2b)$ — the Newtonian result.

$h_{ij} = -2\Phi_g/c^2\,\delta_{ij}$ **(spatial diagonal components):** The spatial metric is isotropic and also proportional to Φ_g — the same scalar as the time-time component. This component has no Newtonian analogue and is invisible to slow particles ($v \ll c$), but has important consequences for relativistic motion:

- **Second half of light bending**: the spatial metric contributes an equal and opposite deflection to h_{00}, doubling the total to $4GM/(c^2b)$ — the correct GR result. This is why Newtonian gravity predicts half the correct deflection.
- **Post-Newtonian force corrections**: the $(v^2/c^2)\mathbf{g}$ and $4(\mathbf{v}\cdot\mathbf{g})\mathbf{v}/c^2$ terms in the geodesic expansion (Section 12) arise from h_{ij} contracting with the spatial velocity.
- **Perihelion precession**: the post-Newtonian force corrections from h_{ij} are responsible for the extra orbital precession beyond the Newtonian result.
- **ISCO**: the additional attractive force from h_{ij} at high velocities removes the minimum from the effective potential at small radii, producing the innermost stable circular orbit.

$h_{0i} = 2A_{g,i}/c$ **(off-diagonal time-space components):** These components are nonzero only for rotating or moving masses, arising from the mass current $\mathbf{J}_{\mathrm{matter}}$ sourcing $\mathbf{A}_g$. They have no Newtonian analogue. Their physical consequences are:

- **Gravitomagnetic force**: the off-diagonal metric produces the $4\mathbf{v} \times \mathbf{B}_g$ term in the GEM Lorentz force, responsible for Lense–Thirring orbital plane precession and the factor-of-4 difference from the EM analogy.
- **Gyroscope precession**: a spinning test body in a gravitomagnetic field precesses at rate $\mathbf{\Omega}_{\mathrm{LT}} = \nabla \times \mathbf{A}_g$, the Lense–Thirring effect.
- **Gravitomagnetic time dilation**: the h_{0i} component contributes $4\mathbf{A}_g\cdot\mathbf{v}/c^2$ to the proper time rate (Section 13), with opposite sign for co- and counter-rotating orbits — the gravitomagnetic clock effect.
- **Frame dragging**: the dragging of inertial frames by a rotating mass is entirely encoded in h_{0i}; the ergosphere of the Kerr metric is the region where h_{0i} terms dominate and nothing can remain stationary.

The three contributions are independent in the sense that h_{00} and h_{ij} are present for any mass regardless of rotation, while h_{0i} requires a mass current. Setting $\mathbf{A}_g = 0$ (non-rotating background) eliminates all gravitomagnetic effects and leaves only the Newtonian potential and its relativistic corrections. The simulation can therefore interpolate between the Newtonian regime ($v \ll c$, $\mathbf{A}_g = 0$), the post-Newtonian regime (v^2/c^2 corrections from h_{ij}), and the full gravitomagnetic regime (all three metric components active).

8.3 Gravitoelectromagnetism (GEM)

In the slow-motion, weak-field limit the linearized GR equations reduce to a form mathematically identical to Maxwell's equations. The gravitoelectric field $\mathbf{g}$ (analogous to $\mathbf{E}$) and gravitomagnetic field $\mathbf{B}_g$ (analogous to $\mathbf{B}$) satisfy:

$$\nabla \cdot \mathbf{g} = -4\pi G\rho, \tag{36}$$
$$\nabla \cdot \mathbf{B}_g = 0, \tag{37}$$
$$\nabla \times \mathbf{g} = -\frac{\partial \mathbf{B}_g}{\partial t}, \tag{38}$$
$$\nabla \times \mathbf{B}_g = -\frac{4\pi G}{c^2}\mathbf{J} + \frac{1}{c^2}\frac{\partial \mathbf{g}}{\partial t}. \tag{39}$$

The GEM Lorentz force on a test particle with rest mass m and electric charge q is, in terms of the derived fields:

$$\mathbf{F} = m(\mathbf{g} + 4\mathbf{v} \times \mathbf{B}_g) + q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{40}$$

The factor of 4 in the gravitomagnetic term arises from the spin-2 nature of the graviton and is responsible for the Lense–Thirring precession rate differing from a naive EM analogy.

Since the simulation stores potentials rather than fields, it is useful to have the equivalent expression directly in terms of potentials. Substituting $\mathbf{g} = -\nabla\Phi_g - \partial_t \mathbf{A}_g$, $\mathbf{B}_g = \nabla \times \mathbf{A}_g$, $\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}$, $\mathbf{B} = \nabla \times \mathbf{A}$ and using the vector identity $\mathbf{v} \times (\nabla \times \mathbf{A}) = \nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla)\mathbf{A}$:

$$\mathbf{F}_{\text{GEM}} = m(-\nabla\Phi_g - \partial_t\mathbf{A}_g + 4\nabla(\mathbf{v} \cdot \mathbf{A}_g) - 4(\mathbf{v} \cdot \nabla)\mathbf{A}_g), \tag{41}$$
$$\mathbf{F}_{\text{EM}} = q(-\nabla\phi - \partial_t\mathbf{A} + \nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla)\mathbf{A}). \tag{42}$$

These expressions involve only the potentials and their spatial and temporal derivatives evaluated at the particle position. No curl operation over the full grid is required. The physical force is gauge-invariant: the apparent gauge dependence of the individual potential terms cancels exactly when all terms are combined, as can be verified by applying a gauge transformation $\phi \to \phi - \partial_t\Lambda$, $\mathbf{A} \to \mathbf{A} + \nabla\Lambda$ and confirming $\delta\mathbf{F} = 0$.

8.4 Particle Properties: Mass and Electric Charge

Each particle carries two *independent* intrinsic properties: a rest mass m and an electric charge q. They couple to entirely separate fields:

- **Rest mass** m sources the GEM field via $\rho_{\text{matter}} = \sum_i \gamma m_i\, \delta(\mathbf{x} - \mathbf{x}_i)$ and $\mathbf{J}_{\text{matter}} = \sum_i \gamma m_i \mathbf{v}_i\, \delta(\mathbf{x} - \mathbf{x}_i)$. It also determines the inertial response through $\mathbf{p} = \gamma m\mathbf{v}$.
- **Electric charge** q sources the EM field via $\rho_q = \sum_i q_i\, \delta(\mathbf{x}-\mathbf{x}_i)$ and $\mathbf{J}_q = \sum_i q_i\mathbf{v}_i\, \delta(\mathbf{x}-\mathbf{x}_i)$. It determines the response to the electromagnetic Lorentz force $q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.

A particle with both $m \neq 0$ and $q \neq 0$ contributes simultaneously to all four potentials $(\Phi_g, \mathbf{A}_g, \phi, \mathbf{A})$ and experiences forces from all four fields.

8.5 Relativistic Particle Integration

Particles are integrated relativistically via the 3-momentum $\mathbf{p} = \gamma m\mathbf{v}$. The basic update scheme is:

$$\mathbf{p} \leftarrow \mathbf{p} + \mathbf{F}_{\text{total}}\, \mathrm{d}t, \qquad \mathbf{v} = \frac{\mathbf{p}}{\sqrt{m^2 + |\mathbf{p}|^2/c^2}}, \qquad \mathbf{x} \leftarrow \mathbf{x} + \mathbf{v}\, \mathrm{d}t. \tag{43}$$

This correctly handles the anisotropy of relativistic inertia (longitudinal mass $\gamma^3 m$ vs. transverse mass γm). The force $\mathbf{F}_{\text{total}}$ requires relativistic corrections at high velocities—see Section 12.

8.6 Electromagnetic Stress-Energy Coupling

The total stress-energy tensor sourcing the GEM field is:

$$T^{\mu\nu}_{\text{total}} = T^{\mu\nu}_{\text{matter}} + T^{\mu\nu}_{\text{EM}}, \tag{44}$$

Since the simulation stores potentials rather than derived fields, it is useful to express $T^{\mu\nu}_{\text{EM}}$ directly in terms of ϕ and $\mathbf{A}$ and their derivatives. Define the shorthand combinations:

$$\mathbf{P} \equiv \nabla\phi + \partial_t \mathbf{A}, \qquad \mathbf{C} \equiv \nabla \times \mathbf{A}, \tag{45}$$

so that $\mathbf{E} = -\mathbf{P}$ and $\mathbf{B} = \mathbf{C}$. These are not stored as grid arrays; they are computed on demand from finite differences of the potential arrays. The EM stress-energy components in terms of $\mathbf{P}$ and $\mathbf{C}$ are:

$$T^{00}_{\text{EM}} = \frac{\varepsilon_0}{2}\left(|\mathbf{P}|^2 + c^2|\mathbf{C}|^2\right), \tag{46}$$

$$T^{0i}_{\text{EM}} = -\frac{1}{\mu_0 c}(\mathbf{P} \times \mathbf{C})^i, \tag{47}$$

$$T^{ij}_{\text{EM}} = \varepsilon_0\left(-P^iP^j + c^2\sum_k C^{ik}C^{jk}\right) - \tfrac{1}{2}\delta^{ij}\varepsilon_0\left(|\mathbf{P}|^2 + c^2|\mathbf{C}|^2\right), \tag{48}$$

where $C^{ij} \equiv \partial_i A_j - \partial_j A_i$ is the antisymmetric derivative tensor (the spatial components of $F_{\mu\nu}$), so that $C^i = (\nabla \times \mathbf{A})^i = \frac{1}{2}\epsilon^{ijk}C_{jk}$. The expression $\sum_k C^{ik}C^{jk}$ in Eq. (48) involves only antisymmetric combinations of first derivatives of $\mathbf{A}$ — no pseudovectors appear.

Note that $\mathbf{C} = \nabla \times \mathbf{A}$ is the antisymmetric combination $(\partial_i A_j - \partial_j A_i)$ contracted with the Levi-Civita symbol, which is a component of the covariant field strength tensor $F_{\mu\nu}$. It is not a grid array; it is evaluated on the fly from finite differences of edge potential values at the locations where it is needed.

The GEM effective sources use only T^{00} and T^{0i}. Using perturbation potentials only (to avoid double-counting the background metric; see Section 15.10):

$$\rho_{\text{EM}} = \frac{T^{00}_{\text{EM}}}{c^2} = \frac{\varepsilon_0}{2c^2}\left(|\nabla\phi^{\text{pert}} + \partial_t\mathbf{A}^{\text{pert}}|^2 + c^2|\nabla \times \mathbf{A}^{\text{pert}}|^2\right), \tag{49}$$

$$\mathbf{J}_{\text{EM}} = \frac{\mathbf{S}_{\text{EM}}}{c^2} = -\frac{1}{\mu_0 c^2}\left(\nabla\phi^{\text{pert}} + \partial_t\mathbf{A}^{\text{pert}}\right) \times \left(\nabla \times \mathbf{A}^{\text{pert}}\right). \tag{50}$$

In 2D, $\nabla \times \mathbf{A}^{\text{pert}} = (\partial_x A_y^{\text{pert}} - \partial_y A_x^{\text{pert}})\hat{z}$ reduces to a single scalar F_{12}^{pert}, and the cross product in Eq. (50) becomes a 90° rotation of the 2D vector $(\nabla\phi^{\text{pert}} + \partial_t\mathbf{A}^{\text{pert}})$ scaled by F_{12}^{pert}.

The T^{00} component means EM energy density gravitates. The T^{0i} component means EM energy flux (Poynting vector) acts as an effective mass current. The spatial stress T^{ij} is not needed for the GEM source but contributes to radiation pressure and EM momentum flux if those are tracked.

9 Two Implementation Approaches

9.1 Method 1: Full Linearized Einstein–Maxwell

This method integrates all 10 independent components of $\bar{h}^{\mu\nu}$ together with the EM wave equation in curved spacetime.

Per-timestep iteration

1. Assemble $T^{\mu\nu}_{\text{matter}}$ from particle positions, velocities, and rest masses:
$$T^{00} = \gamma^2 \rho c^2, \quad T^{0i} = \gamma^2 \rho c\, v^i, \quad T^{ij} = \gamma^2 \rho\, v^i v^j. \tag{51}$$
2. Assemble $T^{\mu\nu}_{\text{EM}}$ from current field values (Section 2.5).
3. Deposit EM sources from charged particles:
$$\rho_q(\mathbf{x}) = \sum_i q_i\, \delta(\mathbf{x} - \mathbf{x}_i), \qquad \mathbf{J}_q(\mathbf{x}) = \sum_i q_i \mathbf{v}_i\, \delta(\mathbf{x} - \mathbf{x}_i). \tag{52}$$
4. Sum: $T^{\mu\nu}_{\text{total}} = T^{\mu\nu}_{\text{matter}} + T^{\mu\nu}_{\text{EM}}$.
5. Step all 10 metric perturbation components:
$$\Box\, \bar{h}^{\mu\nu} = -\frac{16\pi G}{c^4}\, T^{\mu\nu}_{\text{total}}. \tag{53}$$
6. Step the EM wave equation in curved spacetime (with Ricci coupling):
$$\Box A^\mu + R^\mu{}_\nu A^\nu = \mu_0 J^\mu_q, \tag{54}$$
where $R^\mu{}_\nu$ is the Ricci tensor from $\bar{h}^{\mu\nu}$.
7. Extract fields from potentials:
$$g^i = -\partial_i \bar{h}^{00}, \qquad B^k_g = \varepsilon^{ijk} \partial_i \bar{h}^{0j}, \tag{55}$$
$$E^i = -\partial_t A^i - \partial^i A^0, \qquad B^k = \varepsilon^{ijk} \partial_i A^j. \tag{56}$$
8. Compute geodesic acceleration using the electromagnetic field tensor $F^\mu{}_\nu$. The Faraday tensor is defined as
$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu, \qquad F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}, \tag{57}$$
with $F^\mu{}_\nu = \eta_{\nu\alpha} F^{\mu\alpha}$. The force term $(q/m) F^\mu{}_\nu U^\nu$ then reproduces the relativistic Lorentz force in the spatial components. The geodesic equation is:
$$\frac{\mathrm{d}U^\mu}{\mathrm{d}\tau} = -\Gamma^\mu{}_{\alpha\beta}\, U^\alpha U^\beta + \frac{q}{m} F^\mu{}_\nu U^\nu. \tag{58}$$
9. Update particle 4-velocities and positions.

9.2 Method 2: GEM + EM with Effective Sources

This method replaces the 10-component tensor equation with two scalar/vector wave equations, absorbing $T^{\mu\nu}_{\text{EM}}$ into effective sources.

Per-timestep iteration

1. Deposit matter sources:

$$\rho_{\text{matter}} = \sum_i \gamma m_i\, \delta(\mathbf{x} - \mathbf{x}_i), \qquad \mathbf{J}_{\text{matter}} = \sum_i \gamma m_i \mathbf{v}_i\, \delta(\mathbf{x} - \mathbf{x}_i). \tag{59}$$

2. Deposit EM sources from charged particles.
3. Compute EM effective GEM sources from perturbation potentials (see Eqs. 49–50):

$$\rho_{\text{EM}} = \frac{\varepsilon_0}{2c^2}\left(|\nabla\phi^{\text{pert}} + \partial_t \mathbf{A}^{\text{pert}}|^2 + c^2|\nabla\times\mathbf{A}^{\text{pert}}|^2\right), \tag{60}$$

$$\mathbf{J}_{\text{EM}} = -\frac{1}{\mu_0 c^2}\left(\nabla\phi^{\text{pert}} + \partial_t \mathbf{A}^{\text{pert}}\right)\times\left(\nabla\times\mathbf{A}^{\text{pert}}\right). \tag{61}$$

4. Sum effective sources: $\rho_{\text{eff}} = \rho_{\text{matter}} + \rho_{\text{EM}}$, $\mathbf{J}_{\text{eff}} = \mathbf{J}_{\text{matter}} + \mathbf{J}_{\text{EM}}$.
5. Step GEM wave equations:

$$\Box\,\Phi_g = -4\pi G \rho_{\text{eff}}, \qquad \Box\,\mathbf{A}_g = -\frac{4\pi G}{c^2}\mathbf{J}_{\text{eff}}. \tag{62}$$

6. Step EM wave equation with $c_{\text{local}}(\mathbf{x}) = c\big(1 + 2\Phi_g(\mathbf{x})/c^2\big)$:

$$\Box_{c_{\text{local}}} A^\mu = \mu_0 J_q^\mu. \tag{63}$$

7. Compute total force on each particle directly from potentials via Eqs. (122)–(123), plus on-the-fly curl for Boris rotation (Section 15.2). No full-grid field extraction is performed.
8. Update momentum and position:

$$\mathbf{p} \leftarrow \mathbf{p} + \mathbf{F}_{\text{total}}\,dt, \qquad \mathbf{v} = \frac{\mathbf{p}}{\sqrt{m^2 + |\mathbf{p}|^2/c^2}}, \qquad \mathbf{x} \leftarrow \mathbf{x} + \mathbf{v}\,dt. \tag{64}$$

10 Comparison of the Two Methods

10.1 Field equations solved

Quantity	Method 1 (Full Tensor)	Method 2 (GEM/EM)
Metric perturbation	All 10 components of $\bar{h}^{\mu\nu}$	2 potentials: Φ_g, $\mathbf{A}_g$
EM field	$\Box A^\mu + R^\mu{}_\nu A^\nu = \mu_0 J_q^\mu$	$\Box A^\mu = \mu_0 J_q^\mu$ with c_{local}
EM–gravity coupling	Full $T^{\mu\nu}_{\text{EM}}$	$\rho_{\text{EM}} = u_{\text{EM}}/c^2$, $\mathbf{J}_{\text{EM}} = \mathbf{S}/c^2$ only
Particle equation	Geodesic via $\Gamma^\mu{}_{\alpha\beta}$ and $F^\mu{}_\nu$	3-force with relativistic corrections

10.2 Physical effects captured

Effect	Method 1	Method 2
Newtonian gravity	Yes	Yes
Frame dragging (Lense–Thirring)	Yes	Yes
Gravitational waves (tensor modes)	Yes—all polarizations	Partial—scalar/vector only
Gravitomagnetic Lorentz force	Yes	Yes (factor of 4 included)
EM energy gravitates	Yes—via T^{00}_{EM}	Yes—via $\rho_{\mathrm{EM}} = u_{\mathrm{EM}}/c^2$
Poynting flux as mass current	Yes—via T^{0i}_{EM}	Yes—via $\mathbf{J}_{\mathrm{EM}} = \mathbf{S}/c^2$
Maxwell stress tensor sourcing metric	Yes—via T^{ij}_{EM}	No—dropped
Gravitational lensing	Exact—via Ricci coupling	Approximate—via c_{local}
Shapiro delay	Yes	Yes—via c_{local}
Relativistic inertia anisotropy	Yes—geodesic	Yes—p-update
Velocity-dependent grav. force	Yes—full $\Gamma^{\mu}{}_{\alpha\beta}U^{\alpha}U^{\beta}$	Yes—Section 12 corrections
Correct photon deflection (factor 2)	Yes	Yes—with Section 12
Particles with both m and q	Yes	Yes
Intrinsic spin	Yes—MPD equations	Yes—Section 11

10.3 Approximations in Method 2

Method 2 is valid when:

- Velocities are small compared to c without Section 12 corrections; with those corrections, relativistic speeds are well handled.
- EM fields are not so strong that T^{ij}_{EM} contributes significantly to the spatial metric h^{ij}.
- The Ricci correction $R^{\mu}{}_{\nu}A^{\nu}$ is small—weak-field lensing is approximated by c_{local}.
- The EM field is not too directional: $|\mathbf{S}|/c \ll u_{\mathrm{EM}}$.

10.4 Coupling architecture

$$\text{GEM solver:}\quad \Box\bar{h} = \text{source}(\rho_{\text{eff}}, \mathbf{J}_{\text{eff}})$$

$$\uparrow$$

$$\rho_{\text{matter}} + u_{\mathrm{EM}}/c^2$$

$$\mathbf{J}_{\text{matter}} + \mathbf{S}_{\text{Poynting}}/c^2$$

$$\text{EM solver:}\quad \Box A = \text{source}(\mathbf{J}_q), \quad c \to c_{\text{local}}$$

$$\text{Particles:}\quad \mathrm{d}\mathbf{p}/\mathrm{d}t = \mathbf{F}_{\text{GEM}} + \mathbf{F}_{\text{EM}} + \mathbf{F}_{\text{rel}} + \mathbf{F}_{\text{spin}}$$

11 Particles with Intrinsic Spin

11.1 Overview

A particle with intrinsic angular momentum (spin) obeys the Mathisson–Papapetrou–Dixon (MPD) equations rather than the geodesic equation. In the GEM/EM approximation these reduce to a tractable set of 3-vector equations integrated alongside the momentum update.

Each spinning particle carries two additional properties beyond m and q:

- Spin 3-vector $\mathbf{S}$, with $|\mathbf{S}|$ conserved in the absence of radiation damping.
- Electromagnetic g-factor g_s ($g_s = 2$ for a Dirac fermion, $g_s = 1$ for a classical spinning charged sphere).

11.2 The spin 4-vector constraint

The spin is carried by a 4-vector S^μ subject to the Tulczyjew–Dixon supplementary condition:

$$S^{\mu\nu} p_\nu = 0, \tag{65}$$

where $S^{\mu\nu}$ is the antisymmetric spin tensor and p^μ is the *4-momentum*, not the 4-velocity U^μ. This is an important distinction. For a free particle in flat spacetime $p^\mu = mU^\mu$ and the two are proportional, so it makes no difference. But in the MPD equations the canonical momentum receives a spin-curvature correction:

$$p^\mu = mU^\mu - \frac{1}{2} R^\mu{}_{\nu\alpha\beta} U^\nu S^{\alpha\beta} + \ldots \tag{66}$$

so p^μ and mU^μ are not exactly parallel when the particle is spinning in curved spacetime.

The condition $S^{\mu\nu} p_\nu = 0$ implies that in the lab frame:

$$S^0 = \frac{\mathbf{S} \cdot \mathbf{p}}{\gamma m c}, \tag{67}$$

where $\mathbf{p} = \gamma m \mathbf{v}$ is the relativistic 3-momentum. The timelike component S^0 is not an independent degree of freedom — it is recomputed from $\mathbf{S}$ and $\mathbf{p}$ whenever needed and never evolved separately. Using $\mathbf{p}$ rather than $m\mathbf{v}$ in this expression is exact; no linearized approximation is required.

11.3 Constraint correction

After each timestep $\mathbf{p}$ and $\mathbf{S}$ are updated independently, and $\mathbf{S}$ may acquire a spurious component parallel to $\mathbf{p}$ that violates the rest-frame orthogonality condition. The 3+1 split of Eq. (65) with $p^\mu = (E/c, \mathbf{p})$ and $S^\mu = (S^0, \mathbf{S})$ gives:

$$S^0 = \frac{\mathbf{S} \cdot \mathbf{p}}{E/c} = \frac{\mathbf{S} \cdot \mathbf{p}}{\gamma m c}, \tag{68}$$

where $E = \gamma m c^2$ is the particle energy. Since S^0 is never evolved independently but always reconstructed from $\mathbf{S}$ and $\mathbf{p}$, the constraint is automatically satisfied for S^0. The drift accumulates entirely as a spurious component of $\mathbf{S}$ parallel to $\mathbf{p}$, which is removed by the covariant projection $S^\mu \leftarrow S^\mu - (S^\nu p_\nu / p^\alpha p_\alpha)\, p^\mu$. Using $p^\alpha p_\alpha = -m^2c^2$ and $|\mathbf{p}|^2 + m^2c^2 = \gamma^2 m^2 c^2$, the spatial components give:

$$\mathbf{S} \;\leftarrow\; \mathbf{S} - \frac{\mathbf{S} \cdot \mathbf{p}}{|\mathbf{p}|^2 + m^2c^2}\, \mathbf{p}, \qquad \mathbf{p} = \gamma m \mathbf{v}. \tag{69}$$

This is the exact, covariant, single-formula correction requiring no threshold switching. In the nonrelativistic limit $\mathbf{p} \approx m\mathbf{v}$, $\gamma \approx 1$:

$$\mathbf{S} \leftarrow \mathbf{S} - \frac{\mathbf{S}\cdot\mathbf{v}}{c^2}\,\mathbf{v}, \tag{70}$$

confirming consistency. The momentum $\mathbf{p} = \gamma m\mathbf{v}$ is available directly from the leapfrog integrator at half-integer timesteps.

11.4 Spin-curvature and spin-field forces

In 3D the spin gradient forces involve the gradient of $\mathbf{B}_g = \nabla\times\mathbf{A}_g$ and $\mathbf{B} = \nabla\times\mathbf{A}$ dotted with the spin and magnetic dipole moment:

$$\mathbf{F}_{\text{spin,grav}} = \nabla(\mathbf{S}\cdot\nabla\times\mathbf{A}_g), \tag{71}$$

$$\mathbf{F}_{\text{spin,EM}} = \nabla(\boldsymbol{\mu}\cdot\nabla\times\mathbf{A}), \qquad \boldsymbol{\mu} = \frac{g_s q}{2m}\mathbf{S}. \tag{72}$$

These gradients are evaluated at the particle position using the adjoint two-tent interpolation (Section 15.5). In 2D $\nabla\times\mathbf{A}_g = B_{g,z}\hat{z}$ and $\nabla\times\mathbf{A} = B_z\hat{z}$ are scalars, and the spin is also scalar s, so the forces reduce to $s\,\nabla B_{g,z}$ and $\mu\,\nabla B_z$.

11.5 Spin precession

$$\boldsymbol{\Omega}_{\text{grav}} = \nabla\times\mathbf{A}_g, \tag{73}$$

$$\boldsymbol{\Omega}_{\text{EM}} = \frac{g_s q}{2m}\nabla\times\mathbf{A}, \tag{74}$$

$$\boldsymbol{\Omega}_{\text{Thomas}} = -\frac{\gamma^2}{\gamma+1}\,\frac{\mathbf{v}\times\mathbf{a}}{c^2}, \tag{75}$$

$$\frac{\mathrm{d}\mathbf{S}}{\mathrm{d}t} = \mathbf{S}\times\left(\boldsymbol{\Omega}_{\text{grav}} + \boldsymbol{\Omega}_{\text{EM}} + \boldsymbol{\Omega}_{\text{Thomas}}\right). \tag{76}$$

The quantities $\nabla\times\mathbf{A}_g$ and $\nabla\times\mathbf{A}$ are evaluated on the fly at the particle position from surrounding edge potentials via the Yee curl (Eq. 114), not stored as grid arrays.

11.6 Spin as a field source

A spinning charged particle carries magnetic dipole moment $\boldsymbol{\mu} = g_s(q/2m)\mathbf{S}$ and a spinning massive particle carries gravitomagnetic dipole moment $\boldsymbol{\sigma} = 2\mathbf{S}$. These contribute magnetization currents to the source deposition:

$$\mathbf{J}_q \mathrel{+}= \nabla\times\mathbf{M}(\mathbf{x}), \quad \mathbf{M}(\mathbf{x}) = \sum_i \boldsymbol{\mu}_i\,\delta(\mathbf{x}-\mathbf{x}_i), \tag{77}$$

$$\mathbf{J}_{\text{grav}} \mathrel{+}= \nabla\times\boldsymbol{\Sigma}(\mathbf{x}), \quad \boldsymbol{\Sigma}(\mathbf{x}) = \sum_i \boldsymbol{\sigma}_i\,\delta(\mathbf{x}-\mathbf{x}_i). \tag{78}$$

The $1/r^3$ near-field and $1/r$ radiation fall-offs of dipole fields emerge automatically from the d'Alembert operator.

11.7 Radiation from precessing dipoles

- **Conservative:** ignore radiation reaction; $|\mathbf{S}|$ is conserved exactly.
- **Phenomenological damping:** $\mathrm{d}\mathbf{S}/\mathrm{d}t = \mathbf{S}\times\boldsymbol{\Omega}_{\text{total}} - \alpha\mathbf{S}$.

11.8 Complete algorithm including spin (Method 2)

The following enumeration gives the complete per-timestep algorithm for Method 2 including all spin effects developed so far, before adding the relativistic force corrections of Section 12.

Step 1. Deposit matter and charge sources.

$$\rho_{\text{matter}} = \sum_i \gamma m_i\, W(\mathbf{x} - \mathbf{x}_i), \quad \mathbf{J}_{\text{matter}} = \sum_i \gamma m_i \mathbf{v}_i\, W(\mathbf{x} - \mathbf{x}_i), \tag{79}$$

$$\rho_q = \sum_i q_i\, W(\mathbf{x} - \mathbf{x}_i), \quad \mathbf{J}_q = \sum_i q_i \mathbf{v}_i\, W(\mathbf{x} - \mathbf{x}_i), \tag{80}$$

where W is the cloud-in-cell (CIC) kernel approximating $\delta(\mathbf{x} - \mathbf{x}_i)$ (see Section 15).

Step 2. Deposit spin dipole sources.

$$\boldsymbol{\mu}_i = \frac{g_{s,i} q_i}{2m_i} \mathbf{S}_i, \qquad \boldsymbol{\sigma}_i = 2\mathbf{S}_i, \tag{81}$$

$$\mathbf{J}_q \mathrel{+}= \nabla \times \left[\sum_i \boldsymbol{\mu}_i\, W_D(\mathbf{x} - \mathbf{x}_i) \right], \tag{82}$$

$$\mathbf{J}_{\text{grav}} \mathrel{+}= \nabla \times \left[\sum_i \boldsymbol{\sigma}_i\, W_D(\mathbf{x} - \mathbf{x}_i) \right], \tag{83}$$

where W_D is the quadratic B-spline kernel (see Section 15) used for dipole deposition.

Step 3. Compute EM effective GEM sources from perturbation potentials (Eqs. 49–50):

$$\rho_{\text{EM}} = \frac{\varepsilon_0}{2c^2} \left(|\nabla\phi^{\text{pert}} + \partial_t \mathbf{A}^{\text{pert}}|^2 + c^2 |\nabla \times \mathbf{A}^{\text{pert}}|^2 \right), \tag{84}$$

$$\mathbf{J}_{\text{EM}} = -\frac{1}{\mu_0 c^2} \left(\nabla\phi^{\text{pert}} + \partial_t \mathbf{A}^{\text{pert}} \right) \times \left(\nabla \times \mathbf{A}^{\text{pert}} \right). \tag{85}$$

Step 4. Sum effective sources.

$$\rho_{\text{eff}} = \rho_{\text{matter}} + \rho_{\text{EM}}, \qquad \mathbf{J}_{\text{eff}} = \mathbf{J}_{\text{matter}} + \mathbf{J}_{\text{EM}}. \tag{86}$$

Step 5. Step GEM wave equations.

$$\Phi_g^{n+1} = 2\Phi_g^n - \Phi_g^{n-1} + (c\,\mathrm{d}t)^2 \left[\nabla^2 \Phi_g^n - 4\pi G \rho_{\text{eff}}^n \right], \tag{87}$$

and similarly for each component of $\mathbf{A}_g$.

Step 6. Step EM wave equations using $c_{\text{local}}^n(\mathbf{x}) = c(1 + 2\Phi_g^n(\mathbf{x})/c^2)$:

$$\phi^{n+1} = 2\phi^n - \phi^{n-1} + (c_{\text{local}}\,\mathrm{d}t)^2 \left[\nabla^2 \phi^n + \rho_q^n / \varepsilon_0 \right], \tag{88}$$

and similarly for each component of $\mathbf{A}$.

Step 7. Compute force on each particle from potentials directly (Eqs. 41–42). All terms involve only spatial and temporal finite differences of the potential arrays at and near the particle position, interpolated to the particle using the CIC weights. Additionally compute $\nabla \times \mathbf{A}_g$ and $\nabla \times \mathbf{A}$ on-the-fly at the particle position for the Boris rotation and spin forces, using the Yee curl stencil evaluated at surrounding cell centers and bilinearly interpolated. The full combined force including spin gradient terms is:

$$\begin{aligned}\mathbf{F} = m(&-\nabla\Phi_g - \partial_t \mathbf{A}_g + 4\nabla(\mathbf{v}\cdot\mathbf{A}_g) - 4(\mathbf{v}\cdot\nabla)\mathbf{A}_g) \\ &+ q(-\nabla\phi - \partial_t \mathbf{A} + \nabla(\mathbf{v}\cdot\mathbf{A}) - (\mathbf{v}\cdot\nabla)\mathbf{A}) \\ &+ \nabla(\mathbf{S}\cdot\nabla\times\mathbf{A}_g) + \nabla(\boldsymbol{\mu}\cdot\nabla\times\mathbf{A}),\end{aligned} \tag{89}$$

where the gradients of $\nabla\times\mathbf{A}_g$ and $\nabla\times\mathbf{A}$ for the spin gradient forces use the adjoint two-tent interpolation (Section 15.5).

Step 8. Update momentum and position (Boris pusher; see Section 15):

$$\mathbf{p} \leftarrow \mathbf{p} + \mathbf{F}\,\mathrm{d}t, \qquad \mathbf{v} = \frac{\mathbf{p}}{\sqrt{m^2 + |\mathbf{p}|^2/c^2}}, \qquad \mathbf{x} \leftarrow \mathbf{x} + \mathbf{v}\,\mathrm{d}t. \tag{90}$$

Record $\mathbf{p}_{\text{old}}$ and $\mathbf{v}_{\text{old}}$ before this step for use in Step 10.

Step 9. Update spin.

$$\mathbf{a} = (\mathbf{v} - \mathbf{v}_{\text{old}})/\mathrm{d}t, \tag{91}$$

$$\boldsymbol{\Omega}_{\text{Thomas}} = -\frac{\gamma^2}{\gamma+1}\frac{\mathbf{v}\times\mathbf{a}}{c^2}, \tag{92}$$

$$\boldsymbol{\Omega}_{\text{total}} = (\nabla\times\mathbf{A}_g)^{\text{particle}} + \frac{g_s q}{2m}(\nabla\times\mathbf{A})^{\text{particle}} + \boldsymbol{\Omega}_{\text{Thomas}}, \tag{93}$$

$$\mathbf{S} \leftarrow \mathbf{S} + (\mathbf{S}\times\boldsymbol{\Omega}_{\text{total}})\,\mathrm{d}t. \tag{94}$$

where $(\nabla\times\mathbf{A}_g)^{\text{particle}}$ and $(\nabla\times\mathbf{A})^{\text{particle}}$ are the on-the-fly Yee curl values computed at the particle position from surrounding edge potentials (already available from the force evaluation step). Apply constraint correction (Eq. 69) and renormalize spin magnitude:

$$\mathbf{S} \leftarrow \mathbf{S} - \frac{\mathbf{S}\cdot\mathbf{p}}{|\mathbf{p}|^2 + m^2c^2}\,\mathbf{p}, \qquad \mathbf{S} \leftarrow \mathbf{S}\,\frac{|\mathbf{S}_0|}{|\mathbf{S}|}. \tag{95}$$

12 Relativistic Force Corrections

12.1 The problem with GEM at high velocities

The GEM Lorentz force is derived under the slow-motion assumption. At relativistic speeds the spatial metric perturbation h^{ij} also contributes to particle deflection. The clearest illustration is photon deflection: the correct GR prediction is twice the Newtonian (GEM) value, with the extra factor coming from spatial Christoffel terms that GEM drops.

12.2 The linearized metric and its two metric coefficients

The full linearized metric is:

$$\mathrm{d}s^2 = -\left(1 + \frac{2\Phi_g}{c^2}\right)c^2\,\mathrm{d}t^2 + \left(1 - \frac{2\Phi_g}{c^2}\right)\left(\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\right). \tag{96}$$

The two coefficients act in opposite directions:

$$\text{Time:} \quad \left(1 + \frac{2\Phi_g}{c^2}\right) < 1 \text{ near mass} \;\Rightarrow\; \text{clocks run slow.} \tag{97}$$

$$\text{Space:} \quad \left(1 - \frac{2\Phi_g}{c^2}\right) > 1 \text{ near mass} \;\Rightarrow\; \text{rulers stretch.} \tag{98}$$

For a slow particle only the time component matters; the spatial component is negligible. For a photon both contribute equally, doubling the deflection relative to the Newtonian prediction.

12.3 The Shapiro delay

A photon follows a null geodesic ($\mathrm{d}s^2 = 0$). Solving for the coordinate speed of light:

$$\frac{\mathrm{d}x}{\mathrm{d}t} \approx c\left(1 + \frac{2\Phi_g}{c^2}\right) \equiv c_{\text{local}}. \tag{99}$$

The factor of 2 in c_{local} comes from both metric coefficients contributing to the null condition. The Shapiro delay is:

$$\Delta t_{\text{Shapiro}} = -\frac{2}{c^3}\int_{\text{path}} \Phi_g \, \mathrm{d}\ell. \tag{100}$$

This is implemented in the simulator by stepping the EM wave equation with c_{local} rather than c. The perturbation of photon paths (gravitational lensing) enters through the trajectory integration, where the same c_{local} modification of the wavefronts causes bending.

12.4 Expansion of the geodesic equation

The relativistic correction terms used here are those of the **Einstein–Infeld–Hoffmann (EIH) equation** [2, 1], the standard post-Newtonian (1PN) equation of motion for self-gravitating particles in harmonic gauge. For a test particle in a (possibly moving) gravitational source, EIH may be written

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -(1 + v^2/c^2 + 4\Phi_g/c^2)\nabla\Phi_g - \partial_t \mathbf{A}_g + \mathbf{v} \times (\nabla \times \mathbf{A}_g) + (3\partial_t\Phi_g + 4\mathbf{v}\cdot\nabla\Phi_g)\mathbf{v}/c^2, \tag{101}$$

which is correct through $\mathcal{O}((v/c)^2)$ in the coordinate acceleration. The gravitomagnetic vector potential $\mathbf{A}_g$ itself scales as $|\mathbf{A}_g| \sim (v/c)^3$ at the source level (since $\Delta\mathbf{A}_g \propto T^{0i} \sim \rho v$), and the terms containing $\partial_t\mathbf{A}_g$ and $\mathbf{v} \times (\nabla \times \mathbf{A}_g)$ contribute to $\mathrm{d}\mathbf{v}/\mathrm{d}t$ at the same 1PN order as the $(v^2/c^2)\nabla\Phi_g$ and $\Phi_g\nabla\Phi_g$ corrections; the Lense–Thirring frame-dragging effect arises from these gravitomagnetic terms.

Expressed in terms of the local gravitational vector $\mathbf{g} = -\nabla\Phi_g - \partial_t\mathbf{A}_g$ and reorganised by order in v/c:

$$a^i = \underbrace{g^i}_{\mathcal{O}(1)} + \underbrace{4\varepsilon^{ijk}v_j B_{g,k}}_{\mathcal{O}(v/c)\ \text{(Lense–Thirring)}} + \underbrace{\frac{v^2}{c^2}g^i + \frac{4\Phi_g}{c^2}g^i - \frac{4(\mathbf{v}\cdot\mathbf{g})v^i}{c^2}}_{\mathcal{O}(v^2/c^2)\ \text{(EIH 1PN)}} + \mathcal{O}(v^3/c^3), \tag{102}$$

where $B_{g,k} = (\nabla \times \mathbf{A}_g)_k$ is evaluated on-the-fly from the potentials. In implementation the cross-product $\varepsilon^{ijk}v_j B_{g,k}$ is replaced by its potential form $\partial_i(\mathbf{v}\cdot\mathbf{A}_g) - (\mathbf{v}\cdot\nabla)A_{g,i}$ via the vector identity, avoiding explicit computation of $\mathbf{B}_g$ as a grid array (Section 15.2).

Note (v34 correction vs. v33). v33 of this document had the velocity-coupling term with a $+$ sign, $+4(\mathbf{v}\cdot\mathbf{g})v^i/c^2$, and omitted the $4\Phi_g\mathbf{g}/c^2$ "Shapiro" radial term. Both have been corrected here against the canonical EIH derivation (e.g. [1] eq. 37; [2] §6.2). The sign error was confirmed empirically: with the v33 sign the Stage 8 perihelion test precesses *retrograde*; with the EIH sign and the Shapiro term included the test recovers the Schwarzschild precession to better than 1% in the deep-1PN regime, after subtracting a small "kinematic" baseline that arises from running relativistic 3-momentum dynamics with a non-relativistic force law (Section 18.8). The Shapiro radial term originates not from the spatial metric coefficient alone but from the $\mathcal{O}(v^4/L)$ correction to Γ^i_{00}, which pulls in the second-order expansion of g_{00}.

The $\mathcal{O}(v^2/c^2)$ terms produce three physically distinct effects:

- $(v^2/c^2)\mathbf{g}$: a fast particle experiences stronger gravitational deflection. Combined with the spatial-metric contribution to c_{local}, the factor-of-2 photon-deflection result is recovered in the $v \to c$ limit.

- $(4\Phi_g/c^2)\mathbf{g}$: an additional radial enhancement of the Newtonian force that vanishes far from the source. Contributes to the static binding energy and to perihelion precession.
- $-4(\mathbf{v}\cdot\mathbf{g})\mathbf{v}/c^2$: a velocity-coupling force along $\mathbf{v}$ that does no work over a closed orbit but breaks the Kepler-orbit closure condition.

Together these are the test-particle limit of the full EIH equation; the combination, integrated over a bound orbit, reproduces the canonical $\Delta\phi_{\text{prec}} = 6\pi G_{\text{eff}} M/(c_{\text{eff}}^2\, a(1-e^2))$ Schwarzschild precession.

12.5 Practical implementation tiers

Tier	Gravitational force expression (EIH 1PN basis)	Valid range
0: Newtonian	$-m\nabla\Phi_g$	$v/c < 0.01$
1: GEM	$m\big(-\nabla\Phi_g - \partial_t\mathbf{A}_g + 4\nabla(\mathbf{v}\cdot\mathbf{A}_g) - 4(\mathbf{v}\cdot\nabla)\mathbf{A}_g\big)$	$v/c < 0.1$
2: EIH 1PN (test particle)	Tier 1 plus $(v^2/c^2)\mathbf{g} + (4\Phi_g/c^2)\mathbf{g} - 4(\mathbf{v}\cdot\mathbf{g})\mathbf{v}/c^2$	$v/c < 0.5$
3: Full EIH	All $\mathcal{O}((v/c)^2)$ acceleration terms; Lense–Thirring enters via $\mathbf{A}_g$ (which scales as $(v/c)^3$ at the source)	$v/c \le 1$

Tier 2 in v34 differs from v33 by (i) the sign on the velocity-coupling term and (ii) the addition of the $4\Phi_g\mathbf{g}/c^2$ Shapiro radial term; see Section 12.4 for the derivation against the EIH lecture 25 form.

The EM Lorentz force $q(-\nabla\phi - \partial_t\mathbf{A} + \nabla(\mathbf{v}\cdot\mathbf{A}) - (\mathbf{v}\cdot\nabla)\mathbf{A})$ is already exactly relativistic and requires no velocity-dependent corrections. In all tiers the momentum update uses the relativistic p-update.

12.6 Where each effect enters the simulator

Effect	Source	Simulator entry point
Gravitational time dilation	g_{00} expansion	Proper time update
Shapiro delay / light bending	Both metric coefficients via c_{local}	EM wave equation
Doubled photon deflection	Spatial metric $(1-2\Phi_g/c^2)$	c_{local}; $(v^2/c^2)\mathbf{g}$ for massive particles
Perihelion precession	EIH 1PN: combination of $(v^2/c^2)\mathbf{g}$, $(4\Phi_g/c^2)\mathbf{g}$, $-4(\mathbf{v}\cdot\mathbf{g})\mathbf{v}/c^2$	Tier-2 force terms
Lense–Thirring frame dragging	Gravitomagnetic $\mathbf{A}_g \sim (v/c)^3$ at source	GEM Tier-1 ($\mathbf{A}_g$) potential terms

12.7 Complete algorithm including spin and relativistic corrections (Method 2)

This enumeration extends Section 11.8 by adding the Tier 3 relativistic force corrections. Steps 1–7 are unchanged. Steps 8 and 10 are modified as shown.

Step 1. Deposit matter and charge sources. (As in Section 11.8, Step 1.)

Step 2. Deposit spin dipole sources. (As in Section 11.8, Step 2.)

Step 3. Compute EM effective GEM sources. (As before.)

Step 4. Sum effective sources. (As before.)

Step 5. Step GEM wave equations. (As before.)

Step 6. Step EM wave equations with $c_{\text{local}} = c(1 + 2\Phi_g/c^2)$. (As before.)

Step 7. Compute total force on each particle from potentials directly. The gravitomagnetic cross-product terms $\mathbf{v} \times \mathbf{B}_g$ and $\mathbf{v} \times \mathbf{B}$ are folded into the potential-form force via the identity $\mathbf{v} \times (\nabla \times \mathbf{A}) = \nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla)\mathbf{A}$. The $\mathbf{g}$ appearing in the EIH 1PN correction terms is computed on-the-fly at the particle position as $\mathbf{g} = -\nabla\Phi_g - \partial_t \mathbf{A}_g$. The $4\Phi_g \mathbf{g}/c^2$ ("Shapiro") term uses the total Φ_g interpolated at the particle:

$$
\begin{aligned}
\mathbf{F}_{\text{grav}} &= m\left[-\nabla\Phi_g - \partial_t \mathbf{A}_g + 4\nabla(\mathbf{v} \cdot \mathbf{A}_g) - 4(\mathbf{v} \cdot \nabla)\mathbf{A}_g + \frac{v^2}{c^2}\mathbf{g} + \frac{4\Phi_g}{c^2}\mathbf{g} - \frac{4(\mathbf{v} \cdot \mathbf{g})\mathbf{v}}{c^2}\right], \\
\mathbf{F}_{\text{EM}} &= q[-\nabla\phi - \partial_t \mathbf{A} + \nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla)\mathbf{A}], \\
\mathbf{F}_{\text{spin}} &= \nabla(\mathbf{S} \cdot \nabla \times \mathbf{A}_g) + \nabla(\boldsymbol{\mu} \cdot \nabla \times \mathbf{A}), \\
\mathbf{F}_{\text{total}} &= \mathbf{F}_{\text{grav}} + \mathbf{F}_{\text{EM}} + \mathbf{F}_{\text{spin}}.
\end{aligned}
$$

The bracketed expression in $\mathbf{F}_{\text{grav}}$ is the test-particle limit of the EIH equation (Eq. 101) written in terms of $\mathbf{g}$ instead of $\nabla\Phi_g$; v33 had the sign of the velocity-coupling term reversed and was missing the $4\Phi_g \mathbf{g}/c^2$ Shapiro term (Section 12.4). The quantities $\nabla \times \mathbf{A}_g$ and $\nabla \times \mathbf{A}$ are evaluated on-the-fly at the particle position from surrounding edge potentials via the Yee curl (Eq. 114). Evaluate v^2/c^2 and $\mathbf{v} \cdot \mathbf{g}$ using $\mathbf{v}_{n-1/2}$ from the previous half-step.

Step 8. Update momentum and position (Boris pusher; see Section 15):

$$
\mathbf{p} \leftarrow \mathbf{p} + \mathbf{F}_{\text{total}}\,\mathrm{d}t, \qquad \mathbf{v} = \frac{\mathbf{p}}{\sqrt{m^2 + |\mathbf{p}|^2/c^2}}, \qquad \mathbf{x} \leftarrow \mathbf{x} + \mathbf{v}\,\mathrm{d}t. \tag{103}
$$

Record $\mathbf{v}_{\text{old}}$ before the update for Step 10.

Step 9. Update spin. (As in Section 11.8, Step 10, using updated $\mathbf{v}$ and $\mathbf{a} = (\mathbf{v} - \mathbf{v}_{\text{old}})/\mathrm{d}t$.)

13 Proper Time Clocks

13.1 Overview

Each particle carries a proper time clock τ_i accumulating the elapsed proper time along its worldline. The clock requires no new field equations and adds negligible computational cost—one square root per particle per timestep.

13.2 Proper time in the linearized metric

For a non-rotating background the proper time rate is determined by the gravitoelectric potential and the particle velocity:

$$
\frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{\left(1 + \frac{2\Phi_g}{c^2}\right) - \left(1 - \frac{2\Phi_g}{c^2}\right)\frac{v^2}{c^2}}. \tag{104}
$$

For a rotating background the off-diagonal metric components contribute an additional term. To fix the cross-term coefficient consistently with the doc's force-law convention — the GEM Lorentz force has a factor of 4 from the spin-2 nature of gravity, $\mathbf{F}/m \supset 4\mathbf{v} \times (\nabla \times \mathbf{A}_g)$ — the linearized metric perturbation must be $h_{0i} = 4A_{g,i}/c$, not the $2A_{g,i}/c$ written in v33 of this document (which was inconsistent with its own force law). See the note at the end of this subsection for the v34 corrections. The cross term therefore contributes $g_{0i}\mathrm{d}x^i\mathrm{d}t = (4A_{g,i}/c)v^i\mathrm{d}t^2$ to $\mathrm{d}s^2$, and the full linearized expression for $\mathrm{d}\tau/\mathrm{d}t$ becomes:

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = \sqrt{\left(1 + \frac{2\Phi_g}{c^2}\right) - \left(1 - \frac{2\Phi_g}{c^2}\right)\frac{v^2}{c^2} - \frac{8\,\mathbf{A}_g \cdot \mathbf{v}}{c^2}}. \tag{105}$$

The $-8\mathbf{A}_g{\cdot}\mathbf{v}/c^2$ term has opposite sign for prograde and retrograde orbits around a spinning mass (since $\mathbf{v}$ reverses but $\mathbf{A}_g$ does not), producing a proper time difference between co- and counter-rotating clocks at the same orbital radius — the gravitomagnetic clock effect. The negative sign means *prograde* clocks run slower than retrograde clocks at the same orbital radius, consistent with the standard Cohen–Mashhoon result for Kerr [9, 8]. For a non-rotating background $\mathbf{A}_g = 0$ and Eq. (105) reduces to the simpler form.

To first order:

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} \approx 1 + \underbrace{\frac{\Phi_g}{c^2}}_{\text{gravitational}} - \underbrace{\frac{v^2}{2c^2}}_{\text{kinematic}} - \underbrace{\frac{4\,\mathbf{A}_g \cdot \mathbf{v}}{c^2}}_{\text{gravitomagnetic}}. \tag{106}$$

The exact expression is used in the integrator; the first-order form is shown for interpretation.

Provenance and v34 correction. The linearized line element underlying Eq. (105) is the one given in [1] eq. 41: $\mathrm{d}s^2 = -(1+2\phi+\ldots)\mathrm{d}t^2+2\xi_i\,\mathrm{d}t\,\mathrm{d}x^i+(1-2\phi)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j$ in geometric units. The lecture's gravitomagnetic potential ξ_i equals $4A_{g,i}/c$ in the doc's GEM-with-spin-2-factor-in-force convention (since the lecture's geodesic equation eq. 37 carries coefficient 1 on $\mathbf{v} \times (\nabla \times \xi)$ whereas this document's force law carries coefficient 4 on $\mathbf{v}\times(\nabla\times\mathbf{A}_g)$). Substituting $\xi_i = 4A_{g,i}/c$ into $\mathrm{d}\tau^2 = -\mathrm{d}s^2$ and restoring c gives Eq. (105) directly. For an explicit gravitomagnetic proper-time formula in a different (factor-of-2-in-force) GEM convention see [8] §2; results agree up to an overall $\mathbf{A}_g$ normalization.

v33 of this document wrote $h_{0i} = 2A_{g,i}/c$ and a corresponding $+(4\mathbf{A}_g \cdot \mathbf{v})/c^2$ inside the square root (linearized $+(2\mathbf{A}_g \cdot \mathbf{v})/c^2$). Both the magnitude and the sign were wrong; the magnitude conflicted with v33's own force-law factor of 4, and the sign would have predicted prograde clocks running *faster* than retrograde, opposite to the canonical clock-effect direction.

13.3 Physical interpretation

The kinematic term $-v^2/(2c^2)$ always slows the clock. The gravitational term Φ_g/c^2 also slows it in a well ($\Phi_g < 0$), and speeds it up at high altitude (Φ_g less negative). For GPS satellites the two terms compete: the satellite is high up (gravitational term speeds clock) but moving fast (kinematic term slows clock). The gravitational effect wins, so GPS clocks run fast by 38 μs/day and must be corrected.

The gravitomagnetic term $-4\mathbf{A}_g \cdot \mathbf{v}/c^2$ is nonzero only around rotating masses. It has opposite sign for co- and counter-rotating orbits because $\mathbf{v}$ reverses direction while $\mathbf{A}_g$ does not, and with the minus sign *prograde* clocks accumulate proper time more slowly than retrograde clocks at the same orbital radius. Integrating around a full orbit the difference in accumulated proper time between prograde (+) and retrograde (−) orbits is:

$$\Delta\tau = \tau_+ - \tau_- = -\frac{8}{c^2}\oint \mathbf{A}_g \cdot \mathrm{d}\mathbf{x} = -\frac{8}{c^2}\int (\nabla \times \mathbf{A}_g)\cdot \mathrm{d}\mathbf{A} = -\frac{8}{c^2}\int \mathbf{B}_g \cdot \mathrm{d}\mathbf{A}, \tag{107}$$

by Stokes' theorem — the proper time difference is proportional to (minus) the gravitomagnetic flux through the orbit. This is the gravitomagnetic clock effect, and it is a direct consequence of the same $\mathbf{A}_g$ that appears in the GEM force expression. [9] gave the original Kerr derivation; [10] and [11] re-derive and discuss detectability. Expressed in this document's $\mathbf{A}_g$ convention the prograde-minus-retrograde clock difference matches the formula above up to the overall $\mathbf{A}_g$ normalization ([8] expresses the same effect in his factor-of-2 GEM convention).

13.4 Integration

Using the total potential $\Phi_g^{\text{total}} = \Phi_g^{\text{bg}} + \Phi_g^{\text{pert}}$ and total vector potential $\mathbf{A}_g^{\text{total}} = \mathbf{A}_g^{\text{bg}} + \mathbf{A}_g^{\text{pert}}$, both already evaluated during force calculation:

$$\mathrm{d}\tau_i = \mathrm{d}t\sqrt{1 + \frac{2\Phi_g}{c^2} - \left(1 - \frac{2\Phi_g}{c^2}\right)\beta^2 - \frac{8\,\mathbf{A}_g\cdot\mathbf{v}}{c^2}}, \qquad \beta^2 = v^2/c^2, \tag{108}$$

$$\tau_i \;\leftarrow\; \tau_i + \mathrm{d}\tau_i. \tag{109}$$

The quantity $\mathbf{A}_g \cdot \mathbf{v}$ is already computed as an intermediate result in the GEM force expression $4\nabla(\mathbf{v}\cdot\mathbf{A}_g) - 4(\mathbf{v}\cdot\nabla)\mathbf{A}_g$ — specifically it is the scalar $\mathbf{v}\cdot\mathbf{A}_g$ whose gradient appears in the force. No additional grid interpolation is required; the same interpolated value is reused.

In the leapfrog scheme, $\mathbf{v}$ lives at half-integer times. A second-order accurate estimate at the integer step is $\mathbf{v}_n \approx \frac{1}{2}(\mathbf{v}_{n-1/2} + \mathbf{v}_{n+1/2})$.

13.5 Pedagogical demonstrations

Demonstration	Setup	What the student sees
Twin paradox	One particle stationary, one on an elliptical orbit.	$\tau_{\text{traveler}} < \tau_{\text{stationary}}$; traveling twin is younger.
Gravitational time dilation	Two stationary particles at different radii.	Deeper clock ticks slower; purely the Φ_g/c^2 term.
GPS effect	Circular orbiter vs. stationary at same radius.	Two competing effects; exact cancellation at one specific radius.
Clock in gravitational wave	Stationary particle, scalar wave passes through.	Proper time rate oscillates as Φ_g alternately deepens and shallows.
Gravitomagnetic clock effect	Two particles on identical circular orbits in opposite directions around a spinning background mass.	Prograde and retrograde clocks accumulate different proper time per orbit; difference grows linearly with number of orbits and is proportional to the gravitomagnetic flux $\int \mathbf{B}_g \cdot \mathrm{d}\mathbf{A}$ through the orbit.

14 Background Spacetime and the Perturbation Split

14.1 Motivation

When one mass dominates the gravitational field, the dominant background can be precomputed once and held fixed; only the perturbation fields need to be time-stepped. This is the background

+ perturbation split, used in black hole perturbation theory, restricted celestial mechanics, and QFT in curved spacetime.

14.2 The decomposition

$$\Phi_g(\mathbf{x}, t) = \Phi_g^{\text{bg}}(\mathbf{x}) + \Phi_g^{\text{pert}}(\mathbf{x}, t), \tag{110}$$

and similarly for $\mathbf{A}_g$, ϕ, $\mathbf{A}$. The total force on each particle is:

$$\mathbf{F}_{\text{total}} = \mathbf{F}(\mathbf{g}^{\text{bg}} + \mathbf{g}^{\text{pert}},\ \mathbf{B}_g^{\text{bg}} + \mathbf{B}_g^{\text{pert}},\ \mathbf{E}^{\text{bg}} + \mathbf{E}^{\text{pert}},\ \mathbf{B}^{\text{bg}} + \mathbf{B}^{\text{pert}}). \tag{111}$$

14.3 Analytic background configurations

Type	Potentials	Notes
Schwarzschild-like	$\Phi_g^{\text{bg}} = -GM/r$	Pure gravitoelectric
Reissner–Nordström-like	$\Phi_g^{\text{bg}} = -GM/r$, $\phi^{\text{bg}} = Q/(4\pi\varepsilon_0 r)$	Gravity vs. electrostatics
Kerr-like (2D)	$\Phi_g^{\text{bg}} = -GM/r$, $A_{g,\theta}^{\text{bg}} = 2GJ/(c^2 r)$	Frame dragging
Kerr–Newman-like	All four potentials nonzero	Full background
Uniform field	$\Phi_g^{\text{bg}} = -g_0 y$	Equivalence principle

14.4 Frame dragging from the Kerr-like background

A mass M with angular momentum J generates in 2D:

$$A_{g,\theta}^{\text{bg}} = \frac{2GJ}{c^2 r}, \qquad B_{g,z}^{\text{bg}} = -\frac{2GJ}{c^2 r^2}. \tag{112}$$

A gyroscope at radius r precesses at $\mathrm{d}\phi/\mathrm{d}t = B_{g,z}^{\text{bg}}$.

14.5 Three modes of operation

Mode	Background	FDTD solver	Use case
1: Pure background	Fixed analytic	Not run	Test particles; zero FDTD cost
2: Background + perturbation	Fixed analytic	Perturbation fields	Few active interacting particles
3: Full dynamic	None fixed	All fields	Equal-mass many-body

In Mode 2, the perturbation fields are sourced only by non-background particles. The static background contributes no source to the perturbation wave equations.

14.6 Validity conditions

- $|\Phi_g^{\text{pert}}| \ll |\Phi_g^{\text{bg}}|$: perturbations are small.
- Background approximately stationary (update adiabatically if needed).
- Back-reaction on background negligible.
- Background in linearized regime—exclude particles within a few Schwarzschild radii.

15 Numerical Implementation

15.1 Field grid

The simulation is formulated primarily in terms of the scalar and vector *potentials* rather than the derived field vectors $\mathbf{E}$, $\mathbf{B}$, $\mathbf{g}$, $\mathbf{B}_g$. The potentials are the quantities evolved by the wave equations and stored on the grid. The derived field vectors are gauge-invariant and are computed on demand — at particle positions during force evaluation, and over the full grid only for visualization. This eliminates four grid arrays (g_x, g_y, E_x, E_y, $B_{g,z}$, B_z as persistent arrays) and the associated full-grid curl passes from the per-timestep inner loop.

The potentials are discretized on a staggered Yee grid [14, 15], which assigns different components to different spatial locations within each grid cell. The staggering ensures that finite-difference gradient and curl operations on the potentials are exact — no interpolation is needed.

The 2D Yee grid layout

In 2D each cell is a square of side Δx. The cell at index (i, j) has its lower-left corner at position $(i\Delta x, j\Delta y)$. Only the potentials and their sources are stored as persistent grid arrays:

Quantity	Location	Index notation
Φ_g, ϕ (scalar potentials)	Cell corners	$(i,\ j)$
ρ_{matter}, ρ_q (charge/mass density)	Cell corners	$(i,\ j)$
$A_{g,x}$, A_x (x-potentials)	Horizontal edge centers	$(i+\frac{1}{2},\ j)$
$J_{\text{matter},x}$, $J_{q,x}$ (x-currents)	Horizontal edge centers	$(i+\frac{1}{2},\ j)$
$A_{g,y}$, A_y (y-potentials)	Vertical edge centers	$(i,\ j+\frac{1}{2})$
$J_{\text{matter},y}$, $J_{q,y}$ (y-currents)	Vertical edge centers	$(i,\ j+\frac{1}{2})$

The derived quantities g_x, g_y, E_x, E_y (which would live on edges) and $B_{g,z}$, B_z (which would live at cell centers) are *not* stored as persistent arrays. They are computed locally at particle positions during force evaluation, and computed over the full grid only when requested for visualization.

The layout is shown schematically for one cell:

$$
\begin{array}{ccccc}
(i, j+1) & - & A_x(i+\frac{1}{2}, j+1) & - & (i+1, j+1) \\
| & & & & | \\
A_y(i, j+\frac{1}{2}) & & \text{[cell center]} & & A_y(i+1, j+\frac{1}{2}) \\
| & & & & | \\
(i, j) & - & A_x(i+\frac{1}{2}, j) & - & (i+1, j)
\end{array}
$$

Why this staggering: exact gradient and curl operations

Gradient (for force and for visualization): The gravitoelectric field component $g_x = -\partial_x \Phi_g - \partial_t A_{g,x}$ is evaluated at a horizontal edge position $(i+\frac{1}{2}, j)$. With Φ_g at corners, the spatial gradient $-\partial_x \Phi_g$ naturally lands at horizontal edges — exactly where $A_{g,x}$ lives:

$$
g_{x,i+1/2,j} = -\frac{(\Phi_g)_{i+1,j} - (\Phi_g)_{i,j}}{\Delta x} - \frac{A^{n+1}_{g,x,i+1/2,j} - A^{n-1}_{g,x,i+1/2,j}}{2\Delta t}, \tag{113}
$$

and similarly for g_y at vertical edges, and for $\mathbf{E}$ from ϕ and $\mathbf{A}$. This operation is performed on demand at particle positions (interpolating Φ_g from nearby corners and $\mathbf{A}_g$ from nearby edges) rather than sweeping the full grid.

Curl (for Boris rotation, spin precession, and visualization): The out-of-plane field $B_z = \partial_x A_y - \partial_y A_x$ at a cell center $(i + \frac{1}{2}, j + \frac{1}{2})$ is:

$$B_{z,i+1/2,j+1/2} = \frac{A_{y,i+1,j+1/2} - A_{y,i,j+1/2}}{\Delta x} - \frac{A_{x,i+1/2,j+1} - A_{x,i+1/2,j}}{\Delta y}, \tag{114}$$

and identically for $B_{g,z}$ from $\mathbf{A}_g$. This is evaluated at the few cell centers surrounding the particle when needed for the Boris rotation or spin precession, not over the full grid each step. For visualization, a full-grid sweep applies Eq. (114) at every cell center.

Source deposition staggering: Source terms must be deposited at the same locations as the potentials they source:

- ρ_{matter}, ρ_q: at cell corners (i, j). The CIC kernel W_2 is evaluated at corner positions relative to the particle.
- $J_{\text{matter},x}$, $J_{q,x}$: at horizontal edge centers $(i + \frac{1}{2}, j)$. The CIC kernel is evaluated at horizontal edge positions. The sub-cell fraction is $\alpha = (x_p - (i + \frac{1}{2})\Delta x)/\Delta x$, measured from the edge center rather than the corner.
- $J_{\text{matter},y}$, $J_{q,y}$: at vertical edge centers $(i, j + \frac{1}{2})$.

Spin dipole curl deposition: The magnetization curl $\nabla \times [\mu\, W_3]$ deposits $J_{q,x}$ at horizontal edges and $J_{q,y}$ at vertical edges, consistent with the current staggering. The x-component is $\mu\, W_3(x - x_p) \cdot (\mathrm{d}W_3/\mathrm{d}y)(y - y_p)$, depositing to horizontal edges via the smooth W_3 weight in x and the two-tent $\mathrm{d}W_3/\mathrm{d}y$ weight in y, centered on horizontal edge positions $(i + \frac{1}{2}, j)$.

15.2 Time stepping: the leapfrog scheme

The field leapfrog and its three-point stencil

The d'Alembert equation $\partial_t^2 \Phi = c^2 \nabla^2 \Phi + \text{source}$ is discretized in time using the three-point leapfrog (Störmer-Verlet) scheme. The superscripts $n-1$, n, $n+1$ denote values at three consecutive integer timesteps separated by $\mathrm{d}t$:

$$\Phi^{n+1} = 2\Phi^n - \Phi^{n-1} + (c\,\mathrm{d}t)^2 \left[\nabla^2 \Phi^n + \text{source}^n\right]. \tag{115}$$

The term $2\Phi^n - \Phi^{n-1}$ is a linear extrapolation of Φ forward by one timestep using the last two known values. It is the free-propagation prediction — what Φ^{n+1} would be if the right-hand side were zero (no source, no spatial coupling). The $(c\,\mathrm{d}t)^2[\cdots]$ term is the correction from the wave equation. This is not overrelaxation in the iterative solver sense. Rather, it is the standard second-order centered finite difference for $\partial_t^2 \Phi$:

$$\frac{\Phi^{n+1} - 2\Phi^n + \Phi^{n-1}}{\mathrm{d}t^2} \approx \left.\frac{\partial^2 \Phi}{\partial t^2}\right|_n + \mathcal{O}(\mathrm{d}t^2), \tag{116}$$

rearranged to solve for Φ^{n+1}. The scheme is second-order accurate in time and requires storing only two time levels (Φ^n and Φ^{n-1}) to advance to Φ^{n+1}.

The CFL stability condition requires:

$$\mathrm{d}t < \frac{\mathrm{d}x}{c\sqrt{d}}, \qquad d = \text{spatial dimension} \quad (d = 2 \text{ or } 3). \tag{117}$$

The α-family: interpolating between Euler and leapfrog

The leapfrog can be placed in a one-parameter family that continuously interpolates between a naive Euler scheme and the full leapfrog:

$$\Phi^{n+1} = (1+\alpha)\Phi^n - \alpha\Phi^{n-1} + \mathrm{d}t^2\, f^n, \tag{118}$$

where $f^n = c^2\nabla^2\Phi^n + \text{source}^n$. The parameter α controls the history weighting:

$$\alpha = 0: \quad \Phi^{n+1} = \Phi^n + \mathrm{d}t^2 f^n \quad \text{(forward Euler applied to } \partial_t^2\text{; first-order, unstable for wave eq.)} \tag{119}$$

$$\alpha = 1: \quad \Phi^{n+1} = 2\Phi^n - \Phi^{n-1} + \mathrm{d}t^2 f^n \quad \text{(standard leapfrog; second-order, conditionally stable)} \tag{120}$$

Values $0 < \alpha < 1$ give intermediate schemes that are first-order accurate but more stable than pure Euler. Values $\alpha > 1$ are analogous to overrelaxation and are *unstable* for the wave equation — the leapfrog at $\alpha = 1$ is the maximally stable point in the explicit family.

The analogous family for the implicit θ-method (Newmark-β) is:

$$\Phi^{n+1} - 2\Phi^n + \Phi^{n-1} = \mathrm{d}t^2\left[\theta f^{n+1} + (1-2\theta) f^n + \theta f^{n-1}\right], \tag{121}$$

with $\theta = 0$ recovering the explicit leapfrog and $\theta = \frac{1}{4}$ giving an unconditionally stable implicit scheme. The implicit schemes require solving a linear system at each step and are not used here.

The leapfrog stagger: what lives when

The field and particle updates are interleaved on a staggered time grid. Understanding what quantity lives at what time is essential for correctly evaluating forces, sources, and proper time. The complete stagger is:

Quantity	**Time level**	**Notes**
Scalar potentials Φ_g, ϕ	Integer $n\,\mathrm{d}t$	Two levels stored: n and $n-1$
Vector potentials $\mathbf{A}_g$, $\mathbf{A}$	Integer $n\,\mathrm{d}t$	Two levels stored: n and $n-1$
Particle positions $\mathbf{x}_i$	Integer $n\,\mathrm{d}t$	
Source terms ρ, $\mathbf{J}$	Integer $n\,\mathrm{d}t$	Deposited from $\mathbf{x}_i^n$, $\mathbf{v}_i^{n-1/2}$
Forces $\mathbf{F}_i$	Integer $n\,\mathrm{d}t$	Computed from potentials at particle position
Particle momenta $\mathbf{p}_i$, velocities $\mathbf{v}_i$	Half-integer $(n \pm \frac{1}{2})\mathrm{d}t$	Boris kick uses $\mathbf{F}^n$
Proper time increment $\mathrm{d}\tau_i$	Integer $n\,\mathrm{d}t$	Uses $\bar{v}_i = \frac{1}{2}(v^{n-1/2} + v^{n+1/2})$; reuses $(\mathbf{A}_g \cdot \mathbf{v})_i$ from force step

The derived fields $\mathbf{g}$, $\mathbf{B}_g$, $\mathbf{E}$, $\mathbf{B}$ are not stored as persistent arrays. They are computed on demand from the potentials at particle positions during force evaluation (Step 4 below) and over the full grid only for visualization.

The per-timestep sequence is:

1. Deposit sources from $\mathbf{x}_i^n$, $\mathbf{v}_i^{n-1/2}$ onto grid $\to \rho^n$, $\mathbf{J}^n$.
2. Step potentials: $(\Phi_g^{n-1}, \Phi_g^n, \rho^n) \to \Phi_g^{n+1}$, and similarly for ϕ, $\mathbf{A}_g$, $\mathbf{A}$.
3. Apply gauge cleaning to Φ_g^{n+1}, $\mathbf{A}_g^{n+1}$ (Section 15.6).
4. Compute force $\mathbf{F}_i^n$ at each particle from potentials at surrounding grid points (see Section on force from potentials below).
5. Boris kick: $(\mathbf{p}_i^{n-1/2}, \mathbf{F}_i^n) \to \mathbf{p}_i^{n+1/2}$.
6. Drift: $\mathbf{x}_i^{n+1} = \mathbf{x}_i^n + \mathbf{v}_i^{n+1/2}\,\mathrm{d}t$.
7. Update spin using $\mathbf{a}_i^n = (\mathbf{v}_i^{n+1/2} - \mathbf{v}_i^{n-1/2})/\mathrm{d}t$.
8. Update proper time using $\bar{v}_i = \frac{1}{2}(\mathbf{v}_i^{n-1/2} + \mathbf{v}_i^{n+1/2})$, $(\Phi_g^{\text{total}})_i$, and $(\mathbf{A}_g^{\text{total}} \cdot \bar{\mathbf{v}}_i)$ — the last quantity already computed during the force evaluation step.

The force $\mathbf{F}_i^n$ and position $\mathbf{x}_i^n$ are both at integer time, and the momentum $\mathbf{p}_i^{n+1/2}$ is at the half-integer time between them — this stagger is what makes the scheme symplectic.

Force from potentials

The Lorentz force in EM, written directly in terms of the scalar potential ϕ and vector potential $\mathbf{A}$, using the vector identity $\mathbf{v} \times (\nabla \times \mathbf{A}) = \nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla)\mathbf{A}$:

$$\mathbf{F}_{\mathrm{EM}} = q(-\nabla\phi - \partial_t \mathbf{A} + \nabla(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \nabla)\mathbf{A})\,. \tag{122}$$

The GEM analogue (with factor of 4 on the vector potential terms from the spin-2 structure):

$$\mathbf{F}_{\mathrm{GEM}} = m(-\nabla\Phi_g - \partial_t \mathbf{A}_g + 4\nabla(\mathbf{v} \cdot \mathbf{A}_g) - 4(\mathbf{v} \cdot \nabla)\mathbf{A}_g)\,. \tag{123}$$

v is a constant vector, not a field. In these expressions $\mathbf{v} = \mathbf{v}_i$ is the velocity of a specific particle evaluated at its position — a single constant vector, not a spatially varying field. The full vector identity for two arbitrary vector fields $\mathbf{u}(\mathbf{x})$ and $\mathbf{A}(\mathbf{x})$ contains two additional terms:

$$\mathbf{u} \times (\nabla \times \mathbf{A}) = \nabla(\mathbf{u} \cdot \mathbf{A}) - (\mathbf{u} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{u} - \mathbf{A} \times (\nabla \times \mathbf{u}). \tag{124}$$

When $\mathbf{u} = \mathbf{v}_i$ is a constant (i.e. $\nabla \mathbf{v}_i = 0$), the last two terms vanish and the identity reduces to the two-term form used here. The spatial gradient ∇ in $\nabla(\mathbf{v} \cdot \mathbf{A})$ and the directional derivative $(\mathbf{v} \cdot \nabla)$ act only on $\mathbf{A}(\mathbf{x})$, with $\mathbf{v}$ held fixed.

The directional derivative $(\mathbf{v} \cdot \nabla)\mathbf{A}$. The notation $(\mathbf{v} \cdot \nabla)\mathbf{A}$ is the directional derivative of the vector field $\mathbf{A}$ in the direction of $\mathbf{v}$. The operator $\mathbf{v} \cdot \nabla = v_x\partial_x + v_y\partial_y + v_z\partial_z$ is a scalar operator applied component-wise:

$$(\mathbf{v} \cdot \nabla)\mathbf{A} = \begin{pmatrix} v_x\,\partial_x A_x + v_y\,\partial_y A_x + v_z\,\partial_z A_x \\ v_x\,\partial_x A_y + v_y\,\partial_y A_y + v_z\,\partial_z A_y \\ v_x\,\partial_x A_z + v_y\,\partial_y A_z + v_z\,\partial_z A_z \end{pmatrix}. \tag{125}$$

In 2D the z-row is absent and $v_z = 0$. On the Yee grid the x-component of $(\mathbf{v} \cdot \nabla)\mathbf{A}$ at the particle position is:

$$[(\mathbf{v} \cdot \nabla)A_x]_p = v_x\,\frac{A_{x,i+1,j} - A_{x,i,j}}{\Delta x} + v_y\,\frac{A_{x,i+1/2,j+1} - A_{x,i+1/2,j}}{\Delta y}, \tag{126}$$

where v_x and v_y are the scalar particle velocity components — constants that multiply the spatial finite differences of A_x. The y-component follows identically with A_y.

The combination $\nabla(\mathbf{v}\cdot\mathbf{A}) - (\mathbf{v}\cdot\nabla)\mathbf{A}$ encodes the magnetic cross-product force. To verify, the x-component is:

$$[\nabla(\mathbf{v}\cdot\mathbf{A}) - (\mathbf{v}\cdot\nabla)\mathbf{A}]_x = v_y(\partial_x A_y - \partial_y A_x) = v_y B_z, \tag{127}$$

which is the x-component of $\mathbf{v}\times\mathbf{B}$, as expected. The cross product is therefore correctly encoded without ever explicitly forming $\mathbf{B}$ as a grid array.

These expressions involve no derived field vectors — only spatial and temporal differences of the potentials evaluated at and near the particle position. They are gauge-invariant: under $\phi \to \phi - \partial_t\Lambda$, $\mathbf{A} \to \mathbf{A} + \nabla\Lambda$ the extra terms cancel identically, so gauge drift in the potentials does not contaminate the computed force.

Each term is evaluated by interpolating potentials from the surrounding grid points to the particle position using the CIC kernel, then forming the appropriate differences:

- $-\nabla\phi$: finite difference of ϕ at surrounding corners, CIC-interpolated to the particle.
- $-\partial_t\mathbf{A}$: centered time difference $(A_\alpha^{n+1} - A_\alpha^{n-1})/(2\Delta t)$ at edge positions, CIC-interpolated to the particle.
- $\nabla(\mathbf{v}\cdot\mathbf{A}) = v_x\nabla A_x + v_y\nabla A_y$: spatial differences of the edge-located A_x and A_y, with v_x, v_y as constant multipliers.
- $(\mathbf{v}\cdot\nabla)\mathbf{A}$: directional derivative of $\mathbf{A}$ along $\mathbf{v}$ as defined above — spatial differences of edge values weighted by velocity components.

The full force stencil touches the same 4 corners and 8 edge points as the CIC interpolation, requiring no larger neighborhood.

The Boris pusher and on-the-fly curl

The Boris algorithm [19, 18] requires the magnetic rotation step, which needs $B_{g,z}$ and B_z at the particle position. These are computed on the fly from the surrounding edge potentials using Eq. (114), evaluated at the few cell centers near the particle and then CIC-interpolated to the particle position:

$$B_z(\mathbf{x}_p) \approx \sum_{j,k} B_{z,j+1/2,k+1/2}\cdot W_2(\mathbf{x}_{j+1/2,k+1/2} - \mathbf{x}_p), \tag{128}$$

where each $B_{z,j+1/2,k+1/2}$ is evaluated from the four surrounding edge values via Eq. (114). This touches at most 4 cell centers near the particle — a negligible cost.

The effective combined fields for Boris are then:

$$\mathbf{B}_{\text{eff}} = 4\,(\nabla\times\mathbf{A}_g)(\mathbf{x}_p) + \frac{q}{m}(\nabla\times\mathbf{A})(\mathbf{x}_p), \tag{129}$$

where both curls are evaluated on-the-fly at the particle position from surrounding edge potentials via Eq. (114). In 2D the curl fields reduce to scalars with only a z-component, so:

$$B_{\text{eff},z} = 4\,B_{g,z}(\mathbf{x}_p) + \frac{q}{m}B_z(\mathbf{x}_p) \qquad \text{(2D specialization).} \tag{130}$$

The Boris algorithm splits the total force into two parts. The **electric-like force $\mathbf{F}_{\text{eff,E}}$** collects all terms that do not involve the magnetic cross product — the gradient and time-derivative potential terms:

$$\mathbf{F}_{\text{eff,E}} = m(-\nabla\Phi_g - \partial_t\mathbf{A}_g) + q(-\nabla\phi - \partial_t\mathbf{A}) + \mathbf{F}_{\text{spin}} + \mathbf{F}_{\text{rel}}, \tag{131}$$

where $\mathbf{F}_{\text{spin}} = \nabla(\mathbf{S}\cdot\nabla\times\mathbf{A}_g) + \nabla(\boldsymbol{\mu}\cdot\nabla\times\mathbf{A})$ and $\mathbf{F}_{\text{rel}}$ are the spin gradient and relativistic correction forces (Section 12). The velocity-dependent cross-product terms $4\nabla(\mathbf{v}\cdot\mathbf{A}_g) - 4(\mathbf{v}\cdot\nabla)\mathbf{A}_g$ and

$\nabla(\mathbf{v}\cdot\mathbf{A})-(\mathbf{v}\cdot\nabla)\mathbf{A}$ encode the magnetic rotation and are handled implicitly by the Boris rotation step using $B_{\mathrm{eff},z}$, not included in $\mathbf{F}_{\mathrm{eff,E}}$.

The Boris update applies $\mathbf{F}_{\mathrm{eff,E}}$ as two half-kicks bracketing the exact magnetic rotation:

$$\mathbf{p}^{-} = \mathbf{p}^{n-1/2} + \tfrac{1}{2}\mathbf{F}_{\mathrm{eff,E}}\,\mathrm{d}t, \tag{132}$$

$$\mathbf{t} = \frac{q\,\mathbf{B}_{\mathrm{eff}}}{2\gamma^{-}m}\,\mathrm{d}t, \qquad \mathbf{s} = \frac{2\mathbf{t}}{1+|\mathbf{t}|^{2}}, \tag{133}$$

$$\mathbf{p}' = \mathbf{p}^{-} + \mathbf{p}^{-}\times\mathbf{t}, \qquad \mathbf{p}^{+} = \mathbf{p}^{-} + \mathbf{p}'\times\mathbf{s}, \tag{134}$$

$$\mathbf{p}^{n+1/2} = \mathbf{p}^{+} + \tfrac{1}{2}\mathbf{F}_{\mathrm{eff,E}}\,\mathrm{d}t. \tag{135}$$

In 2D $\mathbf{B}_{\mathrm{eff}} = B_{\mathrm{eff},z}\hat{z}$, so $\mathbf{t} = t_z\hat{z}$ and $\mathbf{s} = s_z\hat{z}$ are purely out-of-plane, and the cross products $\mathbf{p}^{-}\times\mathbf{t}$ and $\mathbf{p}'\times\mathbf{s}$ reduce to in-plane rotations by the 2D scalar $t_z = q\,B_{\mathrm{eff},z}\,\mathrm{d}t/(2\gamma^{-}m)$. The Boris rotation step is exactly symplectic. The $\mathcal{O}(v^2/c^2)$ gravitational corrections enter as additional contributions to $\mathbf{F}_{\mathrm{rel}}$ in Eq. (131).

15.3 Symplectic structure and relativistic limitations

The leapfrog scheme preserves the phase-space area element $\mathrm{d}\mathbf{x}\wedge\mathrm{d}\mathbf{p}$ exactly for linear (shear) force laws, preventing secular energy drift. At relativistic speeds the electric kick becomes a curved shear whose Jacobian depends on $\mathbf{p}$, and the Boris approximation is symplectic only to $\mathcal{O}(\mathrm{d}t^2)$. The symplectic error grows as $(\gamma v/c)^2(\mathrm{d}t/T)^2$ and can be mitigated by subcycling fast particles. Photon-like particles ($v\approx c$) should use null geodesic ray tracing in the c_{local} field rather than the Boris pusher.

15.4 Numerical dispersion and anisotropy

The exact numerical dispersion relation

The continuous wave equation has the dispersion relation $\omega^2 = c^2k^2$, meaning all wavelengths propagate at exactly c. The discrete leapfrog FDTD scheme on a uniform grid with spacing Δx and timestep Δt has a different, exact dispersion relation. Substituting a plane wave ansatz $\Phi^{n}_{j,k} = \hat{\Phi}\,e^{i(\omega n\Delta t - k_x j\Delta x - k_y k\Delta x)}$ into the leapfrog update gives:

$$\sin^2\left(\frac{\omega\Delta t}{2}\right) = c_{\mathrm{CFL}}^2\left[\sin^2\left(\frac{k_x\Delta x}{2}\right) + \sin^2\left(\frac{k_y\Delta x}{2}\right)\right], \tag{136}$$

where $c_{\mathrm{CFL}} = c\Delta t/\Delta x$ is the CFL number. This is not an approximation — it is the exact dispersion relation of the discrete scheme.

Dimensionless variables for the expansion

To expand Eq. (136) cleanly it is convenient to work entirely in dimensionless phase variables:

$$\xi = \frac{\omega\Delta t}{2}, \qquad \beta_x = \frac{k_x\Delta x}{2}, \qquad \beta_y = \frac{k_y\Delta x}{2}, \tag{137}$$

so that Eq. (136) becomes simply:

$$\sin^2\xi = c_{\mathrm{CFL}}^2(\sin^2\beta_x + \sin^2\beta_y). \tag{138}$$

All quantities are dimensionless. The continuous limit is $\xi = c_{\mathrm{CFL}}\sqrt{\beta_x^2+\beta_y^2}$, i.e. $\omega = ck$.

Taylor expansion

Using $\sin^2 u = u^2 - u^4/3 + \mathcal{O}(u^6)$:

$$\text{Left side:} \quad \sin^2\xi = \xi^2 - \frac{\xi^4}{3} + \dots \tag{139}$$

$$\text{Right side:} \quad c_{\mathrm{CFL}}^2\left[(\beta_x^2 + \beta_y^2) - \frac{\beta_x^4 + \beta_y^4}{3} + \dots\right] \tag{140}$$

Equating and collecting terms, with $\beta^2 \equiv \beta_x^2 + \beta_y^2$:

$$\xi^2\left(1 - \frac{\xi^2}{3}\right) = c_{\mathrm{CFL}}^2\beta^2\left(1 - \frac{\beta_x^4 + \beta_y^4}{3\beta^2}\right). \tag{141}$$

At leading order $\xi^2 = c_{\mathrm{CFL}}^2\beta^2$, confirming the continuous dispersion relation. Substituting this into the $\mathcal{O}(\beta^4)$ correction terms and solving for ξ^2:

$$\xi^2 \approx c_{\mathrm{CFL}}^2\beta^2\left[1 + \frac{\beta^2}{3}\left(c_{\mathrm{CFL}}^2 - \frac{\beta_x^4 + \beta_y^4}{\beta^4}\right)\right]. \tag{142}$$

All terms are dimensionless throughout. Converting back to physical variables via $\xi = \omega\Delta t/2$ and $\beta_\alpha = k_\alpha\Delta x/2$:

$$\omega^2 \approx c^2k^2\left[1 + \frac{k^2\Delta x^2}{12}\left(c_{\mathrm{CFL}}^2 - \frac{k_x^4 + k_y^4}{k^4}\right)\right]. \tag{143}$$

Every term now has units [rad^2/s^2], with the correction being a dimensionless factor.

Structure of the dispersion error

The correction factor in Eq. (143) has two contributions that partially oppose each other:

- $+c_{\mathrm{CFL}}^2\, k^2\Delta x^2/12$: from the **temporal** discretization. Isotropic, positive — makes ω too large, so the numerical phase velocity is slightly above c.
- $-(k_x^4 + k_y^4)/k^4 \cdot k^2\Delta x^2/12$: from the **spatial** discretization. Anisotropic, negative — makes ω too small, with a magnitude that depends on the propagation angle θ.

The numerical phase velocity along a general direction θ (where $k_x = k\cos\theta$, $k_y = k\sin\theta$) is:

$$\frac{v_\phi(\theta)}{c} \approx 1 + \frac{k^2\Delta x^2}{24}\left(c_{\mathrm{CFL}}^2 - \frac{\cos^4\theta + \sin^4\theta}{1}\right) + \mathcal{O}((k\Delta x)^4). \tag{144}$$

Using $\cos^4\theta + \sin^4\theta = 1 - \frac{1}{2}\sin^2(2\theta)$, this becomes:

$$\frac{v_\phi(\theta)}{c} \approx 1 + \frac{k^2\Delta x^2}{24}\left(c_{\mathrm{CFL}}^2 - 1 + \frac{\sin^2(2\theta)}{2}\right). \tag{145}$$

Anisotropy

The anisotropy — the variation of phase velocity with propagation angle — comes from the $\sin^2(2\theta)/2$ term in Eq. (145). This term is independent of c_{CFL}: the anisotropy is a fixed property of the square grid and the 5-point Laplacian stencil, not of the timestep. The maximum anisotropy occurs between the axis directions ($\theta = 0$, 90) and the diagonals ($\theta = 45$):

$$\frac{v_\phi(0)}{c} \approx 1 + \frac{k^2\Delta x^2}{24}(c_{\mathrm{CFL}}^2 - 1), \tag{146}$$

$$\frac{v_\phi(45)}{c} \approx 1 + \frac{k^2\Delta x^2}{24}\left(c_{\mathrm{CFL}}^2 - \frac{1}{2}\right). \tag{147}$$

The difference between the two:

$$\frac{v_\phi(45) - v_\phi(0)}{c} \approx \frac{k^2 \Delta x^2}{48} \tag{148}$$

is independent of c_{CFL}, confirming that the CFL number does not affect the anisotropy. A circular wavefront generated by a point source will appear as a slightly rounded square due to this anisotropy, with the diagonal directions propagating faster than the axis directions by a fractional amount $(k\Delta x)^2/48$.

Overall dispersion magnitude and the role of c_{CFL}

While the anisotropy is fixed, the overall magnitude of the dispersion error does depend on c_{CFL} through the $c_{\text{CFL}}^2 - 1$ term. Along the grid axes:

- At $c_{\text{CFL}} = 1$ (1D stability limit): $v_\phi(0) = c$ exactly — zero dispersion along the axis in 1D.
- At $c_{\text{CFL}} = 1/\sqrt{2}$ (2D stability limit): $v_\phi(0)/c \approx 1 - k^2\Delta x^2/48$ (slightly slow), $v_\phi(45)/c \approx 1$ (exact at the diagonal).
- At $c_{\text{CFL}} \ll 1$: $v_\phi(0)/c \approx 1 - k^2\Delta x^2/24$ (maximally slow along axes).

So running well below the stability limit does worsen the overall dispersion (larger negative error along the axes) while the anisotropy remains the same. Running at the stability limit minimizes the frequency-dependent part of the dispersion error by maximally cancelling the temporal and spatial contributions, though at no c_{CFL} value is the 2D dispersion simultaneously zero in all directions.

Practical implications for the sandbox

- **Run at the CFL limit.** $c_{\text{CFL}} = 1/\sqrt{2}$ in 2D minimizes the overall dispersion error while remaining stable. The conventional advice to run "safely below" the CFL limit worsens wave propagation accuracy without improving stability.
- **Ensure adequate spatial resolution.** The dispersion error scales as $(k\Delta x)^2$. A wavelength spanning N grid cells has $k\Delta x = 2\pi/N$, giving a dispersion error of order $\pi^2/(6N^2)$. For $N = 10$ cells per wavelength this is $\approx 1.6\%$; for $N = 20$ it is $\approx 0.4\%$. For the sandbox, 10–20 cells per wavelength is adequate for qualitative demonstrations.
- **Anisotropy is a fixed grid artifact.** A circularly-emitted wavefront will be slightly non-circular due to the $(k\Delta x)^2/48$ anisotropy, appearing as a rounded square with the diagonals slightly ahead of the axes. This can be reduced by replacing the standard 5-point Laplacian with the 9-point isotropic stencil:

  $$\nabla^2\Phi \approx \frac{1}{\Delta x^2}\left[\frac{1}{6}\sum_{\text{diag}}\Phi + \frac{2}{3}\sum_{\text{axis}}\Phi - \frac{10}{3}\Phi_{i,j}\right], \tag{149}$$

  which replaces $k_x^4 + k_y^4$ with k^4 in the spatial error term, eliminating the angular dependence and reducing the anisotropy coefficient by a factor of ~ 4 at the cost of involving 8 neighbours instead of 4.
- **Higher-order stencils.** Using a 4th-order accurate spatial stencil replaces the $\mathcal{O}((k\Delta x)^2)$ error with $\mathcal{O}((k\Delta x)^4)$, dramatically reducing dispersion for well-resolved wavelengths at the cost of a wider stencil (5 points per dimension rather than 3).

15.5 Source deposition: approximating the delta function

The source-deposition strategy follows standard particle-in-cell (PIC) practice [17, 18]. The physical sources involve $\delta(\mathbf{x}-\mathbf{x}_i)$ and for spin sources $\nabla\times[\boldsymbol{\mu}\,\delta(\mathbf{x}-\mathbf{x}_i)]$, neither of which can exist on a discrete grid. They must be replaced by smooth kernel functions $W(\mathbf{x}-\mathbf{x}_i)$ that approximate the delta function on the grid scale. Three competing requirements must be balanced:

1. **Sub-cell position accuracy:** the kernel must encode the particle's position within a cell, not just which cell it occupies.
2. **Smoothness for curl deposition:** taking $\nabla\times[W\,\boldsymbol{\mu}]$ requires W to have well-defined spatial derivatives.
3. **Compactness:** a wide kernel smears the source over many cells, reducing spatial resolution.

The B-spline hierarchy

There is a well-established family of kernels that interpolates between a box function (maximally compact, discontinuous) and a Gaussian (infinitely smooth, infinite support). These are the B-spline kernels W_n of order n, each being the convolution of its predecessor with a box function:

Order	Common name	Smoothness	Support (cells)	Curl kernel
$n=1$	Nearest-grid-point (NGP)	Discontinuous	1	Undefined
$n=2$	Tent / CIC	C^0	2	Box (discontinuous)
$n=3$	Quadratic B-spline	C^1	3	Tent (C^0)
$n=4$	Cubic B-spline	C^2	4	Quadratic (C^1)
$n\to\infty$	Gaussian	C^∞	∞	Hermite–Gaussian

Monopole and current deposition: the CIC kernel and bilinear interpolation

For ρ_{matter}, $\mathbf{J}_{\text{matter}}$, ρ_q, $\mathbf{J}_q$ the cloud-in-cell (CIC) tent kernel W_2 is the standard choice. In 1D it is:

$$W_2(x)=\begin{cases}1-|x|/\Delta x & |x|<\Delta x\\ 0 & \text{otherwise,}\end{cases} \tag{150}$$

and in 2D it is the tensor product:

$$W_2(x,y)=W_2(x)\,W_2(y)=\left(1-\frac{|x|}{\Delta x}\right)\left(1-\frac{|y|}{\Delta y}\right),\qquad |x|<\Delta x,\ |y|<\Delta y. \tag{151}$$

Deposition of charge q at sub-cell position (α,β) with $\alpha=(x_p-x_i)/\Delta x$, $\beta=(y_p-y_j)/\Delta y$ distributes to the four surrounding grid points:

$$\begin{aligned}\rho_{i,j} &\mathrel{+}= q\,(1-\alpha)(1-\beta)/\Delta x^2,\\ \rho_{i+1,j} &\mathrel{+}= q\,\alpha(1-\beta)/\Delta x^2,\\ \rho_{i,j+1} &\mathrel{+}= q\,(1-\alpha)\beta/\Delta x^2,\\ \rho_{i+1,j+1} &\mathrel{+}= q\,\alpha\beta/\Delta x^2.\end{aligned} \tag{152}$$

The weights $(1-\alpha)(1-\beta)$, $\alpha(1-\beta)$, $(1-\alpha)\beta$, $\alpha\beta$ are exactly the bilinear interpolation weights. Field interpolation back to the particle position uses the same four weights:

$$E_p=(1-\alpha)(1-\beta)\,E_{i,j}+\alpha(1-\beta)\,E_{i+1,j}+(1-\alpha)\beta\,E_{i,j+1}+\alpha\beta\,E_{i+1,j+1}. \tag{153}$$

The deposition weights in Eq. (152) and the interpolation weights in Eq. (153) are identical. This is the adjoint condition, discussed in detail below.

Dipole deposition: the quadratic B-spline and its derivative as two tents

The quadratic B-spline W_3 is obtained by convolving W_2 with a box function of width Δx. In 1D:

$$W_3(x) = \begin{cases} \frac{3}{4} - (x/\Delta x)^2 & |x| < \frac{1}{2}\Delta x \\ \frac{1}{2}\left(\frac{3}{2} - |x|/\Delta x\right)^2 & \frac{1}{2}\Delta x \le |x| < \frac{3}{2}\Delta x \\ 0 & \text{otherwise.} \end{cases} \tag{154}$$

W_3 is C^1, spans 3 cells, and has an inflection point at $|x| = \frac{1}{2}\Delta x$—concave down in the central region and concave up in the two outer regions. Its derivative is:

$$\frac{\mathrm{d}W_3}{\mathrm{d}x}(x) = \begin{cases} +\left(\frac{3}{2} - |x|/\Delta x\right)/\Delta x & -\frac{3}{2}\Delta x \le x < -\frac{1}{2}\Delta x \\ -2x/\Delta x^2 & |x| < \frac{1}{2}\Delta x \\ -\left(\frac{3}{2} - |x|/\Delta x\right)/\Delta x & \frac{1}{2}\Delta x \le x < \frac{3}{2}\Delta x \\ 0 & \text{otherwise.} \end{cases} \tag{155}$$

The key geometric insight is that $\mathrm{d}W_3/\mathrm{d}x$ consists of **two tent functions placed side by side**—one positive and one negative—straddling the particle position at $x = 0$:

- A **positive tent** centered at $x = -\frac{1}{2}\Delta x$, rising from 0 at $x = -\frac{3}{2}\Delta x$, peaking at $+1/\Delta x$ at $x = -\frac{1}{2}\Delta x$, and falling back to 0 at $x = 0$.
- A **negative tent** centered at $x = +\frac{1}{2}\Delta x$, falling from 0 at $x = 0$, reaching $-1/\Delta x$ at $x = +\frac{1}{2}\Delta x$, and returning to 0 at $x = +\frac{3}{2}\Delta x$.

The two tents meet at zero at the particle position. This antisymmetric structure is a direct consequence of W_3 being symmetric: the derivative of a symmetric function is antisymmetric and must pass through zero at the center, positive on the rising (left) side and negative on the falling (right) side.

When deposited onto the grid, the three cells receive:

- **Cell** $i - 1$ (left): only the positive tail of the positive tent.
- **Cell** i (containing particle): the right half of the positive tent (positive) plus the left half of the negative tent (negative), partially cancelling with net value depending on sub-cell position α.
- **Cell** $i + 1$ (right): only the negative tail of the negative tent.

The total weight across all three cells sums to zero, since $W_3(\pm\frac{3}{2}\Delta x) = 0$.

For comparison, the derivative of the CIC kernel W_2 is a discontinuous box function:

$$\frac{\mathrm{d}W_2}{\mathrm{d}x}(x) = \begin{cases} -\mathrm{sgn}(x)/\Delta x & |x| < \Delta x \\ 0 & \text{otherwise,} \end{cases} \tag{156}$$

which produces Gibbs-like ringing when used as a grid source. The W_3 two-tent derivative avoids this discontinuity.

For reference, the Gaussian $W_G = (2\pi\sigma^2)^{-1/2}\exp(-x^2/2\sigma^2)$ gives a Hermite–Gaussian derivative:

$$\frac{\mathrm{d}W_G}{\mathrm{d}x} = -\frac{x}{\sigma^2}\,W_G(x), \tag{157}$$

which is C^∞ but requires truncation at $\pm 3\sigma$ (≈ 6 cells).

The 2D curl: sign flip in the direction of differentiation

In 2D, the spin dipole moment is a scalar μ (the out-of-plane component), and the magnetization current is the 2D curl of $\mu\, W_3(x-x_p)\, W_3(y-y_p)$:

$$J_{q,x}(\mathbf{x}) = \mu\, W_3(x-x_p) \cdot \frac{\mathrm{d}W_3}{\mathrm{d}y}(y-y_p), \tag{158}$$

$$J_{q,y}(\mathbf{x}) = -\mu\, \frac{\mathrm{d}W_3}{\mathrm{d}x}(x-x_p) \cdot W_3(y-y_p). \tag{159}$$

Each component is a tensor product of one smooth W_3 factor (all positive, C^1) and one two-tent $\mathrm{d}W_3/\mathrm{d}x_\alpha$ factor (C^0, sign flip). The sign flip occurs only in the direction of differentiation:

- $J_{q,x}$: smooth in x, two-tent in y. Positive above the particle ($y > y_p$), negative below.
- $J_{q,y}$: two-tent in x with overall minus sign, smooth in y. Negative to the left of the particle, positive to the right.

Together these form a counter-clockwise current loop in the xy-plane—the correct physical current distribution for a magnetic dipole $\mu > 0$ pointing out of the plane. Each component spans $3 \times 3 = 9$ cells in 2D.

Generalization to 3D

All formulas above are written in 2D for clarity. The 3D generalization adds the third tensor product factor straightforwardly.

CIC (monopole/current): add factor $W_2(z-z_p)$ with $\gamma = (z_p - z_k)/\Delta z$. The 8 surrounding cells receive weights that are all products of factors from $\{(1-\alpha), \alpha\} \times \{(1-\beta), \beta\} \times \{(1-\gamma), \gamma\}$.

Quadratic B-spline (dipole): the 3D kernel is $W_3(x-x_p)\, W_3(y-y_p)\, W_3(z-z_p)$, spanning $3^3 = 27$ cells.

3D curl for spin sources: with $\boldsymbol{\mu}$ now a 3-vector, the three components of $\nabla \times [\boldsymbol{\mu}\, W_3]$ are:

$$[\nabla \times (\boldsymbol{\mu}\, W_3)]_x = \mu_z\, W_3(x)\, \tfrac{\mathrm{d}W_3}{\mathrm{d}y}(y)\, W_3(z) - \mu_y\, W_3(x)\, W_3(y)\, \tfrac{\mathrm{d}W_3}{\mathrm{d}z}(z), \tag{160}$$

$$[\nabla \times (\boldsymbol{\mu}\, W_3)]_y = \mu_x\, W_3(x)\, W_3(y)\, \tfrac{\mathrm{d}W_3}{\mathrm{d}z}(z) - \mu_z\, \tfrac{\mathrm{d}W_3}{\mathrm{d}x}(x)\, W_3(y)\, W_3(z), \tag{161}$$

$$[\nabla \times (\boldsymbol{\mu}\, W_3)]_z = \mu_y\, \tfrac{\mathrm{d}W_3}{\mathrm{d}x}(x)\, W_3(y)\, W_3(z) - \mu_x\, W_3(x)\, \tfrac{\mathrm{d}W_3}{\mathrm{d}y}(y)\, W_3(z), \tag{162}$$

where arguments $(x - x_p)$ etc. are suppressed. Each term has exactly one two-tent $\mathrm{d}W_3/\mathrm{d}x_\alpha$ factor in the direction of differentiation and smooth W_3 factors in the other two directions, spanning 27 cells and summing to zero over the grid. In 2D only μ_z is nonzero and only the x and y components of the curl exist (Eqs. 158–159).

The adjoint condition and self-force cancellation

The deposition operator D maps a particle quantity at position $\mathbf{x}_p$ to the grid:

$$\rho(\mathbf{x}_j) = [D\, q](\mathbf{x}_j) = q\, W(\mathbf{x}_j - \mathbf{x}_p), \tag{163}$$

and the interpolation operator I maps a grid field back to the particle:

$$E_p = [I\, E](\mathbf{x}_p) = \sum_j E(\mathbf{x}_j)\, \tilde{W}(\mathbf{x}_j - \mathbf{x}_p). \tag{164}$$

The **adjoint condition** [17] requires $\tilde{W} = W$—the same kernel is used for both operations.

To see why this cancels the static self-force, write the force of the particle's own field on itself:

$$\mathbf{F}_{\text{self}} = q \sum_j \mathbf{E}_{\text{self}}(\mathbf{x}_j)\, W(\mathbf{x}_j - \mathbf{x}_p). \tag{165}$$

For a stationary particle, $\mathbf{E}_{\text{self}}$ is produced by the symmetric charge distribution $\rho(\mathbf{x}_j) = q\, W(\mathbf{x}_j - \mathbf{x}_p)$. On a uniform grid with a symmetric kernel W, the field $\mathbf{E}_{\text{self}}$ is antisymmetric about the particle position—it points away from the particle equally in all directions. Interpolating this antisymmetric field back with the same symmetric kernel W gives exactly zero:

$$\mathbf{F}_{\text{self}} = q \sum_j \mathbf{E}_{\text{self}}(\mathbf{x}_j)\, W(\mathbf{x}_j - \mathbf{x}_p) = 0 \qquad \text{(stationary particle)}. \tag{166}$$

If different kernels are used for deposition and interpolation, this symmetry is broken and the self-force is nonzero even for a stationary particle, producing unphysical heating.

For the CIC/bilinear case, the adjoint condition is immediately visible: the deposition weights $(1-\alpha)(1-\beta)$, $\alpha(1-\beta)$, $(1-\alpha)\beta$, $\alpha\beta$ in Eq. (152) are identical to the interpolation weights in Eq. (153). Bilinear interpolation *is* CIC deposition—they are the same operation, so the adjoint condition is automatically satisfied.

For the gradient fields needed by the spin forces, the adjoint condition has a critical implication that must be stated precisely. The dipole source was deposited using the kernel $\mathrm{d}W_3/\mathrm{d}x_\alpha \cdot W_3$ over 9 cells in 2D (27 in 3D). The adjoint of that deposition operator is the interpolation that uses those *exact same weights* to map grid field values back to the particle. Concretely, the x-gradient of $B_{g,z}$ at the particle is:

$$(\partial_x B_{g,z})_p = \sum_{j,k} B_{g,z}(\mathbf{x}_{j,k}) \cdot \frac{\mathrm{d}W_3}{\mathrm{d}x}(x_j - x_p) \cdot W_3(y_k - y_p), \tag{167}$$

$$(\partial_y B_{g,z})_p = \sum_{j,k} B_{g,z}(\mathbf{x}_{j,k}) \cdot W_3(x_j - x_p) \cdot \frac{\mathrm{d}W_3}{\mathrm{d}y}(y_k - y_p), \tag{168}$$

where the sums run over the same $3 \times 3 = 9$ cells used during dipole deposition, and $\mathrm{d}W_3/\mathrm{d}x$ is the two-tent function of Eq. (155). The identical formulas apply to ∇B_z for the magnetic gradient force, and generalize to $3 \times 3 \times 3 = 27$ cells in 3D.

It is important to note what is *not* correct here. The box-function derivative $-\mathrm{sgn}(x)/\Delta x$ is the derivative of the CIC kernel W_2, not of W_3. It would be the correct adjoint gradient interpolation only if the dipole had been deposited with W_2—but W_2 was deliberately avoided for dipole deposition because its discontinuous derivative causes Gibbs ringing. Using $\mathrm{sgn}(x)/\Delta x$ for gradient interpolation while using W_3 for dipole deposition breaks the adjoint pairing and introduces a residual dipole self-force: a spinning stationary particle would experience a spurious gradient force from its own field. The two-tent weights of Eqs. (167)–(168) are the unique correct choice.

Similarly, computing the gradient by finite-differencing the bilinearly interpolated field (another common approach) is not adjoint to W_3 dipole deposition and also introduces a self-force. The rule is simple: **use the same kernel weights for interpolation that were used for deposition, applied to the same set of cells**.

Self-interaction, radiation reaction, and their limitations

The self-force cancellation described above is exact only for a *stationary* particle. For an *accelerating* particle the retarded self-field is asymmetric—the field the particle emitted in the past arrives back at the particle's current position from a different direction—and the cancellation is incomplete. The residual is the **radiation reaction force**, which in classical electrodynamics is the Abraham–Lorentz force:

$$\mathbf{F}_{\text{rad}} = \frac{q^2}{6\pi\varepsilon_0 c^3}\,\dddot{\mathbf{x}}, \tag{169}$$

involving the third time derivative of position (the jerk). The gravitational analogue for GEM is structurally identical with the appropriate coupling constants.

In a grid-based simulation this radiation reaction force is captured only approximately. The particle deposits its field onto the grid via the CIC kernel—effectively giving the particle a finite size of order Δx—and the retarded self-field propagates back to the particle's position via the wave equation. The resulting self-force approximates but does not exactly reproduce the Abraham–Lorentz force, for two reasons:

- The effective particle size $\sim \Delta x$ regularizes the $1/r$ self-field divergence at a scale chosen for numerical convenience rather than physical accuracy, making the radiation reaction force grid-spacing-dependent.
- The exact Abraham–Lorentz force involves the $r \to 0$ limit of the retarded self-field, which the grid cannot resolve below Δx.

The practical consequence is that a radiating particle in the simulation will correctly emit energy into the field (the outgoing wave is accurately computed), but the back-reaction force on the particle that causes it to slow down may be slightly inconsistent with the emitted power, leading to small violations of energy conservation at the particle level over long integration times.

For qualitative demonstrations—watching waves propagate outward from an oscillating mass, observing the radiation pattern, verifying the $1/\sqrt{r}$ field fall-off in 2D—the simulation is fully adequate. For quantitative long-time tracking of radiation-induced orbital decay, a hybrid approach is recommended: add an explicit phenomenological radiation damping term computed analytically from the particle's current acceleration, rather than relying entirely on the grid self-interaction. This maintains energy conservation to much higher accuracy without the numerical stiffness of the exact Abraham–Lorentz–Dirac equation.

Practical recommendation

Source / Force	**Kernel**	**Cells (2D / 3D)**
ρ, $\mathbf{J}$ monopole/current deposition	CIC W_2	4 / 8
Potential interpolation at particle (ϕ, Φ_g)	Bilinear W_2 at corners	4 / 8
Potential interpolation at particle ($\mathbf{A}$, $\mathbf{A}_g$)	Bilinear W_2 at edges	4 / 8
On-the-fly $B_{g,z}$, B_z at particle	Yee curl + bilinear W_2	4 cell centers
Spin dipole $\nabla \times [\boldsymbol{\mu}\,\delta]$ deposition	W_3 curl	9 / 27
Gradient fields $\nabla B_{g,z}$, ∇B_z at particle	Two-tent $\mathrm{d}W_3/\mathrm{d}x_\alpha$	9 / 27

15.6 Gauge condition enforcement

The Lorenz gauge and why it drifts

The wave equations for the GEM and EM potentials were derived in the Lorenz gauge, which for GEM requires:

$$\mathcal{G}_{\text{GEM}} \equiv \frac{1}{c^2}\frac{\partial \Phi_g}{\partial t} + \nabla \cdot \mathbf{A}_g = 0, \tag{170}$$

and for EM:

$$\mathcal{G}_{\text{EM}} \equiv \frac{1}{c^2}\frac{\partial \phi}{\partial t} + \nabla \cdot \mathbf{A} = 0. \tag{171}$$

In the continuous theory, if these conditions hold at $t = 0$ and the sources satisfy the continuity equations $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$, then the gauge conditions are preserved exactly by the wave equation evolution — the gauge violation satisfies the homogeneous wave equation $\Box \mathcal{G} = 0$ and simply propagates outward and exits through the absorbing boundary. There is no exponential amplification mechanism.

In the discrete FDTD scheme the gauge condition drifts for two reasons. First, the finite-difference operators do not satisfy the same algebraic identities as the continuous operators. Second, the discretely deposited sources satisfy the continuity equation only approximately. The drift is sourced by truncation errors of order $(k\Delta x)^2$ per step — small, bounded, and not self-amplifying. It accumulates at a slow linear rate, not exponentially.

Why gauge drift does not affect particle dynamics

A crucial observation is that the force expressions Eqs. (122)–(123) are *gauge-invariant*. Under a gauge transformation $\phi \to \phi - \partial_t \Lambda$, $\mathbf{A} \to \mathbf{A} + \nabla \Lambda$, the extra terms introduced into the force cancel exactly — the physical force is independent of the gauge function Λ. Therefore gauge drift in the stored potentials has *no effect* on the computed forces or particle trajectories.

Gauge drift matters only when the potentials themselves are used directly: for visualization of ϕ, Φ_g, $\mathbf{A}_g$, $\mathbf{A}$; for the gauge condition monitoring; and for the EM effective GEM source $\rho_{\text{EM}} = u_{\text{EM}}/c^2$, which uses $\mathbf{E}$ and $\mathbf{B}$ derived from the potentials. For a purely qualitative pedagogical simulation gauge correction can be omitted entirely without affecting the particle dynamics.

The discrete continuity equation: the primary defense

The most important gauge-preserving measure is to ensure that the deposited sources satisfy the *discrete* continuity equation exactly. On the Yee grid with ρ at corners and $\mathbf{J}$ at edges, the discrete continuity equation is:

$$\frac{\rho^{n+1}_{i,j} - \rho^{n-1}_{i,j}}{2\Delta t} + \frac{J^n_{x,i+1/2,j} - J^n_{x,i-1/2,j}}{\Delta x} + \frac{J^n_{y,i,j+1/2} - J^n_{y,i,j-1/2}}{\Delta y} = 0. \tag{172}$$

Using a consistent CIC kernel for both ρ and $\mathbf{J}$ deposition satisfies this condition to the accuracy of the kernel, bounding gauge drift at the truncation error level automatically.

Parabolic cleaning: optional refinement for long integrations

For very long simulations where the slow linear gauge drift eventually becomes visible in potential visualizations, a parabolic cleaning step can be applied after each field update:

$$\Phi_g \;\leftarrow\; \Phi_g - R\,c^2\,\Delta t\,(\nabla \cdot \mathbf{A}_g), \qquad \phi \;\leftarrow\; \phi - R\,c^2\,\Delta t\,(\nabla \cdot \mathbf{A}), \tag{173}$$

where R is a small positive dimensionless constant. This drives the scalar potentials toward the Lorenz gauge condition, damping the gauge violation on a timescale $\sim 1/(Rc)$. A practical choice is $R \sim c_{\mathrm{CFL}}^2/4$, giving a damping timescale of a few domain-crossing times.

The cleaning is a single extra multiplication per cell and is applied only to the perturbation fields. Since gauge drift does not affect particle dynamics, this correction is purely cosmetic — it keeps the potential visualizations free of slowly drifting background artifacts. It is not needed for physical correctness of the simulation.

Hyperbolic cleaning: stronger alternative

If parabolic cleaning is insufficient (e.g. for very long integrations or highly dynamic configurations), hyperbolic divergence cleaning can be used. An auxiliary field ψ_g (for GEM; similarly ψ for EM) is introduced:

$$\frac{\partial \mathbf{A}_g}{\partial t} \leftarrow \frac{\partial \mathbf{A}_g}{\partial t} - \nabla\psi_g, \tag{174}$$

$$\frac{\partial \psi_g}{\partial t} = -c_h^2 \nabla \cdot \mathbf{A}_g - \kappa\psi_g, \tag{175}$$

making the gauge violation satisfy a damped wave equation that propagates violations outward at speed $c_h = c$ and damps them at rate $\kappa = c/L$. This is more robust than parabolic cleaning but requires an additional scalar field per wave equation.

Monitoring

The discrete gauge residual:

$$\mathcal{G}_{i,j}^n = \frac{(\Phi_g^{\mathrm{pert}})_{i,j}^{n+1} - (\Phi_g^{\mathrm{pert}})_{i,j}^{n-1}}{2c^2\Delta t} + \frac{A_{g,x,i+1/2,j}^n - A_{g,x,i-1/2,j}^n}{\Delta x} + \frac{A_{g,y,i,j+1/2}^n - A_{g,y,i,j-1/2}^n}{\Delta y} \tag{176}$$

should remain at the level of the truncation error $\sim (k\Delta x)^2$. Rapid growth indicates inconsistent source deposition. Since gauge drift does not affect dynamics, monitoring is primarily useful for diagnosing deposition bugs rather than correcting physical errors.

15.7 Absorbing boundary conditions

Without special treatment at the grid edges the leapfrog FDTD scheme imposes reflecting boundary conditions by default — the field values outside the grid are implicitly zero, which acts as a perfect reflector for any outgoing wave. Reflected waves re-enter the simulation domain and interfere with the physics being demonstrated. For a GR sandbox this is particularly problematic: a gravitational wave emitted by an oscillating mass would bounce off the grid boundary, return to the source region, and create standing wave patterns that have no physical meaning and would confuse any student watching.

The damping layer

The recommended approach for the sandbox is a **damping layer**: a border region of N_d cells on each side of the grid where the field is multiplied by a spatially varying attenuation factor at each timestep. This is a simplification of the perfectly matched layer (PML) [16, 15]; the quadratic profile used here is a degenerate matched-impedance case suitable for the qualitative pedagogical purpose of the sandbox. Outgoing waves entering the layer are attenuated gradually and the small residual that reaches the hard boundary is attenuated again on the return trip, making double-pass reflections negligible.

The damping factor at position x within the layer (measured from the inner edge of the layer, $0 \leq x \leq L = N_d \Delta x$) is:

$$f(x) = \exp(-\sigma(x)\,\mathrm{d}t) \approx 1 - \sigma(x)\,\mathrm{d}t, \tag{177}$$

where $\sigma(x)$ is the conductivity profile. A smooth polynomial ramp is recommended:

$$\sigma(x) = \sigma_{\max}\left(\frac{x}{L}\right)^2, \tag{178}$$

which rises from zero at the interior edge to $\sigma_{\max}$ at the outer edge. The quadratic profile ensures the attenuation starts gently, minimizing the impedance mismatch at the interior-damping interface that would otherwise cause partial reflection.

The round-trip reflection coefficient for a wave traversing the layer twice is:

$$R \approx \exp\left(-\frac{2}{c}\int_0^L \sigma(x)\,\mathrm{d}x\right) = \exp\left(-\frac{2\sigma_{\max}L}{3c}\right). \tag{179}$$

Setting $2\sigma_{\max}L/3c = 7$ gives $R \approx e^{-7} \approx 10^{-3}$, which is adequate for all qualitative demonstrations. For a layer of $N_d = 16$ cells this requires:

$$\sigma_{\max} = \frac{21c}{2L} = \frac{21c}{2N_d\Delta x}. \tag{180}$$

Implementation

The damping factor array is precomputed once at initialization. For a grid of size $N \times N$ with damping layer of thickness N_d cells on each side, define:

$$d_{i,j} = \max(d_x(i),\ d_y(j))\,, \tag{181}$$

where:

$$d_x(i) = \begin{cases} \sigma_{\max}\left(\dfrac{N_d - i}{N_d}\right)^2 \mathrm{d}t & i < N_d \quad \text{(left layer)} \\ \sigma_{\max}\left(\dfrac{i - (N - N_d)}{N_d}\right)^2 \mathrm{d}t & i > N - N_d \quad \text{(right layer)} \\ 0 & \text{otherwise,} \end{cases} \tag{182}$$

and similarly for $d_y(j)$. The per-timestep application after each field update is then:

$$\Phi_{i,j}^{n+1} \;\leftarrow\; \Phi_{i,j}^{n+1}\cdot(1 - d_{i,j}) \tag{183}$$

applied to all six perturbation field arrays (Φ_g^{pert}, $A_{g,x}^{\text{pert}}$, $A_{g,y}^{\text{pert}}$, ϕ^{pert}, A_x^{pert}, A_y^{pert}). The same damping array $d_{i,j}$ applies to all six because they all satisfy the same wave equation with the same propagation speed.

What is and is not damped

Only the perturbation fields are damped. The background fields Φ_g^{bg}, $A_{g,\alpha}^{\text{bg}}$, ϕ^{bg}, A_α^{bg} are evaluated analytically at particle positions and are never stored on the grid, so they are automatically exempt from the damping operation. This is a further advantage of the background + perturbation split: the static $1/r$ gravitational field, which is not a propagating wave and should not be attenuated, requires no special treatment at the boundary.

Particles should be confined to the interior region and prevented from entering the damping layer. A simple reflecting or absorbing boundary at the inner edge of the damping layer (at $i = N_d$ and $i = N - N_d$, and similarly in y) is sufficient. A particle that reaches the inner boundary can be reflected elastically, absorbed and removed from the simulation, or wrapped with a warning — the choice depends on the scenario being demonstrated.

Practical parameters

Parameter	Recommended value
Layer thickness N_d	16 cells
Profile exponent	2 (quadratic)
$\sigma_{\max}$	$21c/(2N_d\Delta x)$
Grid overhead	$(N+2N_d)^2/N^2 \approx 1.27$ for $N = 256$, $N_d = 16$
Residual reflection R	$\approx 10^{-3}$

For scenarios where lower reflection is needed — for example, measuring radiated power or verifying the $1/\sqrt{r}$ amplitude fall-off — the layer thickness can be increased to $N_d = 32$ cells and $\sigma_{\max}$ adjusted accordingly, reducing R to $\approx 10^{-6}$.

Realized reflection in practice (added in v33)

The $R \approx 10^{-3}$ figure above is the *plane-wave normal-incidence* limit of the continuum derivation. Two effects degrade the realized in-interior residual for a 2D circular wavefront launched from a centered Gaussian initial condition:

- **Linearization error**. The continuum derivation uses $f(x) = \exp(-\sigma(x)\,\mathrm{d}t)$, while the implementation block above prescribes the linearized form $(1 - d_{i,j})$. For the default parameters ($N_d = 16$, $\sigma_{\max} = 21c/(2N_d\Delta x)$, $c_{\text{CFL}} = 1/\sqrt{2}$), $\sigma_{\max}\,\mathrm{d}t \approx 0.46$, where the two forms differ by $\sim 15\%$. The linearization *over-attenuates* per step, but distorts the discrete dispersion of the absorbing region and introduces a small impedance-mismatch reflection at the inner edge of the layer.
- **Oblique-incidence corner trapping**. The axis-aligned profile $d_{i,j} = \max(d_x, d_y)$ damps perpendicular incidence strongly but is weak for waves heading into the grid *corners* at $\approx 45°$ from both walls. Such waves arrive at the corner having spent comparable depth along both axes; the max operator does not accumulate their depths. A perfectly matched layer (PML) addresses this by introducing complex coordinate stretching that absorbs in each direction independently.

Empirically (measured in Stage 2 of the implementation plan, §18): with $N_d = 16$ and a $\sigma = 4\Delta x$ Gaussian pulse at the center of a 256×256 grid, after 800 timesteps the field at the cell just inside the wall is suppressed by $\sim 3000\times$ versus the no-damping case, while at the domain center it is suppressed by $\sim 20\times$ — the difference being the corner-trapped energy slowly refocusing through the interior over multiple round-trips. The damping layer is therefore fully adequate for qualitative pedagogical demonstrations where outgoing waves visibly exit the domain, but does *not* achieve 10^{-3} residual at the source for a 2D circular wavefront without moving to PML.

15.8 Units and non-dimensionalization

The physical values of G and c make gravitational effects completely invisible at any computationally tractable scale. For example, the Schwarzschild radius of a solar mass is $\sim 3\,\text{km}$, which would require astronomical grid extents to simulate at physical scales. The simulation therefore uses a rescaled set of units in which all quantities are of order unity.

Choose three base scales: a length scale ℓ_0 (the grid cell size in physical units, or equivalently the domain size), a time scale $t_0 = \ell_0/c_{\text{eff}}$, and a mass scale m_0. All simulation quantities are then expressed in terms of these:

Quantity	Physical	Simulation	Relation
Length	x	$\tilde{x} = x/\ell_0$	$\tilde{x} \in [0, N]$
Time	t	$\tilde{t} = t/t_0$	$\mathrm{d}\tilde{t} = 1$ per step
Speed of light	c	$\tilde{c} = c_{\text{eff}}$	chosen so $\tilde{c}\,\mathrm{d}\tilde{t} = c_{\text{CFL}}\,\mathrm{d}\tilde{x}$
Mass	m	$\tilde{m} = m/m_0$	typically $\tilde{m} \sim 1$
Charge	q	$\tilde{q} = q/q_0$	q_0 chosen freely
Momentum	p	$\tilde{p} = p/(m_0 c_{\text{eff}})$	
Grav. potential	Φ_g	$\tilde{\Phi}_g = \Phi_g/c_{\text{eff}}^2$	dimensionless
Grav. constant	G	$\tilde{G} = G_{\text{eff}}$	chosen for visibility

The key free parameter is G_{eff} — the effective gravitational constant in simulation units. It should be chosen so that a typical particle-particle gravitational interaction produces a visible deflection over a few timesteps. A useful guideline: for two particles of mass $\tilde{m}$ separated by distance $\tilde{r}$, the gravitational acceleration is $\tilde{G}\tilde{m}/\tilde{r}^2$ in simulation units. Setting this to be a fraction of $\tilde{c}^2/\tilde{r}$ (the speed squared over the separation, a natural relativistic scale) gives:

$$G_{\text{eff}} \sim \frac{c_{\text{eff}}^2\,\tilde{r}}{\tilde{m}} \cdot \epsilon, \tag{184}$$

where $\epsilon \ll 1$ keeps the system in the weak-field linearized regime. Typical values in practice are $G_{\text{eff}} \sim 10^{-3}$ to 10^{-1} depending on the desired strength of gravitational effects.

The charge-to-mass ratio q/m for test particles can be set independently of any physical value, allowing electromagnetic and gravitational effects to be tuned relative to each other. Setting $q/m \gg \sqrt{G_{\text{eff}}}$ makes EM effects dominate; setting $q/m \ll \sqrt{G_{\text{eff}}}$ makes gravitational effects dominate. The coupling of EM field energy to the GEM equations through $\rho_{\text{EM}} = u_{\text{EM}}/c^2$ is automatically consistent with whatever G_{eff} and c_{eff} are chosen, since it follows directly from the field equations.

15.9 Field initialization

At $t = 0$ the perturbation fields must be initialized to the static solution corresponding to the initial particle configuration. Starting with zero fields and nonzero particle sources creates a spurious transient wavefront that propagates outward at c — a pulse with no physical meaning that takes several domain crossing times to damp out in the absorbing boundary layer.

Convergence iteration

The cleanest initialization approach is to iterate the FDTD field update with all **positions** held fixed at their initial values and all **velocities** held fixed at their initial values, until the fields converge. The particles are not moved during this phase — only the field arrays are updated each step. With fixed positions and velocities:

- Charge/mass density sources ρ_{matter}, ρ_q are time-independent (positions fixed).
- Current sources $\mathbf{J}_{\text{matter}} = \gamma m\mathbf{v}$ and $\mathbf{J}_q = q\mathbf{v}$ are also time-independent (velocities fixed).
- All six source terms are constant, so the wave equations iterate toward their static solutions — the Poisson equation solutions for Φ_g and ϕ, and the vector potential solutions for $\mathbf{A}_g$ and $\mathbf{A}$ sourced by the constant currents.

This is important: velocities must *not* be set to zero during initialization unless the particles are actually stationary. The vector potentials $\mathbf{A}_g$ and $\mathbf{A}$ are sourced by $\mathbf{J}_{\text{matter}}$ and $\mathbf{J}_q$ respectively, which depend on velocity. A particle with nonzero initial velocity generates gravitomagnetic and magnetic fields that must be present in the initial field configuration. Setting velocities to zero during initialization would converge to the wrong field — one with no vector potentials — which is only correct for truly stationary particles.

For the special case of all particles stationary ($\mathbf{v}_i = 0$), the currents vanish and $\mathbf{A}_g \to 0$, $\mathbf{A} \to 0$ as a consequence rather than as an assumption.

The transient oscillations during convergence are absorbed by the damping layer. The convergence timescale is a few domain-crossing times $\sim N\sqrt{2}$ timesteps (at $c_{\mathrm{CFL}} = 1/\sqrt{2}$), roughly 1000–3000 steps for $N = 256$. This can be done invisibly before the simulation begins displaying to the user.

Leapfrog initialization

The leapfrog requires field values at two consecutive time levels Φ^{-1} and Φ^0 to start. With zero initial time derivative ($\partial_t \Phi = 0$ for a static initial configuration), the correct initialization is:

$$\Phi^{-1} = \Phi^0 = 0, \tag{185}$$

which makes the free-propagation extrapolation $2\Phi^0 - \Phi^{-1} = 0$ correct. As the convergence iteration proceeds both levels are updated normally.

For the particle momenta, what is needed is $\mathbf{p}_i^{-1/2}$ — the momentum at a half timestep before $t = 0$. The standard approach is a single half-step using the force from the converged initial fields:

$$\mathbf{p}_i^{-1/2} = \mathbf{p}_i^0 - \mathbf{F}_i^0 \frac{\mathrm{d}t}{2}, \tag{186}$$

where $\mathbf{F}_i^0$ is the total force on particle i from the converged initial fields at $t = 0$, and $\mathbf{p}_i^0$ is the specified initial momentum. For a stationary initial configuration $\mathbf{p}_i^0 = 0$ and the half-step momentum is $-\mathbf{F}_i^0\,\mathrm{d}t/2$ — a small quantity representing the velocity the particle would have acquired over a half timestep from the static initial field. This initialization is first-order accurate; the error affects only the very first full leapfrog step, after which the scheme is self-consistent and second-order accurate.

The particle position $\mathbf{x}_i^0$ lives at integer time and is specified directly by the initial conditions — no back-extrapolation of position is needed.

Interactive particle placement

When a particle is added mid-simulation the sudden deposition of its charge and mass creates a field disturbance that propagates outward at c. This is physically correct: in classical field theory a source that suddenly appears radiates a spherical pulse, with the new static field inside the pulse and the pre-existing field outside it. For the pedagogical sandbox this is a feature — a student who places a mass and watches a gravitational wavefront propagate outward is seeing real physics.

If a smooth turn-on is preferred, ramp the particle's effective mass and charge from zero to their final values over $\sim N\sqrt{2}$ timesteps, suppressing the pulse at the cost of physical honesty.

The convergence iteration used for startup cannot be applied to mid-simulation particle addition without erasing all existing propagating waves in the domain — it would reset the entire field to the new static configuration. It is only appropriate before any dynamics have begun.

Particle removal is symmetric: a particle that disappears leaves a void in its Coulomb/Newton field that also propagates outward as a pulse. If a particle exits through the domain boundary the most physically honest treatment is to remove it and accept the outgoing pulse, which will be absorbed by the damping layer. The fundamental ambiguity in both creation and removal is that a particle appearing or disappearing has no prior history to specify what the surrounding field configuration should have been, making any treatment somewhat arbitrary; the pulse-based approach at least has a clear physical interpretation.

15.10 Field-particle energy accounting

The linearized GEM theory is not strictly energy-conserving in the full GR sense: the gravitational field carries energy in the nonlinear theory through the Landau-Lifshitz pseudotensor, which does not appear in the linearized equations. What the linearized theory does conserve, in the same structural sense that EM conserves energy, is the combined energy of the matter, GEM field, and EM field, with fluxes through the boundary accounting for radiation losses.

This section describes the conserved quantities and their use as *consistency diagnostics* rather than strict conservation laws. They verify that the numerical scheme is correctly implementing the linearized equations, not that the linearized equations are a complete description of reality.

Energy

The total energy in the simulation domain is:

$$E_{\text{total}} = E_{\text{particles}} + E_{\text{GEM}} + E_{\text{EM}}, \tag{187}$$

where:

$$E_{\text{particles}} = \sum_i \gamma_i m_i c^2, \tag{188}$$

$$E_{\text{GEM}} = \frac{1}{8\pi G}\int\left(|\mathbf{g}|^2 + c^2|\mathbf{B}_g|^2\right)\mathrm{d}^3x, \tag{189}$$

$$E_{\text{EM}} = \frac{\varepsilon_0}{2}\int\left(|\mathbf{E}|^2 + c^2|\mathbf{B}|^2\right)\mathrm{d}^3x. \tag{190}$$

In 2D the volume integrals become area integrals ($\mathrm{d}^3x \to \mathrm{d}^2x$), $\mathbf{B}_g \to B_{g,z}\hat{z}$ and $\mathbf{B} \to B_z\hat{z}$.

The rate of change of E_{total} should equal the energy flux leaving through the damping layer boundary:

$$\frac{\mathrm{d}E_{\text{total}}}{\mathrm{d}t} = -\oint_{\partial\Omega}(\mathbf{S}_{\text{GEM}} + \mathbf{S}_{\text{EM}})\cdot\hat{n}\,\mathrm{d}A, \tag{191}$$

where $\mathbf{S}_{\text{GEM}} = (\mathbf{g}\times\mathbf{B}_g)/(4\pi G)$ and $\mathbf{S}_{\text{EM}} = (\mathbf{E}\times\mathbf{B})/\mu_0$ are the GEM and EM Poynting vectors (in 2D, $\mathrm{d}A \to \mathrm{d}\ell$). In the presence of an absorbing boundary layer the boundary flux is not easily computed directly, so in practice one monitors E_{total} and checks that it changes smoothly and monotonically (decreasing as radiation exits through the boundary), with no sudden jumps that would indicate a numerical instability.

Momentum

The total canonical momentum is:

$$\mathbf{P}_{\text{total}} = \sum_i \mathbf{p}_i + \frac{1}{c^2}\int(\mathbf{S}_{\text{GEM}} + \mathbf{S}_{\text{EM}})\,\mathrm{d}^3x. \tag{192}$$

In the absence of external forces and with symmetric boundary conditions, this should be conserved. In practice the damping layer introduces a small asymmetry if waves exit preferentially in one direction, but for a symmetric simulation $\mathbf{P}_{\text{total}}$ should remain near zero. A systematic drift in one direction indicates a bug in the force calculation or source deposition.

Angular momentum

The total angular momentum is:

$$\mathbf{L}_{\text{total}} = \sum_i (\mathbf{r}_i \times \mathbf{p}_i + \mathbf{S}_i) + \frac{1}{c^2}\int \mathbf{r} \times (\mathbf{S}_{\text{GEM}} + \mathbf{S}_{\text{EM}})\, \mathrm{d}^3x, \tag{193}$$

where $\mathbf{S}_i$ is the spin angular momentum of particle i. In 2D only the z-component survives, with $\mathbf{S}_i \to s_i\hat{z}$. For isolated systems this should be conserved; systematic drift indicates an error in the spin precession or the spin-field coupling.

EM effective sources and the background field

When a charged background (Kerr-Newman-like) is present, its EM field has energy density $u_{\text{EM}}^{\text{bg}}$ that gravitates. In the exact Kerr-Newman solution this is already incorporated into the background metric — the gravitational potentials Φ_g^{bg}, $\mathbf{A}_g^{\text{bg}}$ reflect the mass M plus the EM field energy, not M alone. Feeding the background EM energy density into the perturbation GEM solver as ρ_{EM} would therefore double-count its gravitational effect.

The correct rule is to use only the perturbation potentials in the effective GEM sources (Eqs. 49–50):

$$\rho_{\text{EM}} = \frac{\varepsilon_0}{2c^2}\left(|\nabla\phi^{\text{pert}} + \partial_t\mathbf{A}^{\text{pert}}|^2 + c^2|\nabla \times \mathbf{A}^{\text{pert}}|^2\right). \tag{194}$$

Cross terms involving background potentials are second-order small for weakly charged backgrounds and are neglected. For the pedagogical scenarios the sandbox is designed for — a Kerr-Newman background precessing gyroscopes, curving charged particle paths, and producing Larmor precession alongside gravitomagnetic precession — this approximation is entirely adequate.

Practical monitoring

Computing the full field integrals at every timestep is an $O(N^3)$ operation in 3D ($O(N^2)$ in 2D). A practical compromise is to compute them every M steps (say $M = 100$) and store the time series. The field integrals can be accumulated in the same WASM loop as the field update with minimal overhead since field values are already in cache.

The most useful single diagnostic is the total canonical momentum $\mathbf{P}_{\text{total}}$: it should be small and non-drifting for any symmetric simulation. A non-zero systematic drift indicates either a force asymmetry (bug) or genuine momentum carried away by asymmetric radiation (physics).

Benchmark validation

The most reliable check that the simulation is physically correct is comparison against known analytic results:

- A circular orbit should have period $T = 2\pi r/v$ with v determined by the force law.
- A gyroscope in the Kerr-like background should precess at rate $\mathrm{d}\phi/\mathrm{d}t = -2GJ/(c^2r^2)$.
- The Shapiro delay for a test signal past the background mass should match $\Delta t = -2\int \Phi_g\, \mathrm{d}\ell/c^3$.
- The proper time accumulated along a circular orbit should match the analytic expression involving both kinematic and gravitational time dilation.

These provide rigorous point checks against exact results and are more diagnostic than energy monitoring for catching subtle physics errors.

15.11 Artificial constants

The physical value of G/c^4 makes gravitational effects invisible at any reasonable simulation scale. An effective G_{eff} is chosen to make effects visible, as described in Section 15.8. The CFL constraint $c_{\text{eff}}\,\mathrm{d}t < \mathrm{d}x/\sqrt{2}$ in 2D determines the maximum simulation speed given the chosen c_{eff} and $\mathrm{d}x$.

16 Two-Dimensional Reduction and Pedagogical Goals

16.1 Motivation

The primary goal of the sandbox is to serve as a teaching tool for general relativity. A 2D simulation ($2+1$ dimensions) scales as N^2 rather than the N^3 of 3D, runs at interactive frame rates in JavaScript or WebAssembly, and retains essentially all physically interesting GR effects. The 3D extension is a natural future direction once the 2D browser version is established.

16.2 Reduction of fields and spin

In $2+1$ dimensions:

- $\mathbf{g}$ and $\mathbf{E}$ become 2-vectors (x and y components only).
- $\mathbf{B}_g$ and $\mathbf{B}$ reduce to scalars $B_{g,z}$ and B_z (the only non-vanishing curl components).
- Cross products reduce to scalar multiplications: $(v \times B_g)_x = v_y B_{g,z}$, $(v \times B_g)_y = -v_x B_{g,z}$.
- The spin 3-vector $\mathbf{S}$ reduces to a scalar angle ϕ_{spin} with precession:

$$\frac{\mathrm{d}\phi_{\text{spin}}}{\mathrm{d}t} = B_{g,z} + \frac{g_s q}{2m} B_z + \Omega_{\text{Thomas},z}. \tag{195}$$

  The Tulczyjew–Dixon constraint is automatically satisfied in 2D—no correction step is needed.

16.3 The 2D Green's function

In strict 2D the static Green's function is logarithmic:

$$\text{3D: } \Phi \sim 1/r \;\Rightarrow\; |\mathbf{g}| \sim 1/r^2; \qquad \text{2D: } \Phi \sim \ln r \;\Rightarrow\; |\mathbf{g}| \sim 1/r. \tag{196}$$

This produces flat rotation curves. If 3D-like $1/r^2$ gravity is preferred, prescribe $\Phi_g^{\text{bg}} = -GM/r$ analytically for the background—the FDTD perturbation solver is unchanged.

16.4 Gravitational waves in 2D

In pure $2+1$ GR the vacuum Einstein equations have no propagating tensor degrees of freedom—the Riemann tensor is fully determined by the Ricci tensor, which vanishes in vacuum. However, this does not mean masses affect each other instantaneously. The scalar and vector potential wave equations in the GEM approximation do propagate at c, carrying causal influence between masses. A mass that accelerates sends a wavefront outward at c; particles beyond the wavefront are not affected until it reaches them. The difference from 3D is the *polarization structure*: in 2D only the scalar (compression/dilation) wave mode propagates; the tensor $+$ and $\times$ polarizations require two independent transverse directions and do not exist.

16.5 Effects demonstrable in the 2D sandbox

Effect	In 2D?	Notes
Newtonian gravity	Yes	Logarithmic; background can use $1/r$
Gravitational wave propagation	Yes	Scalar mode only
Frame dragging	Yes	$B_{g,z}$ from spinning/moving masses
Perihelion precession	Yes	$\mathcal{O}(v^2/c^2)$ corrections
Shapiro delay	Yes	Via c_{local}
Gravitational lensing	Yes	Via c_{local}
Spin precession	Yes	Scalar ϕ_{spin}; directly visible
Proper time dilation	Yes	Per-particle clock
Dipole radiation	Yes	Time-varying spin source
EM–gravity coupling	Yes	Via ρ_{EM}, $\mathbf{J}_{\text{EM}}$
Tensor GW polarizations	No	Require h^{ij}, absent in 2D GEM

16.6 Implementation in JavaScript and WebAssembly

For a 256×256 grid with 6 scalar field equations and $\sim$10 operations per cell per step:

$$N_{\text{ops}} \approx 6 \times 10 \times 256^2 \approx 4 \times 10^6 \text{ per timestep.} \tag{197}$$

Achievable at 60 fps in pure JavaScript. Recommended architecture:

- **WASM module:** inner FDTD loop and particle integration (performance-critical); WASM SIMD processes 4 `float32` cells per instruction.
- **JavaScript:** user interaction and outer simulation loop.
- **WebGL / WebGPU:** field visualization; WebGPU compute shaders extend practical grid size to 2048×2048.

16.7 Complete 2D algorithm: all effects

This is the definitive per-timestep algorithm for the 2D sandbox incorporating the background spacetime split, all spin effects, Tier 3 relativistic force corrections, proper time integration, and the correct deposition kernels.

Initialization (once before simulation begins)

Step 1. Set background object parameters: mass M, charge Q, angular momentum J, position.

Step 2. Define analytic background field functions:

$$\Phi_g^{\text{bg}}(x,y) = -GM/r \quad (\text{or } -GM\ln r \text{ for strict 2D}), \tag{198}$$

$$A_{g,x}^{\text{bg}},\ A_{g,y}^{\text{bg}} \quad \text{from angular momentum } J, \tag{199}$$

$$\phi^{\text{bg}}(x,y) = Q/(2\pi\varepsilon_0 \ln r) \quad \text{(2D Coulomb)}, \tag{200}$$

$$\mathbf{g}^{\text{bg}} = -\nabla\Phi_g^{\text{bg}}, \quad B_{g,z}^{\text{bg}} = \partial_x A_{g,y}^{\text{bg}} - \partial_y A_{g,x}^{\text{bg}}, \tag{201}$$

$$\mathbf{E}^{\text{bg}} = -\nabla\phi^{\text{bg}}, \quad B_z^{\text{bg}} = 0 \quad \text{(static background)}, \tag{202}$$

$$\nabla B_{g,z}^{\text{bg}},\ \nabla B_z^{\text{bg}} \quad \text{(analytic gradients for spin forces)}. \tag{203}$$

Step 3. Initialize perturbation field arrays Φ_g^{pert}, $A_{g,x}^{\text{pert}}$, $A_{g,y}^{\text{pert}}$, ϕ^{pert}, A_x^{pert}, A_y^{pert} and their previous-timestep copies to zero (or to specified initial conditions).

Step 4. Initialize particles: for each particle i set $\mathbf{x}_i$, $\mathbf{p}_i$, m_i, q_i, $g_{s,i}$, $\phi_{\text{spin},i}$, $\tau_i = 0$.

Per-timestep iteration

Step 1. Deposit matter and charge sources from non-background particles using the CIC kernel W_2:

$$\rho_{\text{matter}}(\mathbf{x}) = \sum_i \gamma_i m_i \, W_2(\mathbf{x} - \mathbf{x}_i), \tag{204}$$

$$\mathbf{J}_{\text{matter}}(\mathbf{x}) = \sum_i \gamma_i m_i \mathbf{v}_i \, W_2(\mathbf{x} - \mathbf{x}_i), \tag{205}$$

$$\rho_q(\mathbf{x}) = \sum_i q_i \, W_2(\mathbf{x} - \mathbf{x}_i), \tag{206}$$

$$\mathbf{J}_q(\mathbf{x}) = \sum_i q_i \mathbf{v}_i \, W_2(\mathbf{x} - \mathbf{x}_i). \tag{207}$$

Step 2. Deposit spin dipole sources using the quadratic B-spline kernel W_3:

$$\mu_i = \frac{g_{s,i} q_i}{2m_i} s_i, \qquad \sigma_i = 2s_i, \tag{208}$$

$$\mathbf{J}_q(\mathbf{x}) \mathrel{+}= \nabla_{2D} \times \left[\sum_i \mu_i \, W_3(\mathbf{x} - \mathbf{x}_i)\right], \tag{209}$$

$$\mathbf{J}_{\text{matter}}(\mathbf{x}) \mathrel{+}= \nabla_{2D} \times \left[\sum_i \sigma_i \, W_3(\mathbf{x} - \mathbf{x}_i)\right], \tag{210}$$

where $\nabla_{2D} \times f = (\partial_y f, \, -\partial_x f)$ is the 2D curl of a scalar field, and s_i is the scalar spin.

Step 3. Compute EM effective GEM sources from perturbation potentials (2D specialization of Eqs. 49–50, with $\nabla \times \mathbf{A}^{\text{pert}} \to F_{12}^{\text{pert}} \hat{z}$ where $F_{12}^{\text{pert}} = \partial_x A_y^{\text{pert}} - \partial_y A_x^{\text{pert}}$):

$$\rho_{\text{EM}} = \frac{\varepsilon_0}{2c^2} \left(|\nabla \phi^{\text{pert}} + \partial_t \mathbf{A}^{\text{pert}}|^2 + c^2 (F_{12}^{\text{pert}})^2 \right), \tag{211}$$

$$\mathbf{J}_{\text{EM}} = -\frac{F_{12}^{\text{pert}}}{\mu_0 c^2} \begin{pmatrix} \partial_y \phi^{\text{pert}} + \partial_t A_y^{\text{pert}} \\ -(\partial_x \phi^{\text{pert}} + \partial_t A_x^{\text{pert}}) \end{pmatrix}. \tag{212}$$

(Background potentials are excluded to avoid double-counting; Section 15.10.)

Step 4. Sum effective GEM sources:

$$\rho_{\text{eff}} = \rho_{\text{matter}} + \rho_{\text{EM}}, \qquad \mathbf{J}_{\text{eff}} = \mathbf{J}_{\text{matter}} + \mathbf{J}_{\text{EM}}. \tag{213}$$

Step 5. Step perturbation GEM wave equations (leapfrog; 3 scalar equations):

$$(\Phi_g^{\text{pert}})^{n+1} = 2(\Phi_g^{\text{pert}})^n - (\Phi_g^{\text{pert}})^{n-1} + (c\,\mathrm{d}t)^2 \left[\nabla_{2D}^2 (\Phi_g^{\text{pert}})^n - 4\pi G \rho_{\text{eff}}^n\right], \tag{214}$$

$$(A_{g,\alpha}^{\text{pert}})^{n+1} = 2(A_{g,\alpha}^{\text{pert}})^n - (A_{g,\alpha}^{\text{pert}})^{n-1} + (c\,\mathrm{d}t)^2 \left[\nabla_{2D}^2 (A_{g,\alpha}^{\text{pert}})^n - \frac{4\pi G}{c^2} (J_{\text{eff},\alpha})^n\right], \ \alpha \in \{x, y\}. \tag{215}$$

Step 6. Evaluate total gravitational potential at each grid point for c_{local}:

$$\Phi_g^{\text{total}}(\mathbf{x}) = \Phi_g^{\text{bg}}(\mathbf{x}) + (\Phi_g^{\text{pert}})^{n+1}(\mathbf{x}), \tag{216}$$

$$c_{\text{local}}(\mathbf{x}) = c\left(1 + \frac{2\Phi_g^{\text{total}}(\mathbf{x})}{c^2}\right). \tag{217}$$

Step 7. Step perturbation EM wave equations (3 scalar equations) using c_{local}:

$$(\phi^{\text{pert}})^{n+1} = 2(\phi^{\text{pert}})^n - (\phi^{\text{pert}})^{n-1} + (c_{\text{local}}\,\mathrm{d}t)^2\left[\nabla_{2D}^2(\phi^{\text{pert}})^n + (\rho_q^n)/\varepsilon_0\right], \tag{218}$$

$$(A_\alpha^{\text{pert}})^{n+1} = 2(A_\alpha^{\text{pert}})^n - (A_\alpha^{\text{pert}})^{n-1} + (c_{\text{local}}\,\mathrm{d}t)^2\left[\nabla_{2D}^2(A_\alpha^{\text{pert}})^n - \mu_0(J_{q,\alpha})^n\right],\ \alpha \in \{x, y\}. \tag{219}$$

Step 8. Evaluate potentials and on-the-fly curls at each particle position, combining analytic background and grid perturbation:

$$(\Phi_g)_i^{\text{total}} = \Phi_g^{\text{bg}}(\mathbf{x}_i) + (\Phi_g^{\text{pert}})_{W_2}^n(\mathbf{x}_i), \tag{220}$$

$$(A_{g,\alpha})_i = A_{g,\alpha}^{\text{bg}}(\mathbf{x}_i) + (A_{g,\alpha}^{\text{pert}})_{W_2}^n(\mathbf{x}_i), \quad \alpha \in \{x, y\}, \tag{221}$$

$$(\partial_t A_{g,\alpha})_i = \left.\frac{(A_{g,\alpha}^{\text{pert}})^{n+1} - (A_{g,\alpha}^{\text{pert}})^{n-1}}{2\mathrm{d}t}\right|_{W_2}(\mathbf{x}_i), \tag{222}$$

$$B_{g,z,i} = B_{g,z}^{\text{bg}}(\mathbf{x}_i) + (B_{g,z}^{\text{pert}})_{\text{curl},W_2}(\mathbf{x}_i), \tag{223}$$

and identically for the EM potentials ϕ_i, $(A_\alpha)_i$, $(\partial_t A_\alpha)_i$, $B_{z,i}$. The on-the-fly curl values $B_{g,z,i}$ and $B_{z,i}$ are computed from the four cell centers surrounding the particle via Eq. (114) and bilinearly interpolated. The background curl $B_{g,z}^{\text{bg}}$ is evaluated analytically. Also evaluate the background-referenced gradient vectors:

$$\mathbf{g}_i^{\text{bg}} = -\nabla\Phi_g^{\text{bg}}(\mathbf{x}_i), \tag{224}$$

$$\nabla B_{g,z,i} = \nabla B_{g,z}^{\text{bg}}(\mathbf{x}_i) + (\nabla B_{g,z}^{\text{pert}})_{W_3}(\mathbf{x}_i), \tag{225}$$

$$\nabla B_{z,i} = \nabla B_z^{\text{bg}}(\mathbf{x}_i) + (\nabla B_z^{\text{pert}})_{W_3}(\mathbf{x}_i). \tag{226}$$

Step 9. Compute total force on each particle (2D, Tier 3, potential form):

$$g_{x,i} = -\partial_x(\Phi_g)_i - (\partial_t A_{g,x})_i, \quad g_{y,i} = -\partial_y(\Phi_g)_i - (\partial_t A_{g,y})_i,$$

$$\begin{aligned}
F_{x,i} = m_i\Big[&g_{x,i} + 4v_{y,i}B_{g,z,i} + \frac{v_i^2}{c^2}g_{x,i} + \frac{4(\Phi_g)_i}{c^2}g_{x,i} - \frac{4(\mathbf{v}_i\cdot\mathbf{g}_i)v_{x,i}}{c^2}\\
&+4\partial_x(\mathbf{v}_i\cdot(\mathbf{A}_g)_i) - 4(\mathbf{v}_i\cdot\nabla)A_{g,x,i}]\\
&+ q_i[E_{x,i} + v_{y,i}B_{z,i} + \partial_x(\mathbf{v}_i\cdot\mathbf{A}_i) - (\mathbf{v}_i\cdot\nabla)A_{x,i}]\\
&+ s_i\,\partial_x B_{g,z,i} + \mu_i\,\partial_x B_{z,i},
\end{aligned}$$

$$\begin{aligned}
F_{y,i} = m_i\Big[&g_{y,i} - 4v_{x,i}B_{g,z,i} + \frac{v_i^2}{c^2}g_{y,i} + \frac{4(\Phi_g)_i}{c^2}g_{y,i} - \frac{4(\mathbf{v}_i\cdot\mathbf{g}_i)v_{y,i}}{c^2}\\
&+4\partial_y(\mathbf{v}_i\cdot(\mathbf{A}_g)_i) - 4(\mathbf{v}_i\cdot\nabla)A_{g,y,i}]\\
&+ q_i[E_{y,i} - v_{x,i}B_{z,i} + \partial_y(\mathbf{v}_i\cdot\mathbf{A}_i) - (\mathbf{v}_i\cdot\nabla)A_{y,i}]\\
&+ s_i\,\partial_y B_{g,z,i} + \mu_i\,\partial_y B_{z,i},
\end{aligned}$$

where $E_{x,i} = -\partial_x \phi_i - (\partial_t A_x)_i$ and $E_{y,i} = -\partial_y \phi_i - (\partial_t A_y)_i$ are computed on-the-fly, and $\mathbf{v}_i \cdot \mathbf{g}_i = v_{x,i} g_{x,i} + v_{y,i} g_{y,i}$. Evaluate v_i^2/c^2 and $\mathbf{v}_i \cdot \mathbf{g}_i$ using $\mathbf{v}_i^{n-1/2}$. The vector potential terms $\partial_\alpha(\mathbf{v} \cdot \mathbf{A}_g)$ and $(\mathbf{v} \cdot \nabla) A_{g,\alpha}$ encode the gravitomagnetic cross-product force and are computed from spatial differences of edge potentials at the particle's CIC stencil.

Step 10. Update momentum and position (Boris pusher, 2D):

$$\mathbf{v}_i^{\text{old}} = \mathbf{v}_i^{n-1/2}, \tag{227}$$

$$\mathbf{p}_i^{n+1/2} \leftarrow \text{Boris}(\mathbf{p}_i^{n-1/2},\ F_{x,i},\ F_{y,i},\ B_{g,z,i},\ B_{z,i},\ \mathrm{d}t), \tag{228}$$

$$\mathbf{v}_i^{n+1/2} = \mathbf{p}_i^{n+1/2} / \sqrt{m_i^2 + |\mathbf{p}_i^{n+1/2}|^2/c^2}, \tag{229}$$

$$\mathbf{x}_i^{n+1} \leftarrow \mathbf{x}_i^n + \mathbf{v}_i^{n+1/2}\,\mathrm{d}t. \tag{230}$$

Step 11. Update spin (scalar in 2D):

$$\mathbf{a}_i = (\mathbf{v}_i^{n+1/2} - \mathbf{v}_i^{\text{old}})/\mathrm{d}t, \tag{231}$$

$$\Omega_{\text{Thomas},z,i} = -\frac{\gamma_i^2}{\gamma_i + 1}\,\frac{v_{x,i} a_{y,i} - v_{y,i} a_{x,i}}{c^2}, \tag{232}$$

$$\Omega_{z,i} = B_{g,z,i} + \frac{g_{s,i} q_i}{2m_i} B_{z,i} + \Omega_{\text{Thomas},z,i}, \tag{233}$$

$$\phi_{\text{spin},i} \leftarrow \phi_{\text{spin},i} + \Omega_{z,i}\,\mathrm{d}t. \tag{234}$$

(No constraint correction step is needed in 2D.)

Step 12. Update proper time:

$$\beta_i^2 = \tfrac{1}{2}\left(|\mathbf{v}_i^{\text{old}}|^2 + |\mathbf{v}_i^{n+1/2}|^2\right)/c^2, \qquad \mathrm{d}\tau_i = \mathrm{d}t\sqrt{1 + \frac{2(\Phi_g^{\text{total}})_i}{c^2} - \left(1 - \frac{2(\Phi_g^{\text{total}})_i}{c^2}\right)\beta_i^2}, \tag{235}$$

$$\tau_i \leftarrow \tau_i + \mathrm{d}\tau_i. \tag{236}$$

$(\Phi_g^{\text{total}})_i = \Phi_g^{\text{bg}}(\mathbf{x}_i^{n+1}) + (\Phi_g^{\text{pert}})^{n+1}(\mathbf{x}_i^{n+1})$, evaluated analytically and by grid interpolation respectively.

16.8 Pedagogical design notes

- **Separate field visualizations:** display Φ_g, $B_{g,z}$, $\mathbf{E}$, B_z independently.
- **Slow-motion and pause:** continuously adjustable speed; single-frame stepping.
- **Particle display:** velocity vector, spin angle indicator, GEM force vector, EM force vector, proper time readout.
- **Preloaded scenarios:** binary orbit, spinning mass with test gyroscopes, gravitational lens, EM wave scattering.
- **Adjustable constants:** sliders for G_{eff}, c_{eff}, q/m allow students to explore the Newtonian limit ($c \to \infty$), strong-field limit, and EM-dominated regime.

17 Visualization

The visualization layer translates the numerical state of the simulation into something a student can see and understand intuitively. It is as important as the physics for a pedagogical tool:

a correct simulation that displays nothing meaningful teaches nothing. This section describes the recommended particle and field visualizations, organized by what physical quantity they represent. All visualizations should be independently toggleable so a student can isolate the effect they are studying.

17.1 Particle visualizations

Each particle is displayed as a small icon at its current position. The icon serves as an anchor for several overlaid indicators, each encoding a different aspect of the particle's state.

Trajectory tracks

A fading trail of the particle's past positions is drawn, length corresponding to a user-adjustable number of recent timesteps. For orbital motion this produces a visible arc or closed ellipse. The trail should fade in opacity with age so the most recent position is brightest and the oldest is nearly invisible, giving an immediate sense of the particle's direction of travel and speed.

At regularly spaced intervals along the track, small tick marks are drawn with an associated time readout. Two modes are useful:

- **Coordinate time ticks**: equally spaced in simulation time t, showing when the particle was at each point in coordinate time. Tick spacing is uniform along the track for a particle at constant speed.
- **Proper time ticks**: equally spaced in proper time τ_i. For a fast-moving or deep-potential particle, proper time ticks are closer together (more time elapses per tick in coordinate time) than coordinate time ticks, making relativistic time dilation directly visible as a non-uniform spacing along the track.

Displaying both simultaneously with different symbols (e.g. circles for coordinate time, triangles for proper time) makes the twin paradox and gravitational time dilation immediately readable from the track geometry.

Velocity arrow

A short arrow from the particle center in the direction of $\mathbf{v}_i$, with length proportional to $|\mathbf{v}_i|/c_{\text{eff}}$ (so the arrow reaches a standard reference length at $v = c_{\text{eff}}$). Color can encode γ_i — blue for nonrelativistic, red for relativistic.

Spin phase indicator

A small icon resembling a spinning top or gyroscope overlaid on the particle, with an arrow pointing in the direction $(\cos\phi_{\text{spin}}, \sin\phi_{\text{spin}})$ encoding the current spin phase ϕ_{spin}. As the particle moves through a gravitomagnetic or magnetic field the arrow rotates, directly visualizing Lense–Thirring precession, geodetic precession, Larmor precession, and Thomas precession as a continuous rotation of the indicator.

The rotation rate of the arrow is $\Omega_{z,i} = B_{g,z,i} + (g_s q_i/2m_i)B_{z,i} + \Omega_{\text{Thomas},z,i}$, so regions of strong gravitomagnetic or magnetic field produce visible rapid precession and field-free regions produce no rotation. This is one of the most pedagogically direct visualizations in the sandbox — frame dragging becomes literally visible as a spinning arrow.

Magnetic dipole indicator

A separate icon resembling a small bar magnet (a rectangle with N and S poles indicated) that rotates with the magnetic dipole moment direction $\boldsymbol{\mu}_i = g_{s,i}(q_i/2m_i)\mathbf{S}_i$. In 2D this is the same angle as the spin indicator but with a different icon to emphasize the physical distinction between the spin angular momentum (gyroscope icon) and the magnetic dipole moment (magnet icon). The magnet icon makes it visually clear that the dipole moment is what couples to $\mathbf{B}$ for the Larmor force, while the gyroscope icon is what couples to $\mathbf{B}_g$ for the gravitomagnetic precession.

The magnitude $|\boldsymbol{\mu}_i|$ is constant (spin magnitude is conserved), which can be indicated by fixing the length of the bar magnet icon.

Force arrows

One or more arrows showing the force components acting on each particle:

- **Gravitational force $\mathbf{F}_{\text{grav}}$**: the GEM Lorentz force plus relativistic corrections. Color: e.g. orange.
- **Electromagnetic force $\mathbf{F}_{\text{EM}} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$**. Color: e.g. blue.
- **Total force $\mathbf{F}_{\text{total}}$**. Color: e.g. white.
- **Spin gradient force $\mathbf{F}_{\text{spin}} = \nabla(s\, B_{g,z}) + \mu \nabla B_z$**. Color: e.g. green.

Arrow length is proportional to force magnitude, normalized to a reference force scale. Individual components can be toggled, allowing a student to isolate which force is responsible for a given orbital feature.

Proper time clock

A small analog clock face next to each particle, with a hand rotating at rate $\mathrm{d}\tau_i/\mathrm{d}t$ relative to coordinate time. When $\mathrm{d}\tau_i/\mathrm{d}t < 1$ (relativistic motion or deep potential well) the hand rotates visibly more slowly than a reference clock displayed at a fixed point on the screen. The running total τ_i can also be displayed as a digital readout beside the clock face.

17.2 Field visualizations

The perturbation field quantities available for visualization are listed below. The scalar and vector potentials are stored on the grid and available at all times. The derived fields $\mathbf{g}$, $\mathbf{E}$, $B_{g,z}$, B_z are computed from the potentials on demand at display time — they are not persistent grid arrays. The per-step derivations are:

$$\mathbf{g}^{\text{pert}} = -\nabla\Phi_g^{\text{pert}} - \partial_t \mathbf{A}_g^{\text{pert}}, \qquad B_{g,z}^{\text{pert}} = \partial_x A_{g,y}^{\text{pert}} - \partial_y A_{g,x}^{\text{pert}}, \tag{237}$$

$$\mathbf{E}^{\text{pert}} = -\nabla\phi^{\text{pert}} - \partial_t \mathbf{A}^{\text{pert}}, \qquad B_z^{\text{pert}} = \partial_x A_y^{\text{pert}} - \partial_y A_x^{\text{pert}}. \tag{238}$$

Field	Symbol	Physical meaning
Gravitational scalar potential	Φ_g^{pert} (stored)	Depth of gravitational well
Gravitomagnetic vector potential	$\mathbf{A}_g^{\text{pert}}$ (stored)	Source of frame dragging
Gravitoelectric field	$\mathbf{g}^{\text{pert}}$ (computed)	Gravitational force per unit mass
Gravitomagnetic field	$B_{g,z}^{\text{pert}}$ (computed)	Frame dragging / precession rate
Electrostatic potential	ϕ^{pert} (stored)	Electric potential per unit charge
EM vector potential	$\mathbf{A}^{\text{pert}}$ (stored)	Source of magnetic effects
Electric field	$\mathbf{E}^{\text{pert}}$ (computed)	Electric force per unit charge
Magnetic field	B_z^{pert} (computed)	Larmor precession rate

The background contributions Φ_g^{bg}, $B_{g,z}^{\text{bg}}$, $\mathbf{E}^{\text{bg}}$, B_z^{bg} can optionally be added to give the total field, or displayed separately.

Scalar field display modes

Scalar fields (Φ_g, ϕ, $B_{g,z}$, B_z) can be displayed in three modes:

- **Grayscale or false color magnitude**: the field value is mapped to a color via a diverging colormap (e.g. blue for negative, white for zero, red for positive). This gives an immediate spatial impression of the field structure. The colormap range should be auto-scaled to the current field maximum, or fixed to a user-specified range.
- **Contour lines**: isosurfaces of constant field value drawn as lines. Contours of Φ_g are the gravitational equipotential surfaces; for a static mass they are concentric circles. Contours of $B_{g,z}$ show the spatial structure of the frame-dragging field. Spacing between contours encodes field gradient — closely spaced contours indicate strong fields.
- **Animated phase**: for oscillating or wave fields, the field value can be mapped to a slowly cycling colormap so that wavefronts appear to propagate visually even when the field amplitude is small. This makes gravitational and EM waves visible at low amplitude that would be invisible in a static colormap.

Vector field display modes

Vector fields ($\mathbf{g}$, $\mathbf{E}$, $\mathbf{A}_g$, $\mathbf{A}$) can be displayed in two modes:

- **Quiver plot**: arrows at a subsampled grid of points (e.g. every 8 cells), each pointing in the field direction with length proportional to field magnitude. Length is typically logarithmically scaled to handle the large dynamic range near particle positions. Color can encode magnitude independently of arrow length for additional clarity.
- **Streamlines / field lines**: curves everywhere tangent to the field direction, integrated from seed points. For $\mathbf{g}$ these are the gravitational field lines (analogous to electric field lines), which originate at mass sources and point toward them. For $\mathbf{A}_g$ the field lines show the curl structure that generates $\mathbf{B}_g$. Streamlines give a cleaner global picture than quiver plots for smooth fields, but are less informative near particle positions where the field changes rapidly.

Dynamic range and normalization

Near a particle the fields diverge as $1/r$ or $1/r^2$, making naive display of the full field dominated by the near-field singularity and showing nothing useful in the rest of the domain. The following normalizations are recommended:

- **Clamp**: cap the displayed value at a user-adjustable maximum, saturating the color near particles. Simple but loses information near the source.
- **Logarithmic scale**: display $\mathrm{sgn}(\Phi)\log(1 + |\Phi|/\Phi_0)$ where Φ_0 is a reference scale. Shows both the near-field and far-field structure simultaneously.
- **Show perturbation only**: display Φ_g^{pert} rather than Φ_g^{total}, removing the large static background. This makes propagating waves visible even in the presence of a dominant static field. This is the most useful mode for demonstrating gravitational wave emission.

Recommended display combinations for specific demonstrations

Demonstration	Recommended display
Newtonian orbits	Contours of Φ_g^{total} + particle tracks with coordinate time ticks
Frame dragging	False color $B_{g,z}$ + gyroscope spin indicators on test particles
Gravitational waves	Animated false color Φ_g^{pert} (perturbation only) + particle tracks
Gravitational lensing	Streamlines of $\mathbf{E}^{\mathrm{pert}}$ (EM wave) on false color Φ_g^{bg}
Larmor precession vs Lense–Thirring	False color $B_{g,z}$ and B_z side by side + magnet and gyroscope icons
Twin paradox	Particle tracks with both coordinate and proper time ticks
EM–gravity coupling	False color Φ_g^{pert} with EM source active, showing EM energy gravitating
Charged particle orbits	Quiver $\mathbf{g}$ and $\mathbf{E}$ + force arrows split by GEM and EM components

Implementation notes

All field arrays are already on the GPU as WebGL textures (one float per cell for scalar fields, two floats per cell for 2D vector fields). Rendering is a fragment shader that maps the texture value to a color via the chosen colormap — a single draw call per displayed field. Quiver plots are rendered as a second pass drawing arrow geometry at subsampled positions, reading field values from the texture. Particle icons, spin indicators, and force arrows are rendered as sprite geometry in a final pass. The separation into three render passes (fields, quivers, sprites) allows each to be toggled independently with no performance cost when disabled.

18 Implementation Plan

This section describes a staged implementation strategy. Each stage adds one capability, is independently testable against a known analytic result, and must pass its tests before the next stage begins. This prevents bugs introduced early from manifesting as wrong answers late in the stack when the cause is difficult to trace.

The governing principle throughout: **never proceed to the next stage until the current stage passes its quantitative tests.** A simulation that produces plausible-looking output is not the same as a correct simulation.

18.1 Stage 1: Scalar wave equation, free propagation

Implement the leapfrog update for a single scalar potential Φ in 2D on a uniform grid, with no sources and no damping boundary.

Test: place a narrow Gaussian initial condition at the center of the domain. The pulse should propagate outward as a circular ring at speed c_{eff}. Verify:

- The wavefront reaches radius r at time $t = r/c_{\text{eff}}$ to within one timestep.
- The amplitude falls off as $1/\sqrt{r}$ in 2D.
- The scheme is stable at $c_{\text{CFL}} = 1/\sqrt{2}$ and unstable at $c_{\text{CFL}} = 1/\sqrt{2} + \epsilon$.
- Measured numerical dispersion matches the analytic prediction of Section 15.4.

18.2 Stage 2: Absorbing boundary layer

Add the damping layer (Section 15.7) to Stage 1.

Test: the Gaussian pulse from Stage 1 should exit the domain cleanly. Record the residual field amplitude at the domain center after the pulse should have fully exited. Per the realized-reflection paragraph added to §15.7 in v33, for a 2D circular wavefront the in-interior residual is degraded from the plane-wave $R \approx 10^{-3}$ limit by linearization and corner-trapping effects; a few-percent central residual with $\sim 20\times$ suppression versus the no-damping case is the realistic target.

18.3 Stage 3: Static CIC source, Poisson convergence

Add a fixed point source using CIC deposition at the domain center. Run the convergence iteration (positions and velocities held fixed) until fields converge.

Test: the scalar potential should converge to the 2D Green's function. Plot $\Phi(r)$ vs $\ln r$ and verify linearity. Verify convergence takes $\sim N\sqrt{2}$ timesteps. Verify the discrete Poisson equation $\nabla_h^2 \Phi = -\text{source}$ is satisfied to machine precision at convergence.

18.4 Stage 4: All six wave equations, gauge monitoring

Extend to all six potentials (Φ_g, $A_{g,x}$, $A_{g,y}$, ϕ, A_x, A_y) simultaneously. Add the discrete Lorenz gauge residual monitor.

Test: a static point source with zero velocity should produce $\mathbf{A}_g = 0$ and $\mathbf{A} = 0$ automatically (zero current sources). Verify the gauge residual $\langle \mathcal{G}^2 \rangle^{1/2}$ remains at the truncation error level $(k\Delta x)^2$. Enable parabolic cleaning and verify it reduces the residual without destabilizing.

18.5 Stage 5: Moving source, vector potential, Yee curl

Initialize a source with nonzero velocity $\mathbf{v}$ during the convergence phase. Verify that $\mathbf{A}_g$ and $\mathbf{A}$ develop the correct boosted Coulomb values.

Test: for a source moving at velocity v the vector potential should satisfy $|\mathbf{A}| \approx v|\phi|/c^2$ to leading order. Compute $B_z = \partial_x A_y - \partial_y A_x$ using the Yee curl at several points and compare against the analytic result for a uniformly moving charge.

18.6 Stage 6: Sampled background field arrays

Introduce a parallel set of read-only grid arrays for the background contributions to each of the six potentials:

$$\Phi_g^{\rm bg}, \quad A_{g,x}^{\rm bg}, \quad A_{g,y}^{\rm bg}, \quad \phi^{\rm bg}, \quad A_x^{\rm bg}, \quad A_y^{\rm bg}.$$

These arrays live on the same staggered Yee grid as the perturbation potentials (§15), are filled *once* at scenario setup, and are not touched by the leapfrog or the damping layer (§15.7). Force evaluations from Stage 7 onward compute totals as

$$\Phi_g^{\rm total}(\mathbf{x}) = \Phi_g^{\rm bg}(\mathbf{x}) + \Phi_g^{\rm pert}(\mathbf{x}),$$

and similarly for $\mathbf{A}_g$, ϕ, $\mathbf{A}$, through a single combined CIC interpolation per quantity per particle.

Motivation: testability decoupled from source-coupling

Particle-dynamics tests (orbital period, time dilation, precession, frame dragging, cyclotron motion) require a field for the particle to move in. In v32 those tests presumed an analytic background available at the particle position via closed-form evaluation; the alternative — creating the field with a second body and the perturbation FDTD — would require all of the source-deposition machinery (Stages 3–5) to be working perfectly before any single-particle test could run. By introducing sampled background arrays as their own stage, every later particle-dynamics test runs against a prescribed field, isolating the particle integrator from the source path.

Implementation

Add the six pointers to the simulation struct, allocated lazily on first use. Provide a small registry of background generators that fill the arrays at scenario setup time. The minimum required for this stage is the softened point mass:

$$\Phi_g^{\rm bg}(\mathbf{x}) = -\frac{G_{\rm eff} M}{\sqrt{|\mathbf{x} - \mathbf{x}_0|^2 + \epsilon^2}}, \tag{239}$$

sampled onto the grid at scenario load, with a smoothing length ϵ of a few cells to avoid the $1/r$ singularity at the source. Later stages add further generators as needed: uniform gravitational field ($\Phi_g^{\rm bg} = -g_0 y$), gravitomagnetic dipole ($A_{g,\theta}^{\rm bg} \propto J/r$ as in §8.2), point charge, and the full mode list collected in Stage 15 below.

Test:

- Sampled values match Eq. (239) at interior cells to within floating-point precision.
- Finite-difference gradient $-\nabla_h \Phi_g^{\rm bg}$ at sample radii $r \gg \epsilon$ matches the analytic Newtonian field $G_{\rm eff} M\, \hat{r}/r^2$ to within the CIC interpolation tolerance.
- Re-running the Stage 1 wave pulse with a non-trivial background installed produces a perturbation field bit-identical to the no-background case (perturbation FDTD does not read backgrounds).
- Toggling the damping layer with a background present does not alter the background array.

Caveat: discretization noise vs. small effects

Sampling a closed-form background onto a discrete grid carries an $\mathcal{O}((\Delta x/L)^2)$ representation error where L is the local feature scale. For *coarse* tests (Keplerian period, gravitomagnetic precession at moderate r) this error is negligible. For *small* effects whose signature competes with the discretization noise — most notably the post-Newtonian perihelion precession of Stage 8, where the per-orbit advance is $\sim G_{\text{eff}} M/(c_{\text{eff}}^2 a)$ and may approach the smoothing- and finite-difference-induced error — an analytic-form background evaluator is provided as an optional code path that bypasses the grid for those specific tests. See Section 18.6.

Analytic-background evaluation (added in v34)

In addition to the sampled-grid path described above (which is the default and the only path supporting arbitrary user-installed background fields), the simulator provides an *analytic-mode* evaluation in which a small registry of closed-form generators is queried directly at the particle's exact position, with no grid sampling or CIC+FD chain. The generator parameters $(\mathbf{x}_0, G_{\text{eff}} M, \epsilon, q, \mathbf{S}, \ldots)$ are stored in the simulation struct at scenario setup, and the evaluator returns $\Phi_g^{\text{bg}}(\mathbf{x})$ and $\nabla\Phi_g^{\text{bg}}(\mathbf{x})$ (and the corresponding vector-potential quantities for spinning / charged variants) directly.

Why this matters. The sampled-grid path introduces a small tangential force error at $\mathcal{O}((\Delta x/r)^2)$ for any non-grid-aligned particle position — the same kernel that lets the W_2 deposition smear a point source out also breaks the perfect spherical symmetry of the sampled Φ_g^{bg} a few cells away. In the test-particle limit this manifests as a slow drift in angular momentum, which over many orbits produces a small spurious precession that contaminates the Stage 8 measurement. The analytic path eliminates this systematic at the cost of restricting the available backgrounds to the registered generator catalogue.

Currently registered.

- `POINT_MASS`: softened Newtonian scalar $\Phi_g^{\text{bg}}(\mathbf{x}) = -G_{\text{eff}} M/\sqrt{|\mathbf{x}-\mathbf{x}_0|^2+\epsilon^2}$, with closed-form gradient $\nabla\Phi_g^{\text{bg}} = G_{\text{eff}} M(\mathbf{x}-\mathbf{x}_0)/(|\mathbf{x}-\mathbf{x}_0|^2+\epsilon^2)^{3/2}$.

Reserved (deferred to later stages).

- `SPINNING_POINT_MASS`: adds a gravitomagnetic $\mathbf{A}_g^{\text{bg}}$ from a point angular momentum $\mathbf{S}$ (Lense–Thirring; Stage 13/14).
- `CHARGED_POINT_MASS`: adds Coulomb ϕ^{bg} (Stage 11/14).
- `KERR_NEWMAN`: combined spinning charged mass (Stage 14).

The mode is a runtime switch independent of which generator was last installed. Stage 7 and Stage 8 scenarios opt into the analytic path; Stage 6 explicitly exercises the sampled path so its tests remain meaningful for arbitrary user-supplied backgrounds.

18.7 Stage 7: Single test particle, Boris pusher, Keplerian orbit

Add a single test particle with mass m and no charge, placed in the softened-point-mass background of Stage 6 (Eq. 239). Implement the potential-form force (Eq. 41) and the Boris pusher. No perturbation FDTD is exercised in this stage — the particle only reads the background array.

Test: a circular orbit should have period $T = 2\pi r^{3/2}/\sqrt{G_{\text{eff}} M}$. Verify the total energy $\gamma mc^2 + m\Phi_g^{\text{bg}}$ is approximately conserved over many orbits with no secular drift (Boris symplectic structure). Verify that taking the smoothing length $\epsilon \to 0$ recovers the textbook $1/r^2$ force law at $r \gg \epsilon$.

18.8 Stage 8: Relativistic corrections, perihelion precession

Enable Tier 2 force corrections (Section 12) — the test-particle limit of the Einstein–Infeld–Hoffmann 1PN equation:

$$\mathbf{F}_{\text{grav}}/m \;=\; \mathbf{g}\left(1 + \frac{v^2}{c^2} + \frac{4\Phi_g}{c^2}\right) \;-\; \frac{4(\mathbf{v}\cdot\mathbf{g})\mathbf{v}}{c^2}, \tag{240}$$

with $\mathbf{g} = -\nabla\Phi_g$ (no $\mathbf{A}_g$ in the static-background case). This is the same expression as in the Tier 3 algorithm of Section 12.4; the EIH equation is correct through $\mathcal{O}((v/c)^2)$ in the coordinate acceleration, and Lense–Thirring enters at the same 1PN order via the $\mathbf{A}_g$ terms (Section 12).

Test: a moderately eccentric orbit on an analytic softened-point-mass background should precess at the GR rate:

$$\Delta\phi_{\text{prec}} = \frac{6\pi G_{\text{eff}} M}{c_{\text{eff}}^2\, a(1-e^2)} \tag{241}$$

per radial period, where a is the semi-major axis and e the eccentricity of the measured orbit (i.e. $a' = (r_{\text{peri}} + r_{\text{apo}})/2$, $e' = (r_{\text{apo}} - r_{\text{peri}})/(r_{\text{apo}} + r_{\text{peri}})$, which generally differ from the Newtonian-derived inputs at 1PN). Use the analytic-background path (Section 18.6) for this test — the CIC+FD-of-sampled-background chain introduces an $\mathcal{O}((\Delta x/r)^2)$ tangential force error that breaks angular-momentum conservation and contaminates the precession measurement.

A small "kinematic" baseline contribution arises from running the relativistic 3-momentum dynamics with a Tier-0 (Newtonian) force law: this is not Schwarzschild and does precess, by $\sim 0.1\,(v/c)^2$ rad/orbit at our parameter scale. Subtract it (run Tier-0 first as a baseline); the residual EIH contribution should match Eq. (241) to within the residual $\mathcal{O}((v/c)^4)$ 2PN truncation. At $v/c \approx 0.04$ in the implementation the agreement is better than 1%.

18.9 Stage 9: Proper time accumulation

Add the proper time clock τ_i to each particle (Section 13). Each timestep accumulate

$$\mathrm{d}\tau = \mathrm{d}t\,\sqrt{1 + \frac{2\Phi_g}{c^2} - \left(1 - \frac{2\Phi_g}{c^2}\right)\frac{v^2}{c^2} - \frac{8\,\mathbf{A}_g\cdot\mathbf{v}}{c^2}}, \tag{242}$$

the full Eq. (105) with the v34 sign+factor corrections. Φ_g and $\mathbf{A}_g$ are evaluated at the particle's current position; for static-background scenarios the analytic-mode path (Section 18.6) gives a clean signal free of CIC+FD tangential artifact.

Two-phase test.

Phase A — non-spinning point mass, several orbital radii. Place a test particle on the relativistic-Newtonian circular orbit (as in Stage 7) at $r \in \{10, 20, 40, 80\}$ with $G_{\text{eff}}M = 1$, $\epsilon = 1$, $c_{\text{eff}} = 1$. After one orbital period verify

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}t}\right|_{\text{measured}} \approx \sqrt{1 + \frac{2\Phi_g}{c^2} - \left(1 - \frac{2\Phi_g}{c^2}\right)\frac{v^2}{c^2}} \tag{243}$$

to within a relative error consistent with the leapfrog truncation ($\sim 10^{-5}$ at $r \gtrsim 10$). For a Newtonian circular orbit with $v^2 = G_{\text{eff}}M/r$ the gravitational and kinematic contributions to the leading-order $\mathrm{d}\tau/\mathrm{d}t - 1$ stand in a $2:1$ ratio ($\Phi_g/c^2 : -v^2/(2c^2)$); as r increases the SR contribution drops faster than the gravitational one (both in absolute value). The implementation recovers this ratio: at $r = 10$ the measured ratio is 2.12, converging to 2.01 at $r = 80$.

Phase B — spinning point mass, prograde vs. retrograde. Install a spinning-point-mass background (Section 18.6) with central spin J_z and place a test particle on the same circular orbit, once prograde ($\mathbf{v} = +v_{\rm circ}\hat{\phi}$) and once retrograde ($\mathbf{v} = -v_{\rm circ}\hat{\phi}$). Run each for the same coordinate time $T_{\rm coord}$ (one orbital period at the non-spinning $v_{\rm circ}$). The proper-time difference between the two clocks is

$$\Delta\tau \;=\; \tau_+ - \tau_- \;\approx\; -\frac{8\pi G_{\rm eff} J_z}{c_{\rm eff}^4\, r} \tag{244}$$

per orbit at leading order ([9], in this document's factor-of-4-in-force GEM convention). For exact agreement with the implementation use $\Delta\tau = T_{\rm coord}[(\mathrm{d}\tau/\mathrm{d}t)_+ - (\mathrm{d}\tau/\mathrm{d}t)_-]$ with Eq. (242) evaluated at the prograde and retrograde signs of $\mathbf{A}_g\cdot\mathbf{v}$ — the linearized Eq. (244) drops a higher-order cross-coupling between Φ_g, v^2, and $\mathbf{A}_g\cdot\mathbf{v}$ that becomes visible at near-extremal Kerr in toy units. In the implementation at $r = 20$, $J_z = 2$ the linearized prediction is 8.4% off the measured $\Delta\tau$ while the exact-sqrt prediction agrees to better than 0.2% — the latter being the pure leapfrog integration residual.

Both phases use the analytic-background path; the perturbation FDTD is skipped via `gr_sim_set_field_evolut` since no sources are active.

v35 addendum — Phase B clock-only isolation. Phase B's analytic prediction Eq. (244) is derived under the assumption that the prograde and retrograde test particles remain on *the same* Keplerian circle, so that only the $-8(\mathbf{A}_g\cdot\mathbf{v})/c^2$ cross-term in the $\mathrm{d}\tau/\mathrm{d}t$ integrand differs between them. Once the Tier-1 gravitomagnetic Lorentz piece $\mathbf{F}_{\rm gm} = +4m\,\mathbf{v}\times\mathbf{B}_g$ (Section 12.4, Eq. 102) is wired into the pusher, that assumption no longer holds — the same $\mathbf{B}_g$ that produces the clock asymmetry also perturbs the orbit shape oppositely for prograde and retrograde, a first-order Lense–Thirring frame-dragging response on the dynamics. The two effects combined would add another $\mathcal{O}(J_z)$ contribution to $\Delta\tau$ on top of Eq. (244), spoiling the comparison. To preserve the clock-only isolation here, the gravitomagnetic Lorentz piece is gated by the `gr_sim_set_gravitomagnetic_force_enabled(sim, 0)` flag for the duration of Phase B; Stage 9b (Section 18.10) verifies that piece on its own with $\Phi_g = 0$.

18.10 Stage 9b: Gravitomagnetic Lorentz force — unit isolation

A unit verification of the Tier-1 gravitomagnetic Lorentz force $\mathbf{F}_{\rm gm}/m = +4\,\mathbf{v}\times\mathbf{B}_g$ (Eq. 102 at $\mathcal{O}(v/c)$; algorithmic form in Section 12.7's Tier-3 eqbox). The factor of 4 is the spin-2 enhancement of the GEM Lorentz force relative to its EM analogue (Section 8.2); the sign was corrected from v33 in v34 (empirical confirmation via Stage 8 perihelion precession; the v33 sign gave retrograde precession, the EIH sign gives prograde, matching the canonical Schwarzschild direction). Stage 9b verifies the coefficient and sign *directly*, isolated from the scalar gravity dynamics, by setting $\Phi_g = 0$ everywhere and prescribing a spatially uniform gravitomagnetic field.

Construction. A new analytic background generator `gr_sim_set_background_uniform_gravitomagnetic` installs the symmetric-gauge vector potential

$$\Phi_g^{\rm bg} = 0, \qquad A_{g,x}^{\rm bg}(x,y) = -\tfrac{1}{2}B_0\,(y-y_0), \qquad A_{g,y}^{\rm bg}(x,y) = +\tfrac{1}{2}B_0\,(x-x_0), \tag{245}$$

yielding $B_{g,z} = \partial_x A_{g,y}^{\rm bg} - \partial_y A_{g,x}^{\rm bg} = B_0$ identically. The gauge origin (x_0, y_0) sets the zero of $\mathbf{A}_g$ and is otherwise physically irrelevant.

Analytic prediction. With $\Phi_g = 0$ the force on a test particle reduces to $\mathbf{F} = m(4\,\mathbf{v}\times B_0\hat{z})$; the particle traces a circle (cyclotron orbit) at

$$\omega_{\rm gm} = \frac{4\,|B_0|}{\gamma}, \qquad T = \frac{2\pi}{\omega_{\rm gm}} = \frac{\pi\gamma}{2\,|B_0|}, \qquad r_L = \frac{v_0}{\omega_{\rm gm}} = \frac{v_0\,\gamma}{4\,|B_0|}. \tag{246}$$

For $B_0 > 0$ and initial velocity $\mathbf{v}(0) = v_0\hat{x}$, the initial force is $\mathbf{F}_y = -4mv_0B_0 < 0$, so the particle gyrates clockwise (viewed from $+\hat{z}$). Reversing the sign of B_0 reverses the gyration direction.

Test (stage20_uniform_bg_gyration.c). At $v_0 = 0.1c$, $B_0 = 10^{-3}$ (so $\gamma \approx 1.005$, $\omega_{\rm gm} \approx 3.98 \times 10^{-3}$, $T \approx 1579$, $r_L \approx 25.13$ cells on a 128×128 grid):

- Closure (particle returns to within 1% of r_L of its starting point after T): 0.10% (well under threshold; residual is the $dt \cdot \omega_{\rm gm}$ phase-rounding from integerizing the period in timesteps).
- $|\mathbf{v}|^2$ drift (the Lorentz force does no work): 3×10^{-6}, essentially round-off.
- Measured Larmor radius vs. Eq. (246): 3×10^{-6} fractional error.
- Gyration direction flips under $B_0 \to -B_0$: **verified both signs**.

The Larmor-radius agreement to a part in $\sim 10^6$ directly fixes the coefficient $+4$ on $\mathbf{v}\times\mathbf{B}_g$: an erroneous coefficient α would give $r_L = v_0\gamma/(\alpha|B_0|)$, shifting the measured radius linearly.

Notation note — factor of 1 vs. 4 between Eq. (101) and the Tier table. Eq. (101) is written in the textbook (lecture-25) convention where the gravitomagnetic 4-potential carries coefficient 1 on $\mathbf{v}\times(\nabla\times\mathbf{A}_g)$. The Tier table at Section 12.7, Eq. (102), and the Tier-3 implementation eqbox use the doc's own convention where the same physical quantity carries coefficient 4. The two conventions are related by the identification $\xi_i^{(\rm lecture)} = 4\,A_{g,i}^{(\rm doc)}/c$ (see the provenance discussion immediately after Eq. 105). The factor-of-4 form is consistent with the spin-2 metric $h_{0i} = 4A_{g,i}/c$, the proper-time formula $d\tau/dt \supset -8(\mathbf{A}_g\cdot\mathbf{v})/c^2$, and the implementation; the lecture-1 form is preserved at Eq. (101) only for textual consistency with the cited derivation. All implementations use the factor-of-4 form.

Cross-references. This stage isolates the $\mathcal{O}(v/c)$ Lense–Thirring force piece in a *uniform* $\mathbf{B}_g$; Stage 9c (Section 18.11) extends the verification to a *spatially-varying* $\mathbf{B}_g(r)$ around a spinning point mass. Together with Section 18.9 Phase B (clock effect) they exhaust the leading-order $\mathcal{O}(J_z)$ tests of the GEM coupling.

18.11 Stage 9c: Lense–Thirring orbital period asymmetry

Stage 9b verified the $+4\,\mathbf{v}\times\mathbf{B}_g$ Lorentz piece against a spatially uniform $\mathbf{B}_g$. Stage 9c upgrades the verification to a *spatially varying* $\mathbf{B}_g(r)$ around a spinning point mass — the configuration of physical interest for Kerr-like dynamics — and measures the leading-order Lense–Thirring response: the prograde vs. retrograde orbital-period asymmetry.

Setup. A non-spinning test particle on a circular orbit of radius r around the spinning-point-mass background of Section 18.6. At the orbit's radius the gravitomagnetic field is the equatorial value of the dipole:

$$B_{g,z}(r) \;=\; \frac{G_{\rm eff}\,J_z}{2\,c^2}\,\frac{2\epsilon^2 - r^2}{(r^2+\epsilon^2)^{5/2}} \;\xrightarrow{r\gg\epsilon}\; -\frac{G_{\rm eff}\,J_z}{2\,c^2\,r^3}. \tag{247}$$

The negative sign is the equatorial-plane value of a magnetic-type dipole (antiparallel to the dipole moment by a factor 1/2 of the polar value). The Lorentz force $\mathbf{F} = 4m\,\mathbf{v}\times\mathbf{B}_g$ on a tangential velocity is purely radial: inward for prograde ($\mathbf{v}\times B_{g,z}\hat{z} = +v\,B_{g,z}\,\hat{r}$, with $B_{g,z} < 0$ making this $-|B_{g,z}|\hat{r}$), outward for retrograde.

Analytic prediction. The steady-state circular velocity satisfies the relativistic centripetal balance $\gamma v^2/r = a(r) \pm 4v|B_{g,z}|$ where $a(r) = G_{\rm eff}Mr/(r^2+\epsilon^2)^{3/2}$ and $\gamma = (1-v^2/c^2)^{-1/2}$. Linearising about the relativistic *scalar* circular velocity $v_{\rm rel}$ (defined by $\gamma_{\rm rel}v_{\rm rel}^2 = a(r)r$; the same $v_{\rm circ}$ used in the Stage 7 kepler_orbit scenario) gives

$$\delta v \;=\; \frac{4\,|B_{g,z}|\,r}{\gamma_{\rm rel}\big(\,2+\gamma_{\rm rel}^2\,v_{\rm rel}^2/c^2\,\big)}, \tag{248}$$

so $v_{\rm pro} = v_{\rm rel} + \delta v$ and $v_{\rm ret} = v_{\rm rel} - \delta v$, giving the period asymmetry

$$T_{\rm ret} - T_{\rm pro} \;\approx\; \frac{4\pi r\,\delta v}{v_{\rm rel}^2}. \tag{249}$$

Using the relativistic $v_{\rm rel}$ (not Newtonian) as the linearisation point is essential: the Newtonian $v^2 = a(r)r$ over-predicts v by $\mathcal{O}(v^2/c^2)$ (1.3% at our test parameters), which would put the integrator on a small ellipse rather than a circle and add an $\mathcal{O}(v^2/c^2)$ contamination to the measured period.

Test (`stage21_lense_thirring_period.c`). At $GM = 1$, $r = 20$, $\epsilon = 1$, $J_z = 2$, $c = 1$: $B_{g,z} = -1.236\times10^{-4}$, $v_{\rm rel} = 0.2204$, $\gamma_{\rm rel} = 1.025$, $\delta v = 4.70\times10^{-3}$, analytic $T_{\rm ret} - T_{\rm pro} = +24.33$ (4.27% of $T_{\rm rel} = 570.09$). Measured over four orbits:

- Prograde: $T_{\rm pro} = 557.55$ ($r \in [19.98, 20.00]$).
- Retrograde: $T_{\rm ret} = 582.13$ ($r \in [19.99, 20.00]$).
- $T_{\rm ret} - T_{\rm pro} = +24.57$, agreeing with Eq. (249) to **0.97%**.
- Baseline (GM force disabled): prograde and retrograde periods agree at 570.10 vs. 569.93, matching the relativistic scalar prediction; the residual is the dt-rounded period-detection error.

The orbits stay nearly perfectly circular ($\Delta r/r \lesssim 10^{-3}$), confirming that the perturbative δv correction Eq. (248) captures the steady-state circular condition in the GEM-corrected dynamics.

Interpretation. This is the leading-order Lense–Thirring *orbital* response: the spinning background's frame-dragging torques prograde vs. retrograde orbits differently, shifting their steady-state radii (and hence their periods) by $\mathcal{O}(J_z/(Mc^2))$. It is distinct from (i) the proper-time clock effect of Section 18.9 Phase B, which is the $\mathcal{O}(J_z)$ contribution to $\mathrm{d}\tau/\mathrm{d}t$ at fixed orbit, and (ii) the Schwarzschild perihelion precession from the Tier-2 EIH terms, which is direction-symmetric. The three signals — clock effect, GM Lorentz force on the orbit, EIH precession — are independent at $\mathcal{O}(J_z)$ and $\mathcal{O}((v/c)^2)$, and the verification stages isolate each in turn.

18.12 Stage 9d: Sampled-mode curl of A_g

Stages 9b and 9c verified the gravitomagnetic Lorentz force with *analytic-mode* B_g — closed-form curls of the uniform and spinning-point-mass backgrounds. Stage 9d extends the verification to the *sampled-mode* path: a 2-point Yee curl of the X_EDGE and Y_EDGE potential arrays. This is the machinery the pusher will use once moving masses generate a *self-consistent* perturbation $\mathbf{A}_g^{\rm pert}$ that has no closed form — the same code path stages 9b/9c exercise indirectly via the background route.

Kernel. $B_{g,z}$ lives naturally on the cell-center sublattice $(i+\frac12, j+\frac12)\Delta x$. The discrete curl is

$$B_{g,z}\big[i+\tfrac12, j+\tfrac12\big] = \frac{A_{g,y}[i+1,j] - A_{g,y}[i,j]}{\Delta x} - \frac{A_{g,x}[i,j+1] - A_{g,x}[i,j]}{\Delta x}, \tag{250}$$

using the X_EDGE indexing of $A_{g,x}$ (sample at $(i+\frac12, j)\Delta x$) and the Y_EDGE indexing of $A_{g,y}$ (sample at $(i, j+\frac12)\Delta x$). The particle's $B_{g,z}$ is the bilinear interpolation of the four surrounding cell-center values. The total $B_{g,z}$ at the particle is the sum of (i) the background contribution (analytic via `gr_bg_eval_B_g` when `bg_mode` is ANALYTIC, sampled via Eq. 250 when SAMPLED), and (ii) the perturbation contribution (always sampled, reading the `.prev` buffer when field evolution is active to match the deposit half-step time — the same read-prev convention as for Φ_g in Stage 18's energy-conservation fix).

Test (`stage22_sampled_bg_curl.c`).

- *Phase A — uniform* $\mathbf{B}_g$*:* Symmetric-gauge $\mathbf{A}_g$ is linear in (x, y), so the 2-point Yee forward difference is *exact*. The sampled $B_{g,z}$ matches B_0 at every interior sample point to floating-point round-off ($\sim 10^{-10}$ relative).
- *Phase B — dipole* $\mathbf{B}_g(r)$*:* At $J_z = 2$, $\epsilon = 1$, $\Delta x = 1$, the relative error against the analytic curl is 0.62% at $r = 20$, 0.16% at $r = 40$, 0.04% at $r = 80$. Convergence ratios: 3.97× and 4.00× — exactly the expected 2nd-order behaviour (err $\propto (\Delta x/r)^2$).
- *Phase C — Stage 9b cyclotron repeated with sampled* $\mathbf{B}_g$*:* Larmor-radius error is 5×10^{-6}, identical to the analytic-mode result — linear $\mathbf{A}_g$ gives bit-clean agreement.

Implication. The same curl kernel will compute $B_{g,z}$ from the perturbation $\mathbf{A}_g^{\text{pert}}$ that arises when a moving mass deposits the GEM current $\mathbf{J}_m$ and the wave equation propagates the response. Stage 22 confirms that path is correct *before* the dynamics get self-consistent, so any later discrepancy at the PIC level can be localized to deposition, smoothing, or the wave-equation evolution, rather than to the curl evaluator itself.

18.13 Stage 9e: EM Lorentz force — unit isolation

EM analog of Stage 9b/9d. The Tier-3 force-law eqbox of Section 12.7 contains the EM Lorentz piece

$$\mathbf{F}_{\text{em}} = q\big(-\nabla\phi - \partial_t\mathbf{A} + \mathbf{v}\times\mathbf{B}\big), \qquad \mathbf{B} = \nabla\times\mathbf{A}, \tag{251}$$

with coefficient $+1$ on $\mathbf{v}\times\mathbf{B}$ — the EM analogue of the GEM coefficient $+4$, but without the spin-2 enhancement. Until the scalar $-q\nabla\phi$ and inductive $-q\partial_t\mathbf{A}$ pieces land (planned with the static-Coulomb stage), the implementation realizes the $q\,\mathbf{v}\times\mathbf{B}$ piece alone. Stage 9e verifies that piece in isolation, mirroring Stage 9b but for the EM sector.

Construction. The new analytic background generator `gr_sim_set_background_uniform_magnetic` installs the same symmetric-gauge pattern as the GEM case, applied to the EM potential arrays:

$$\phi^{\text{bg}} = 0, \qquad A_x^{\text{bg}}(x, y) = -\tfrac{1}{2}B_0\,(y - y_0), \qquad A_y^{\text{bg}}(x, y) = +\tfrac{1}{2}B_0\,(x - x_0), \tag{252}$$

producing $B_z = B_0$ everywhere. The same Yee-curl evaluator introduced in Stage 9d serves the EM side identically — the X_EDGE / Y_EDGE staggering of A_x, A_y is structurally identical to the GEM case, so the curl kernel in `particle.c:curl_Agz_sampled_at` is shared between both.

Analytic prediction. With $\phi = 0$ and $\partial_t\mathbf{A} = 0$ the force reduces to $\mathbf{F} = q\,(\mathbf{v}\times B_0\hat{z})$, giving the standard EM cyclotron:

$$\omega_c = \frac{q\,|B_0|}{\gamma\,m}, \qquad T = \frac{2\pi}{\omega_c}, \qquad r_L = \frac{v_0}{\omega_c} = \frac{\gamma\,m\,v_0}{q\,|B_0|}. \tag{253}$$

For $q > 0$, $B_0 > 0$, and $\mathbf{v}(0) = v_0\hat{x}$, the particle gyrates clockwise (viewed from $+\hat{z}$); sign reversal of either q or B_0 flips the direction.

Test (`stage23_em_cyclotron.c`). At $q = 1$, $m = 1$, $v_0 = 0.1c$, $B_0 = 4 \times 10^{-3}$ (chosen so $r_L \approx 25$ cells, matching the Stage 9b/9d orbit geometry on a 128×128 grid):

- ANALYTIC bg: r_L error 3×10^{-6}, $|\mathbf{v}|^2$ drift 3×10^{-6}, closure 0.10% (the same $dt \cdot \omega_c$ phase-rounding floor as Stage 9b). Gyration direction is correct.
- SAMPLED bg: r_L error 5×10^{-6}, agreeing with ANALYTIC to $\sim 5 \times 10^{-6}$ in absolute r_L. Linear $\mathbf{A}^{\text{bg}}$ -¿ Yee curl is exact, as Stage 9d showed.

- Sign reversal: $B_0 \to -B_0$ flips gyration from CW to CCW correctly.
- Neutral particle ($q = 0$): feels no force (drifts linearly), confirming the EM Lorentz piece is correctly gated by charge.

Status of the rest of $\mathbf{F}_{\rm em}$. Eq. (251) has three pieces. Stage 9e covers the $q\,\mathbf{v}\times\mathbf{B}$ term. Stage 9f (Section 18.14) adds the $-q\nabla\phi$ scalar (electrostatic) term. Only the $-q\,\partial_t\mathbf{A}$ inductive piece remains unwired; it will land with the radiative-EM stage (where a time-varying perturbation $\mathbf{A}$ is the primary observable).

18.14 Stage 9f: Electrostatic force — unit isolation

Wires the $-q\nabla\phi$ piece of Eq. (251) into the pusher. Stage 9e covers $q\mathbf{v}\times\mathbf{B}$; Stage 9f covers $-q\nabla\phi$; together they handle the full EM Lorentz force *for static* $\mathbf{A}$ *backgrounds* (the $-q\,\partial_t\mathbf{A}$ inductive piece is still pending and only matters when $\mathbf{A}$ is time-varying).

Construction. A new analytic background generator `gr_sim_set_background_uniform_electric` installs the linear EM scalar potential

$$\phi^{\rm bg}(x,y) = -\big(E_x(x-x_0) + E_y(y-y_0)\big), \tag{254}$$

producing $-\nabla\phi^{\rm bg} = (E_x, E_y)$ uniformly. No $\mathbf{A}^{\rm bg}$ is installed; the magnetic Lorentz piece contributes nothing.

In the pusher, the gradient $\nabla\phi_{\rm em}^{\rm total}$ at the particle is computed by `particle.c:phi_em_grad_at_total`, which mirrors the grav_grad_at structure exactly: analytic background via `gr_bg_eval_phi_em` (when `bg_mode` is ANALYTIC), centered finite-difference of $\phi^{\rm bg}$ on the CORNER sublattice plus CIC interpolation (when SAMPLED), and the perturbation contribution from `fields[PHI_EM]` read with the same `.prev` half-step convention as Φ_g. The TSC and Lewis–Birdsall variants of the EM gradient kernel are not yet ported; they will mirror the GEM-side (Section 18.17) when self-consistent PIC dynamics arrive on the EM side.

Analytic prediction. With a uniform $\mathbf{E} = E_0\hat{x}$ and no $\mathbf{B}$ or scalar gravity, a particle of charge q and rest mass m starting at rest follows

$$p_x(t) = qE_0\,t, \qquad p_y(t) = 0, \tag{255}$$

$$x(t) - x(0) = \frac{mc^2}{qE_0}\left[\sqrt{1 + \left(\frac{qE_0 t}{mc}\right)^2} - 1\right], \tag{256}$$

the exact relativistic kinematic integral. The Boris half-step convention stores $p^{N-1/2}$ at simulation step N, so the recorded momentum at step N is $p_x^{N-1/2} = qE_0(N - \frac{1}{2})\Delta t$.

Test (`stage24_uniform_e_field.c`). At $q = 1$, $m = 1$, $E_0 = 10^{-3}$, $c = 1$, $\Delta t = 1/\sqrt{2}$, $N = 200$:

- ANALYTIC bg: p_x matches Eq. (255) to a relative error of 1.1×10^{-6} (float32 precision floor of the kinematic integral); $p_y = 0$ exactly; position drift matches the exact relativistic formula to 2.5×10^{-6}.
- SAMPLED bg: p_x *bit-identical* to ANALYTIC mode (linear $\phi^{\rm bg} \Rightarrow$ centered FD is exact, as established by Stage 9d for the GEM curl analogue).
- Sign reversal: $|p_x(E_0 > 0) + p_x(E_0 < 0)| < 10^{-8}$ — exact antisymmetry to round-off.
- Neutral particle ($q = 0$): zero momentum and zero drift after 200 steps, confirming the electrostatic force is correctly gated by charge.

18.15 Stage 9g: Inductive force — unit isolation

Wires the $-q\,\partial_t\mathbf{A}$ piece of Eq. (251) into the pusher. With Stages 9e ($q\,\mathbf{v}\times\mathbf{B}$) and 9f ($-q\nabla\phi$) already in place, Stage 9g completes the full EM Lorentz force

$$\mathbf{F}_{\rm em} = q\big(-\nabla\phi - \partial_t\mathbf{A} + \mathbf{v}\times\mathbf{B}\big)$$

on the pusher side.

Temporal derivative on the leapfrog buffer rotation. The field leapfrog in `sim.c:gr_sim_step` cycles three time buffers per field. After each step's rotation,

$$\texttt{fields[*].prev} = \mathbf{A}^n, \qquad \texttt{fields[*].curr} = \mathbf{A}^{n+1}, \qquad \texttt{fields[*].next} = \mathbf{A}^{n-1}.$$

The recycled `.next` slot (which the next leapfrog will overwrite) therefore holds the *previous* time slice. This gives a clean 2nd-order centered difference at t^n:

$$\partial_t\mathbf{A}^n \;=\; \frac{\mathbf{A}^{n+1}-\mathbf{A}^{n-1}}{2\,\Delta t} \;=\; \frac{\texttt{curr}-\texttt{next}}{2\,\Delta t}, \tag{257}$$

evaluated at the same time as $\mathbf{B}^n = \nabla\times\mathbf{A}^n$ (`.prev`) and $\Phi^n_{\rm em}$ (`.prev`). Spatial interpolation uses the same sublattice-aware kernels that serve the magnetic curl path: $\mathbf{A}_x$ on X_EDGE, $\mathbf{A}_y$ on Y_EDGE.

Test (`stage25_inductive_e.c`). The cleanest isolation manually pre-populates the time buffers with a known temporal profile. With `field_evolution` OFF (so the buffers don't rotate), set

$$\texttt{fields[A\_X].curr}[k] = +A_0\,\Delta t, \qquad \texttt{fields[A\_X].next}[k] = -A_0\,\Delta t,$$

uniformly over the grid. The centered difference is then A_0 at every position, mimicking a linear-in-time $A_x(t) = A_0\,t$. No scalar EM background, no magnetic field, so the only force is the inductive piece: $F_x = -q\,\partial_t A_x = -qA_0$. A charged particle at rest then accelerates uniformly along $-x$ at $|qA_0|/(\gamma m)$; under the Boris half-step convention $p_x^{N-1/2} = -qA_0(N-\frac{1}{2})\Delta t$. This is exactly the kinematics of Stage 9f (electrostatic, $-q\nabla\phi$), but driven by the inductive force-law piece instead.

Results at $q=1$, $m=1$, $A_0=10^{-3}$, $N=200$:

- p_x matches the analytic prediction to 1.1×10^{-6} relative (the float32 precision floor, identical to Stage 9f's measurement at the same kinematic point).
- $p_y = 0$ exactly (no transverse force).
- Sign reversal $A_0 \to -A_0$: $|p_x(+) + p_x(-)| < 10^{-8}$ — exact antisymmetry to round-off.
- Neutral particle ($q=0$): zero force, as expected.

Status of the EM Lorentz force after Stages 9e/9f/9g.

Piece of $\mathbf{F}_{\rm em}$	Implemented	Verified by	Evaluator
$+q\,\mathbf{v}\times\mathbf{B}$ (magnetic Lorentz)	yes	Stage 9e	`B_em_z_at_total`
$-q\nabla\phi$ (electrostatic)	yes	Stage 9f	`phi_em_grad_at_total`
$-q\,\partial_t\mathbf{A}$ (inductive)	yes	Stage 9g	`dt_A_em_at_total`

All three pieces share the same three-path structure: ANALYTIC background (closed form via gr_bg_eval_*_em), SAMPLED background (Yee curl or centered FD of the *_bg arrays), and perturbation (always sampled, using the leapfrog buffer rotation for the centered-time-difference in the inductive case). A charged particle in any combination of these three sources — static analytic background, sampled-grid background, and a time-evolving PIC-deposited perturbation — now feels the complete EM Lorentz force. The closed PIC loop (*moving charge* $\to$ *Esirkepov current* $\to$ *wave-equation* $\to$ $\mathbf{A}^{\text{pert}}$, ϕ^{pert} $\to$ *Lorentz force on a probe charge*) is now end-to-end functional and yields a CFL-convergent physical radiation-reaction inspiral (Section 18.16, with CFL convergence verified in Section 18.16.1). Closed-loop stability required reinterpreting the $\mathbf{A}$ buffers as half-integer time slices (the v36 architecture fix); the wave-equation code itself did not change.

18.16 Stage 9h: EM Kepler analog and PIC orbit (closed-loop)

The EM analog of Stage 10 (Section 18.17): a charged test particle orbits a softened analytic point-charge background, with the perturbation EM fields evolving via the wave equation under sources deposited by the particle itself. This is the closed-loop PIC verification on the EM side, exposing any feedback instabilities in the combined deposit +wave-equation +force-interp chain.

EM Kepler baseline (Stage 26). Before the closed loop, a basic verification: charged particle in an analytic softened-Coulomb background $\phi^{\text{bg}} = +k_e Q/\sqrt{r^2+\epsilon^2}$ with opposite-sign Q and q_{test} (attractive). With $|q_{\text{test}}Q|k_e/m_{\text{test}} = G_{\text{eff}}M$, the dynamics are mathematically identical to gravity's Stage 7 Kepler test. The implementation reproduces the analytic orbital period to 0.03% and the orbit stays circular to 4×10^{-5} over one period (`stage26_em_kepler.c`). This verifies the EM force chain POINT_CHARGE BG $\to$ `gr_bg_eval_phi_em` $\to$ `phi_em_grad_at_total` $\to$ `em_force_at` on a closed-orbit observable, without any perturbation FDTD.

Closed-loop PIC orbit (Stage 27). Same setup, but `field_evolution` and `particle_source_deposition` both on. At $Q=+1$, $q_{\text{test}} = -10^{-3}$, $m_{\text{test}} = 10^{-3}$ (same coupling as gravity's Phase E sweet spot), $r=20$, CIC+LEGACY+smoothing = 4, over 4 orbital periods, the radius drifts:

Variant	Drift after 4 orbits	Mechanism
Baseline (`field_evolution` OFF)	+0.003%	— (analytic bg only)
PIC, all three EM Lorentz pieces, centered $\partial_t\mathbf{A}$	**+18.6%**	PIC heating dominates
PIC, inductive ($-q\,\partial_t\mathbf{A}$) disabled	**−7.0%**	radiation-reaction-style inspiral
PIC, backward-diff $\partial_t\mathbf{A}$	**unbound by orbit 4**	worse heating (see below)

The classical-atom-collapse inspiral is present in this sandbox when the inductive piece is disabled. The −7.0%/4 orbits inward drift (no inductive) is the same order as gravity's Phase E result at the same coupling (−2.4% per orbit), consistent with radiation-reaction physics: an orbiting charge radiates EM waves into the absorbing boundary, loses orbital energy, and spirals in.

But *with the inductive piece enabled, PIC numerical heating overwhelms radiation reaction by an order of magnitude*, driving an outward +18.6%/4 orbits instead of an inspiral. The mechanism is specific to the inductive piece: the centered-time derivative

$$\partial_t\mathbf{A}^n = (\mathbf{A}^{n+1} - \mathbf{A}^{n-1})/(2\,\Delta t)$$

reads $\mathbf{A}^{n+1}$, which contains the particle's own current deposit at this step. Although the time-CENTER is at t^n (matching the Stage 18 read-prev fix's timing principle for the spatial gradient), the discrete propagation of the self-field into $\mathbf{A}^{n+1}$ appears to inject a net positive-feedback torque on the orbit.

Investigation and resolution. The initial closed-loop result (with all three EM Lorentz pieces enabled, A stored at integer time) showed +18.6% outward drift over 4 orbits at $q = m = 10^{-3}$ — runaway PIC heating, not the expected radiation-reaction inspiral. A symmetric gravity-side diagnostic (Stage 28, wiring the analogous $-m\,\partial_t \mathbf{A}_g$ piece into the gravity Lorentz force) exhibited the SAME heating (+23%/4 orbits), confirming that the phenomenon was structural to the discretization rather than EM-specific. Three further diagnostic experiments — sign-flip on the inductive term, backward-difference discretization, and source-time correction $J^{n-1/2} \to J^n$ via linear extrapolation — all merely rotated the direction of the closed-loop bias without taming its magnitude, ruling out simple sign-convention or single-step source-timing bugs.

The resolution was a structural insight: in the integer-time storage convention, $\partial_t \mathbf{A}^n = (\mathbf{A}^{n+1} - \mathbf{A}^{n-1})/(2\Delta t)$ spans a 2-step stencil and necessarily reads $\mathbf{A}^{n+1}$, which contains the particle's own current-step deposit (asymmetric self- deposit visibility relative to the spatial-derivative reads at $\mathbf{A}^n$). Furthermore $J^{n-1/2}$ from Esirkepov is at a half-step offset from the operator's integer-time-center, an independent but related inconsistency.

Half-step A architecture (v36 fix). Reinterpret the **A** buffers (both $\mathbf{A}_g$ and $\mathbf{A}_{\rm em}$) as living at half-integer times: `.prev` $= \mathbf{A}^{n-1/2}$, `.curr` $= \mathbf{A}^{n+1/2}$. The wave-equation leapfrog code does not change — only the time-interpretation shifts. This collapses BOTH issues at once:

- $J^{n-1/2}$ from Esirkepov now naturally matches $\mathbf{A}^{n-1/2}$ in the source/operator pairing of the wave-equation update for $\mathbf{A}^{n+1/2}$.
- The pusher's $\partial_t \mathbf{A}^n$ becomes (`.curr` − `.prev`)$/\Delta t = (\mathbf{A}^{n+1/2} - \mathbf{A}^{n-1/2})/\Delta t$ — a tight 1-step centered difference at t^n, with no 2-step asymmetric self-deposit reach.
- The pusher's $\mathbf{B}^n$ becomes the average of curl on the two surrounding half-steps: $\mathbf{B}^n \approx \frac{1}{2}(\nabla\times\mathbf{A}^{n+1/2} + \nabla\times\mathbf{A}^{n-1/2})$.

The scalar ϕ stays at integer time (sourced by ρ at integer time, no inconsistency). This mirrors standard Yee+Boris EM PIC's half-step interleaving of **E** and **B**, transposed to the potentials.

Result after the half-step fix. Rerunning Stage 27 (EM) and Stage 28 (gravity, with the GEM inductive piece enabled):

Variant	**Pre-fix (integer A)**	**Post-fix (half-step A)**
EM, no inductive	−7.0%	−6.9% (unchanged)
EM, full Lorentz	+18.6%	**−9.2%** (inspiral!)
Gravity, no inductive	−0.59%	−0.54% (unchanged)
Gravity, full Tier-3	+23.0%	**−9.1%** (inspiral!)

The catastrophic heating is gone in both sectors. Both sectors show matching inspiral rates ($\sim -9\%$ per 4 orbits, $\approx -2.3\%$ per orbit), consistent with a 2D radiation-reaction signal. Spatial-piece behaviour is unchanged (the static-background reads are bit-clean because averaging two identical static buffers gives the same value).

18.16.1 CFL convergence (Stages 29 and 30)

The half-step fix yields an inspiral, but is the inspiral physical radiation reaction or residual numerical truncation? The test is a CFL convergence sweep: a physical effect should be CFL-independent at small Δt, while a numerical truncation residual should scale as $\mathcal{O}(\Delta t^2)$ (ratio 4 between consecutive CFL halvings).

Gravity (Stage 29) at $m_{\text{test}} = 10^{-3}$, 4 orbits:

CFL	No-inductive drift	Inductive-on drift	Contribution
$1/\sqrt{2} \approx 0.707$	−0.535%	−9.05%	−8.52%
$1/(2\sqrt{2}) \approx 0.354$	−0.391%	−18.29%	−17.90%
$1/(4\sqrt{2}) \approx 0.177$	−0.276%	−19.07%	−18.80%
$1/(8\sqrt{2}) \approx 0.088$	−0.175%	−19.16%	−18.99%

Consecutive-CFL contribution ratios: 0.476, 0.952, 0.990. The ratio approaches 1.0 (CFL-independent plateau) as $\Delta t \to 0$ — the signature of a *physical* effect. The inductive contribution converges to $\sim -19\%/4\text{orbits} \approx -4.75\%$ per orbit.

EM (Stage 30) at $q_{\text{test}} = -10^{-3}$, $m = 10^{-3}$, 4 orbits:

CFL	No-inductive drift	Inductive-on drift	Contribution
$1/\sqrt{2} \approx 0.707$	−6.93%	−9.19%	−2.25%
$1/(2\sqrt{2}) \approx 0.354$	−5.97%	−19.15%	−13.18%
$1/(4\sqrt{2}) \approx 0.177$	−5.40%	−19.73%	−14.33%
$1/(8\sqrt{2}) \approx 0.088$	−4.97%	−20.47%	−15.50%

EM contribution ratios: 0.171, 0.919, 0.925. Same qualitative pattern — ratio approaches 1.0, plateau emerges. The EM no-inductive drift was larger and slower-to-converge than gravity's in this (pre-v36) table, an asymmetry that was tracked down to two distinct issues:

- The EM scalar gradient lacked TSC/LB variants. v36 ports these from `grav_grad_at` onto `phi_em_grad_at_total`; Stage 33 confirms the four (shape × interp) combinations give consistent drift post-fix.
- More fundamentally, the EM source-coupling sign was inverted relative to standard Maxwell (`c_em` was negative, like gravity's $-4\pi G$, instead of $+4\pi k_e$). v36 fixes this; the EM PDE now reads $\Box\phi = -\rho/\epsilon_0$ and $\Box\mathbf{A} = -\mathbf{J}/(\epsilon_0 c^2)$ (standard Maxwell). Stage 34 verifies the full chain end-to-end: Poisson sign, $\mathbf{J} = \rho\mathbf{v}$ deposition, $\mathbf{A}$ wave-equation sign, and Lorentz force direction (including like-repel + opposite-attract).

Open item – inductive-piece sign mismatch. The post-fix inductive contribution at the same CFL sweep reads +3.5%, +23.4%, +25.4%, +22.3% (drift, not contribution-only) — an *outspiral* that is also CFL-converged (ratio approaching 1.0). Pre-v36 the same setup gave inspiral. The sign flip on the field convention also flips the sign of the inductive-piece self-coupling, reversing the discrete-PDE radiation-reaction direction. Energy conservation demands inspiral for a bound orbit, so the outspiral indicates a sign error in the inductive force-law term relative to the new field convention.

Diagnostic chain (Stage 34–36). The v36 sign convention is verified end-to-end on the static EM machinery: Poisson sign, $\mathbf{J} = \rho\mathbf{v}$ deposition, $\mathbf{A}$ wave-equation sign, all four Lorentz-force direction tests pass standard Maxwell (Section **??**, Stage 34). The pure $q\mathbf{v}\times\mathbf{B}$ piece gives Ampère attraction for parallel like-currents (Stage 35) with the expected $(v/c)^2$ scaling. When the inductive piece is OFF, the EM Kepler orbit is essentially stable ($\sim +0.3\%$ drift). Stage 36 confirms that flipping the sign on $-q\partial_t\mathbf{A}$ (via the diagnostic `gr_sim_set_em_inductive_sign(-1)`) turns outspiral into inspiral:

inductive_sign	drift over 4 orbits
+1 (default standard Lorentz)	+3.73% (*outspiral*)
−1 (flipped)	−3.45% (*inspiral*)

So the surgical fix is a one-sign change on the $-q\partial_t \mathbf{A}$ term in `em_force_at`. The deeper question — why the discrete PIC self-force interaction needs the flipped inductive sign in the standard-Maxwell field convention (but not in the gravity-style anti-Maxwell convention used pre-v36) — is documented in `memory/grlite-em-inductive-sign-open.md` as the next investigation step.

Interpretation. The inductive piece's inspiral is a genuine physical radiation reaction in the 2D-softened-Newton+GEM toy system. The magnitude ($\sim 5\%$ per orbit at $v/c \approx 0.22$) is much larger than a 3D Larmor estimate ($\sim 0.01\%$ per orbit at the same parameters) because:

- The 2D wave equation has a logarithmic Green's function vs. $1/r$ in 3D; radiation scaling with source motion differs fundamentally.
- Gravity is softened-Newtonian here, not full GR.
- The toy is self-consistent for the equations it actually solves; those equations are not Maxwell-3D, so quantitative agreement with 3D Larmor is not expected.

The pedagogical conclusion holds: when the full force law is active, the orbit slowly spirals in. The classical-atom-collapse picture is faithfully demonstrated in this sandbox, with a CFL-convergent inspiral rate that emerges as a real prediction of the discrete PDE.

18.16.2 Stage 31: Shapiro delay in the EM wave equation

The doc's headline EM physics addition: light slows in a gravitational potential, as derived from the null geodesic condition (Section 12.3). The implementation replaces the uniform wave speed c in the EM-field leapfrog with a per-cell $c_{\text{local}}(x) = c(1 + 2\Phi_g(x)/c^2)$ on the Laplacian term; the source-coupling term keeps the bare c, since Shapiro modifies free propagation rather than source coupling.

Only the three EM fields (ϕ, A_x, A_y) are affected; the GEM fields keep the uniform c (gravity propagates at c regardless of background gravity, by construction of the toy). Implementation: three per-sublattice arrays c^2_{local} on CORNER, X-EDGE, Y-EDGE are precomputed at enable-time from the installed Φ_g background, sampled at each EM-field's own Yee node positions.

Verification (Stage 31). An EM pulse (Gaussian initial condition on ϕ with zero initial time derivative, $\sigma = 4$ cells, amplitude 1) is launched at $(c_x - L/2,\ c_y)$ for $L = 120$ cells, with a softened point-mass background ($GM = 0.5$, $\epsilon = 4$) at the grid center. The arrival of the wavefront at the diametrically opposite probe is timed via parabolic-interpolated peak amplitude.

	No mass (baseline)	**With mass (Shapiro)**
Peak $\lvert\phi\rvert$	0.0652	0.0802
Arrival time	117.84	124.10

Measured Shapiro delay: $\Delta t_{\text{meas}} = +6.26$. Analytic prediction (Eq. 100 integrated along the axial ray through a softened point mass at impact parameter $b = 0$):

$$\Delta t_{\text{ana}} = \frac{4GM}{c^3}\,\text{asinh}\left(\frac{L}{2\epsilon}\right) = +6.80. \tag{258}$$

Agreement within $\sim 8\%$ — well within the pedagogical tolerance for a 2D-radial wavefront whose path samples a band of Φ_g values, not strictly the axial ray. The amplitude focusing ($0.065 \to 0.080$) is the real signature of gravitational lensing: c_{local} is smaller near the central well, bunching the wavefront on the slow side of the obstacle. See `core/tests/stage31_shapiro_delay.c`.

Stability. For attractive $\Phi_g < 0$ backgrounds, $c_{\text{local}} < c$ everywhere, so the CFL bound stays at $c\,\Delta t/\Delta x \le 1/\sqrt{2}$ (2D, 5-point Laplacian). A defensive floor clamps $1 + 2\Phi_g/c^2 \ge 10^{-3}$ to guard against pathological strong-field probes outside the weak-field regime in which the linearized c_{local} formula is valid; for the Stage 31 configuration $\min(1 + 2\Phi_g/c^2) \approx 0.75$, far from the floor.

18.16.3 Stage 32: EM binary PIC – two-body mutual EM dynamics

Two charged particles mutually orbit via the FDTD perturbation field alone (no analytic background). Each deposits (ρ_q, J_{qx}, J_{qy}) via Esirkepov; each feels the other's ϕ_{em} and $\mathbf{A}_{\text{em}}$ perturbation through the full EM Lorentz force. This is the EM analog of the Stage 12 gravity binary; together with Stage 27 (single test charge in a softened-charge background) it exercises the entire EM Lorentz machinery end-to-end with two active sources.

Sign convention (v36 fix). The original code shared a single source-coupling sign between gravity and EM ($\text{sc}_{em} = -4\pi k_e$, matching $\text{sc}_{\text{grav}} = -4\pi G$). For gravity this is the standard Newton Poisson sign (positive masses attract); but for EM it produced an "anti-Maxwell" channel in which PIC same-sign charges attracted. The analytic point-charge background hand-coded with the standard $+k_e Q/r$ shape was inconsistent with the PIC sign, an asymmetry that didn't matter for Stages 23–27 (which use analytic backgrounds and tiny PIC perturbations) but did show up once Stage 32 promoted both bodies to active PIC sources. v36 flips sc_{em} to $+4\pi k_e$ (and the corresponding A_{em} coupling), so the PIC channel now obeys standard Maxwell: *like charges repel, opposite charges attract.* The analytic $1/r$ background remains standard-EM-signed and is now sign-consistent with the PIC channel; the two paths still differ in spatial shape ($1/r$ vs. box-bounded 2D log), so they should not be mixed in the same simulation. Stage 32 uses opposite-sign charges ($q_1 = +Q$, $q_2 = -Q$) for the attractive orbit.

Centripetal balance. Identical structure to the gravity binary with $G_{\text{eff}} m \to k_e Q^2$ and v_{orb} independent of r:

$$v_{\text{orb}} = Q\sqrt{k_e/m}, \qquad T = \frac{2\pi r}{v_{\text{orb}}}. \tag{259}$$

Verification (Stage 32). Three checks mirror Stage 12:

- *Free-fall:* two opposite-sign charges released at rest attract, falling together from $r = 8$ to $r_{\min} \approx 0$ within 100 steps. Mutual attraction has the right sign. Esirkepov violations = 0 throughout.
- *Short-horizon orbit:* at $r = 8$, $v_{\text{orb}} = 0.1$, the empirical ω matches v_{orb}/r to $\lesssim 5\%$ in the first wave-crossing time.
- v_{orb} *independent of* r*:* at $r = 6$ and $r = 12$, $v_{\text{meas}} = r\omega$ values match each other and the analytic $Q\sqrt{k_e/m}$ to within a few percent — the 2D-log signature on the EM channel.

Stage 32b: sign-convention verification. Three direct sign tests on the PIC EM channel:

- Two like-sign charges at rest *repel* (Coulomb sign).
- Two opposite-sign charges at rest *attract.*
- ϕ_{em} from a single $+Q$ source is positive and decreases monotonically with distance (the expected box-bounded 2D Green's function shape).

The clean Ampère PIC-dynamics test (parallel like-currents attract via $\mathbf{v} \times \mathbf{B}$ at $\mathcal{O}((v/c)^2)$) is deferred: at the $v/c \gtrsim 0.1$ needed to see a measurable magnetic effect, the moving-source

PIC self-heating (Cherenkov-like artifacts at the discrete grid scale) dominates over the small Ampère correction. The $\mathbf{v} \times \mathbf{B}$ *force law* is independently verified in Stage 23 (cyclotron in the analytic uniform-B background).

Long-time orbit stability is left for a future test, with the same caveats as Stage 12: sub-cell PIC heating and the inductive radiation-reaction inspiral act on overlapping timescales here; teasing them apart will need either a weak-coupling regime ($v/c \ll 1\%$) or a CFL sweep analogous to Stages 29/30. Stage 32 is the basic end-to-end EM binary validity check. See `core/scenarios/pic_binary_em.c` and `core/tests/stage32_em_binary_pic.c`.

18.17 Stage 10: Perturbation FDTD with moving particle source

Enable the full perturbation FDTD: the particle deposits ρ_{matter} and $\mathbf{J}_{\text{matter}}$ onto the grid each step (new `gr_sim_set_particle_source_deposition` flag, set on by the Stage 10 scenarios), the wave equations evolve the perturbation potentials, and the particle feels both background and perturbation forces via the same force evaluator used in Stages 7–9.

Weak-coupling regime

A subtle issue with this stage: the Hockney–Eastwood adjoint condition guarantees zero self-force for a *stationary* particle in the statically-converged field, but for a moving particle the residual self-force is a finite mixture of physical radiation reaction and numerical grid-heating. At the Stage 7/8 setup ($G_{\text{eff}}M = 1$, $r = 20$, $m_{\text{test}} = 1$), the grid-heating term is strong enough to unbind a circular orbit on a timescale much shorter than the orbital period.

The doc's Stage 10 specification implicitly assumes the weak-coupling regime $G_{\text{eff}}m_{\text{test}}/(c^2 r) \ll 1$, where both the radiation-reaction and the grid-heating contributions are proportional to m_{test} and become negligible at low m_{test}. In the implementation, the `pic_orbiting` scenario defaults to $m_{\text{test}} = 10^{-3}$ for the orbiter; the Stage 10 test uses 10^{-6} for the orbit-regression phase. This keeps the dynamics in the test-particle limit while still exercising the deposition path.

Test (single-particle suite)

A second particle is not needed to verify the deposition+FDTD chain; four phases on a single particle suffice:

- *(A) Static-source field shape.* Stationary particle in vacuum. Time-averaged $\Phi_g^{\text{pert}}(r)$ should follow the 2D Green's function $\Phi_g = 2G_{\text{eff}}m \ln r + C$ (the slope of Φ_g vs $\ln r$ equals $2G_{\text{eff}}m$). Parallel to Stage 3, but exercising the auto-deposit path.
- *(B) Boosted-Coulomb shape.* Particle moving at constant v in vacuum. $(\mathbf{A}_g)_x^{\text{pert}}$ and Φ_g^{pert} at the same lab points should have the same sign and the same Green's-function support. The 3D-style leading-order ratio $|\mathbf{A}_g|/|\Phi_g| \approx v/c^2$ does *not* hold instantaneously in 2D because of the non-local retardation structure of the 2D wave kernel and the fact that the auto-deposit moves with the particle. Stage 5 already validates vector-potential generation from a moving source via the curl/B_z test; this phase is informational.
- *(C) Self-force cancellation.* Stationary particle in vacuum, 10^4 steps. The particle's position drift should be $\ll 1$ cell. This is the Hockney–Eastwood theorem operating: with CIC deposition paired with CIC-of-FD force interpolation, a static particle exerts zero net force on itself in the statically-converged field.
- *(E) Orbit regression.* Particle in a circular orbit around an analytic softened-point-mass background (Stage 7 setup) with the perturbation FDTD on and the test-particle mass in the weak-coupling regime. Orbital period should match Stage 7's result to within $\sim 1\%$;

the residual is the sum of the (physical) radiation-reaction shift and the (small, $\propto m_{\text{test}}$) numerical heating. In the implementation at $m_{\text{test}} = 10^{-6}$ the rel. diff is $\approx 0.1\%$.

Outgoing-wave 1/√r falloff (deferred)

The original v33 spec for this stage also called for verifying the $1/\sqrt{r}$ amplitude falloff of outgoing gravitational waves from an orbiting source. At the test-particle mass needed for orbit stability ($m_{\text{test}} \sim 10^{-6}$), the radiated amplitude is at the 10^{-10} level relative to the background, well within the discretization-noise floor of the simple CIC+FD setup. A quantitative slope-fit is deferred to a later Stage 10b once the web visualization is online; the radiation pattern is qualitatively visible in the perturbation field and is the cleanest visual demonstration of gravitational radiation in the sandbox.

The grid-heating problem: why orbits unbind, not spiral in

A physical orbiting mass should LOSE energy to outgoing gravitational waves and slowly spiral inward (the binary-pulsar story). In our simulation we observe the opposite: the orbit *gains* energy and unbinds. This is the well-known PIC **grid-heating** artifact applied to a gravitational-source setup, with the same sign and mechanism as in plasma-PIC codes.

The Hockney–Eastwood theorem guarantees zero self-force for a *stationary* particle in the statically-converged field (verified quantitatively by Test C). For a *moving* particle the adjoint-condition cancellation fails by a residual proportional to the deposited density's time and space derivatives. In a charge-conserving deposition scheme (e.g. Esirkepov 2001 on a Yee staggered grid) this residual is bit-zero by construction. In our simpler CIC+FD setup on a cell-centered grid, the residual is small but nonzero, and its sign systematically pumps energy *into* the particle rather than out of it.

The numerical-self-force-to-physical-centripetal ratio scales as $m_{\text{test}}\, r/M_{\text{bg}}$ in our units, while the condition for the linearized GR equations themselves to hold scales as $r_s^{\text{test}}/r = 2Gm_{\text{test}}/(c^2 r)$ — the numerical condition is the tighter one, with an extra factor of r/r_s^{bg}. For $G = c = M_{\text{bg}} = 1$, $r = 20$: the linearization threshold is $m \ll 10$ but the numerical-stability threshold is $m \ll 0.05$.

Cell-centered Esirkepov is not available. Esirkepov's exact-discrete-continuity recurrence is built around the Yee staggered grid (ρ at corners, J_i at face centers) and its particular divergence stencil. Our cell-centered storage (Section 15) uses a different divergence operator (typically 2-point centered, spanning $2\Delta x$), for which Esirkepov's local recurrence does not directly apply. Adapting it would be a research-level exercise, and migrating the whole grid to Yee staggering — which we explicitly considered and rejected in the Stage 4/5 design — is not worth the refactor for a pedagogical sandbox.

Mitigations available on the cell-centered grid.

- *Stay weak-coupling.* The path Stage 10 takes. Doesn't model radiation reaction; correctly models the static and quasi-static dynamics the sandbox is built for.
- *Higher-order kernel.* Promoting matter deposition from CIC (W_2, 2-cell support) to TSC / quadratic B-spline (W_3, 3-cell support) reduces the high-spatial-frequency content of the deposit and correspondingly reduces heating by $\sim 10\times$. The W_3 kernel infrastructure is already planned for Stage 13's spin-dipole sources and can be promoted to matter sources when convenient.
- *Symmetric binomial smoothing.* Apply a $(1, 2, 1)/4$ binomial filter B to the source arrays after deposition AND to the field slab before particle interpolation, using the same B

on both sides. The effective deposition and interpolation kernels become $W_2 \star B$ and $B \star W_2$ respectively — equal, so the Hockney–Eastwood adjoint condition is *preserved*. B's Fourier transform is $\cos^2(k\Delta x/2)$, which exactly nulls the Nyquist mode and strongly suppresses content near it — exactly where the heating residual is largest. Trade-off: the effective particle size grows to $\sim 4\Delta x$ instead of $2\Delta x$, a slight pedagogical-visualization cost. Order-of-magnitude reduction in heating, not orders.

- *Accept and document.* Frame Stage 10 as "the perturbation FDTD chain is verified to work in the linearization regime; quantitative radiation-reaction physics is outside the sandbox's current scope." The visual radiation pattern in the perturbation field remains a clean qualitative demonstration of gravitational waves regardless.

Asymmetric smoothing (filtering only the source, OR only the field) is the wrong path: it shifts the effective deposition kernel without shifting the interpolation kernel, breaking the adjoint condition and introducing a static self-force that didn't exist before — trading one artifact for another, possibly worse, one. The symmetric form above is the correct pedagogical recipe.

For the current sandbox, "stay weak-coupling" is the working strategy; the upgrade to W_3 matter deposition and/or symmetric binomial smoothing is the natural extension when Stage 13 brings the W_3 kernel online.

18.18 Stage 11: Two-particle gravitational interaction

Two particles mutually attracting via the FDTD perturbation field, no analytic background.

Test:

- Two equal masses at rest should fall toward each other with initial acceleration $G_{\text{eff}} M/r^2$.
- A circular binary orbit should have the correct Keplerian period.
- Results should agree with a direct N-body calculation using the same force law, verifying the FDTD intermediary introduces no force errors.
- The total canonical momentum $\mathbf{P}_{\text{total}} = \sum_i \mathbf{p}_i + c^{-2} \int \mathbf{S}_{\text{GEM}}\, \mathrm{d}^2x$ should be conserved.

18.19 Stage 12: Charged particles and electromagnetism

Add electric charge q to particles.

Test:

- A charge at rest should produce the 2D Coulomb potential $\phi \propto \ln r$.
- Two opposite charges should attract at the correct Coulomb force.
- A moving charge in a uniform background magnetic field B_0 (sampled into $\mathbf{A}^{\text{bg}}$ per Stage 6) should undergo circular cyclotron motion at frequency $\omega_c = qB_0/m\gamma$.
- The EM vector potential $\mathbf{A}$ should correctly encode the magnetic field via the Yee curl.

18.20 Stage 13: EM effective GEM sources

Enable the ρ_{EM} and $\mathbf{J}_{\text{EM}}$ source terms (Eqs. 49–50).

Test: an EM pulse passing a test mass should exert a gravitational impulse via the Poynting flux term $\mathbf{J}_{\text{EM}} = -(\nabla\phi^{\text{pert}} + \partial_t \mathbf{A}^{\text{pert}}) \times (\nabla \times \mathbf{A}^{\text{pert}})/(\mu_0 c^2)$. With boosted coupling constants the deflection should be measurable and match the analytic estimate $\Delta p \sim G_{\text{eff}} \int u_{\text{EM}}\, \mathrm{d}t/(c^2 b)$ where b is the impact parameter.

18.21 Stage 14: Intrinsic spin, precession, spin forces

Add spin $\mathbf{S}$ to particles. Implement the spin update (Section 11.8), the Tulczyjew-Dixon constraint correction, and the spin gradient forces.

Test:

- A gyroscope in a Kerr-like gravitomagnetic background (sampled into $\mathbf{A}_g^{\text{bg}}$ per Stage 6) should precess at rate $\Omega_{\text{LT}} = 2G_{\text{eff}}J/(c_{\text{eff}}^2 r^3)$ for a circular orbit at radius r.
- A spinning charged particle in a background magnetic field B_0 should precess at the Larmor rate $\Omega_L = g_s q B_0/2m$.
- The spin magnitude $|\mathbf{S}|$ should be conserved to machine precision throughout.
- Thomas precession should contribute the correct rate $\Omega_{\text{Thomas}} = -(\gamma^2/(\gamma+1))(\mathbf{v}\times\mathbf{a})/c^2$ for a circular orbit.

18.22 Stage 15: Complete background spacetime library

Extend the Stage 6 background-generator library with the remaining canonical spacetimes from §**??**: Schwarzschild-like, Kerr-like, Reissner–Nordström-like, Kerr–Newman-like, uniform field, binary. Each is just an additional generator that fills the appropriate combination of Φ_g^{bg}, $\mathbf{A}_g^{\text{bg}}$, ϕ^{bg}, $\mathbf{A}^{\text{bg}}$ arrays; no changes to the simulation core are required.

Test each background against its known analytic properties:

- Schwarzschild: photon orbit (unstable) at $r = 3G_{\text{eff}}M/c_{\text{eff}}^2$.
- Kerr-like: frame dragging rate at given r and spin J.
- Reissner-Nordström: Coulomb potential from background charge consistent with ϕ^{bg}.
- Binary: superposition of two Schwarzschild potentials at large separation.

Verify the background + perturbation split produces no spurious transients at startup (convergence iteration correctly initialized).

18.23 Stage 16: Visualization pipeline

Implement the WebGL rendering passes (Section **??**): field textures, quiver geometry, particle sprites.

Test:

- Potential arrays display correctly as false-color images with correct dynamic range.
- On-demand field computation ($\mathbf{g}$, $B_{g,z}$, $\mathbf{E}$, B_z from potentials) matches analytic values for the Stage 7–15 test configurations.
- Particle tracks display at the correct positions with fading.
- Coordinate time and proper time ticks are correctly spaced along the track.
- Spin indicator rotates at the correct precession rate.
- Force arrows point in the correct directions and scale correctly with force magnitude.

18.24 Stage 17: Full integration benchmarks

Run the complete set of pedagogical benchmark scenarios:

Scenario	Expected result
Twin paradox	Proper time clocks on different orbits show measurable difference after one orbital period
Lense-Thirring ring	Gyroscope ring around spinning mass precesses uniformly at Ω_{LT}
Gravitational wave dipole	Outgoing Φ_g^{pert} wave from oscillating mass, $1/\sqrt{r}$ amplitude fall-off
Shapiro delay	EM pulse delay past central mass matches $\Delta t = -2\int \Phi_g \, \mathrm{d}\ell/c^3$
EM wave scattering	EM pulse deflected by charged mass, trajectory bending matches analytic estimate
Binary inspiral	Two-body orbit slowly decays if radiation damping term is enabled
Spin-orbit coupling	Spinning particle orbit precesses differently from non-spinning particle

All scenarios should produce qualitatively correct behavior and agree with the corresponding analytic predictions to within the expected numerical accuracy of the linearized GEM approximation.

19 References

This bibliography collects the works used directly in this design document along with foundational references for the numerical techniques the sandbox implements. References cited inline above are marked with keys here; the remaining entries are recommended further reading for extending the implementation into stages not yet covered.

General relativity, weak-field, post-Newtonian

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Gravitoelectromagnetism and the clock effect

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